

Digital Design & Computer Arch.

Lecture 3: Combinational Logic II and Sequential Logic

Prof. Onur Mutlu

ETH Zürich

Spring 2026

26 February 2026

Extra Credit Assignment: Talk Analysis

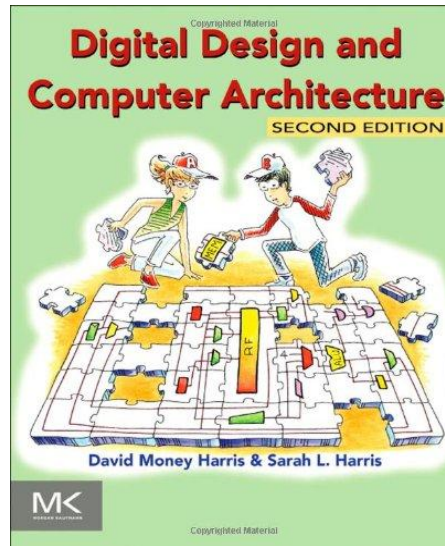
- The Story of RowHammer, RowPress & Beyond
- **Watch and analyze this short lecture (30 minutes)**
 - ❑ <https://www.youtube.com/watch?v=U1EcqXlclKU> (June 2024)



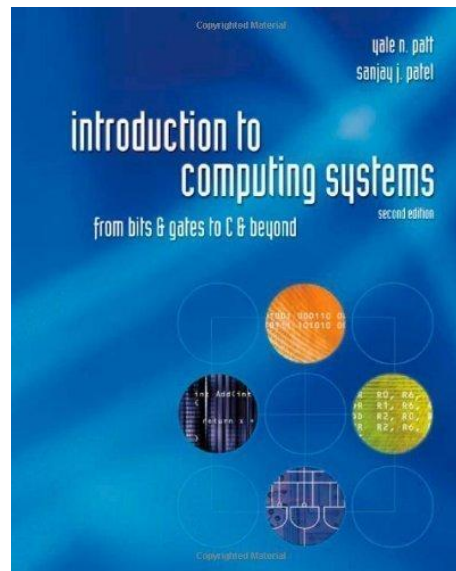
- **Assignment – for 1% extra credit**
 - ❑ **Write a good 1-page individualized summary (no AI use)**
 - What are your key takeaways? What did you learn?
 - What surprised you about the content presented? What excited you?
 - What do you think solutions should be like?
 - Submit your summary to Moodle – deadline March 21

Reading Assignments for This/Next Week

- Chapters 1-2 in Harris & Harris



- Chapters 1,2,3 in Patt and Patel



- Supplementary Lecture Slides on Binary Numbers

Reading Assignments for This/Next Week

■ This week

□ Introduction

■ P&P Chapters 1 & 2 + H&H Chapter 1

□ Combinational Logic

■ P&P Chapter 3 until 3.3 + H&H Chapter 2

■ Next week

□ Hardware Description Languages and Verilog

■ H&H Chapter 4 until 4.3 and 4.5

□ Sequential Logic

■ P&P Chapter 3.4 until end + H&H Chapter 3 in full

■ Within 2-3 weeks, we will be done with

□ **P&P Chapters 1-3 + H&H Chapters 1-4**

What Will We Learn Today?

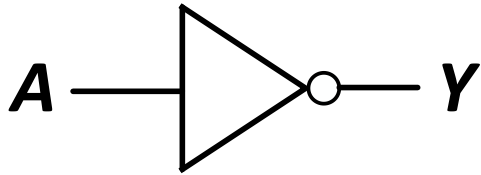
- Complete Combinational Logic
- Start Sequential Logic

Combinational Logic II

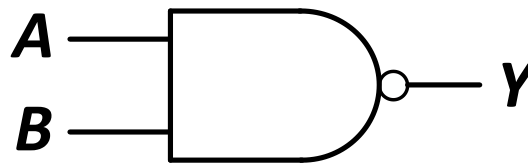
Combinational Logic Lecture Outline

- Building blocks of modern computers
 - Transistors
 - Logic gates
- Combinational logic circuits
- Boolean algebra
- Using Boolean algebra to represent combinational circuits
- Basic combinational logic blocks
- Simplifying combinational logic circuits

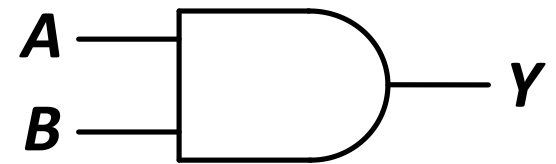
CMOS NOT, NAND, AND Gates



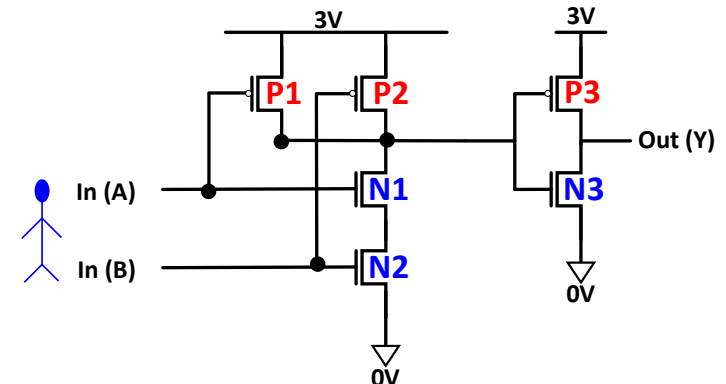
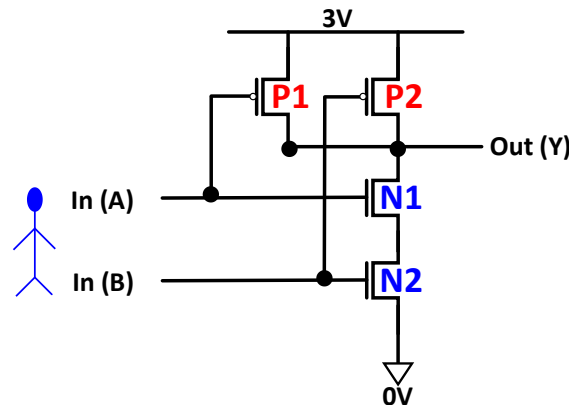
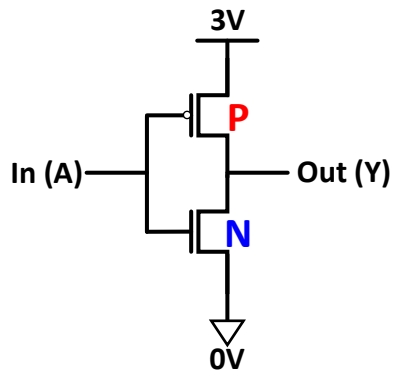
A	Y
0	1
1	0



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



Combinational Logic Circuits

Boolean Logic Equations

Using Boolean Equations to Represent a Logic Circuit

Recall: Boolean Equations Enable Us To...

- Represent the function of a combinational logic block
 - Functional Specification
- Methodically transform the function into simpler functions
 - which lead to different hardware realizations
 - Logic Minimization or Logic Simplification
 - We can automate this process → Computer-Aided Design or Electronic Design Automation
- Different Boolean expressions lead to different logic gate implementations
 - Different hardware area, cost, latency, energy properties

Recall: Standardized SOP and POS Forms

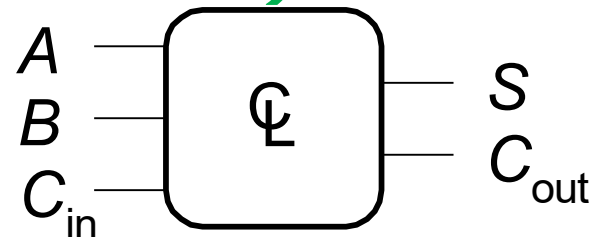
- Enable a single, universally-agreed-on way of representing a Boolean function starting from its truth table
 - Also called “canonical representations”
- Sum of Products (SOP) form
- Product of Sums (POS) form

Recall: Logic Simplification (Minimization)

- Using Boolean Algebra, we can simplify the SOP or POS form of any function in a methodical way
- Starting with the canonical SOP or POS form enables convenience and automation
 - Truth table → SOP/POS form → Boolean Simplification Rules
- **Example (full 1-bit adder – more later):**

$$S = F(A, B, C_{in})$$

$$C_{out} = G(A, B, C_{in})$$



$$S = A \oplus B \oplus C_{in} \quad \text{3-input XOR}$$

$$C_{out} = AB + AC_{in} + BC_{in}$$

3-input majority

Combinational Building Blocks used in Modern Computers

Recall: Common Logic Gates

Buffer



A	Z
0	0
1	1

AND



A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

OR



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

XOR



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

Inverter



A	Z
0	1
1	0

NAND



A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

NOR



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

XNOR



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

We Will Cover Many Building Blocks

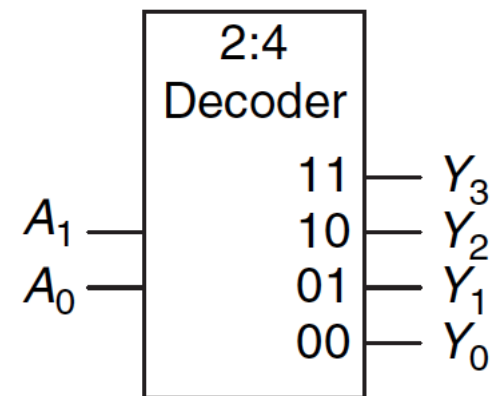
- Basic logic gates (AND, OR, NOT, NAND, NOR, XOR)
 - Decoder
 - Multiplexer
 - Full Adder
 - Programmable Logic Array (PLA)
 - Comparator
 - Arithmetic Logic Unit (ALU)
 - Tri-State Buffer
-
- Standard form representations: SOP & POS
 - Logic simplification via Boolean Algebra
 - Logical completeness

Decoder

Recall: Decoder

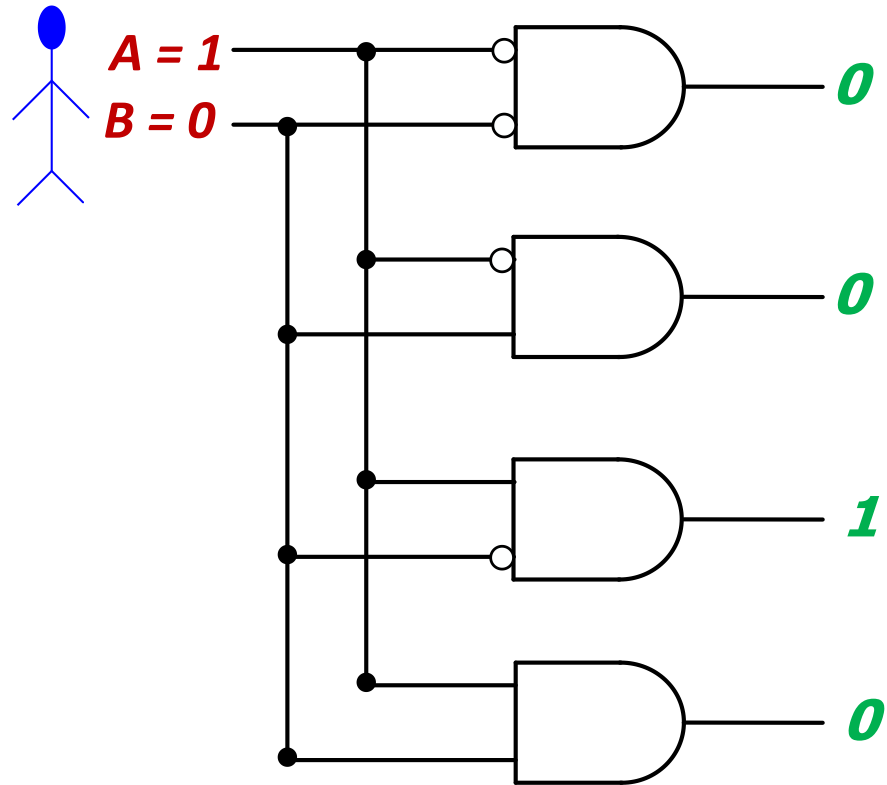
- “Input pattern detector”
- n inputs and 2^n outputs
- Exactly one of the outputs is 1 and all the rest are 0s
- The **output** that is logically 1 is the output corresponding to the input **pattern** that the logic circuit is expected to detect
- Example: 2-to-4 decoder

A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Recall: Decoder (II)

- The decoder is useful in determining how to interpret a bit pattern
 - **It could be the address of a location in memory, that the processor intends to read from**
 - **It could be an instruction in the program and the processor needs to decide what action to take (based on *instruction opcode*)**

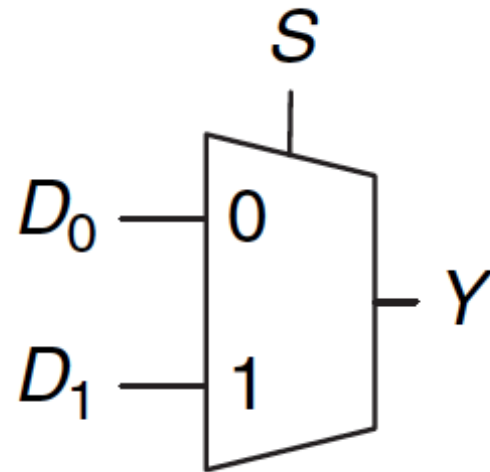


Multiplexer (MUX)

Recall: Multiplexer (MUX), or Selector

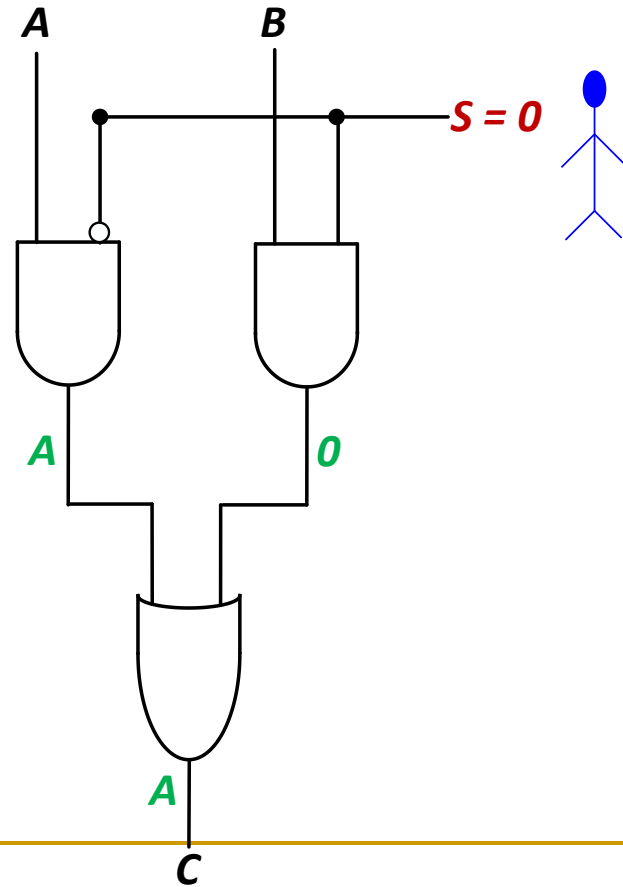
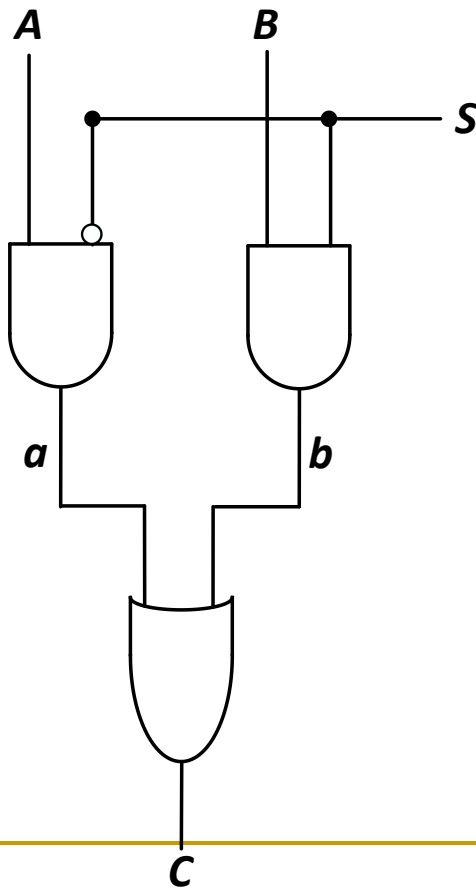
- **Selects** one of the N inputs to connect it to the output
 - based on the value of a $\log_2 N$ -bit control input called **select**
- Example: 2-to-1 MUX

S	D_1	D_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



Recall: Multiplexer (MUX), or Selector (II)

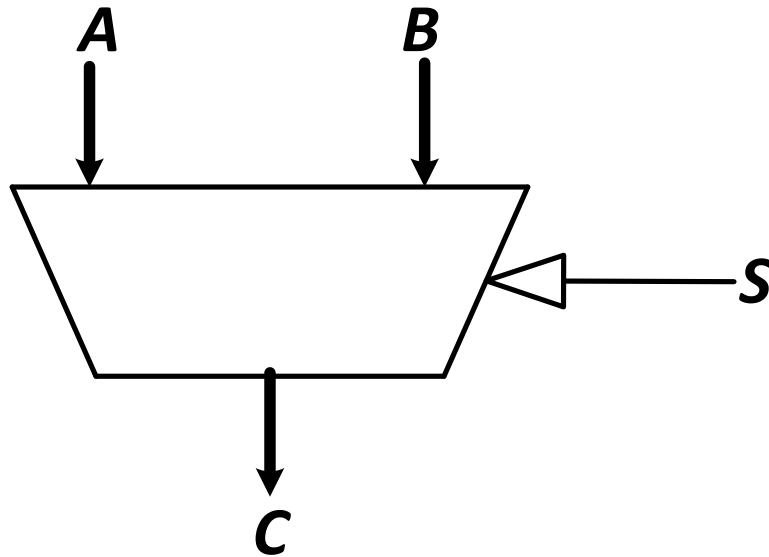
- **Selects** one of the N inputs to connect it to the output
 - based on the value of a $\log_2 N$ -bit control input called **select**
- Example: 2-to-1 MUX



Recall: Multiplexer (MUX), or Selector (III)

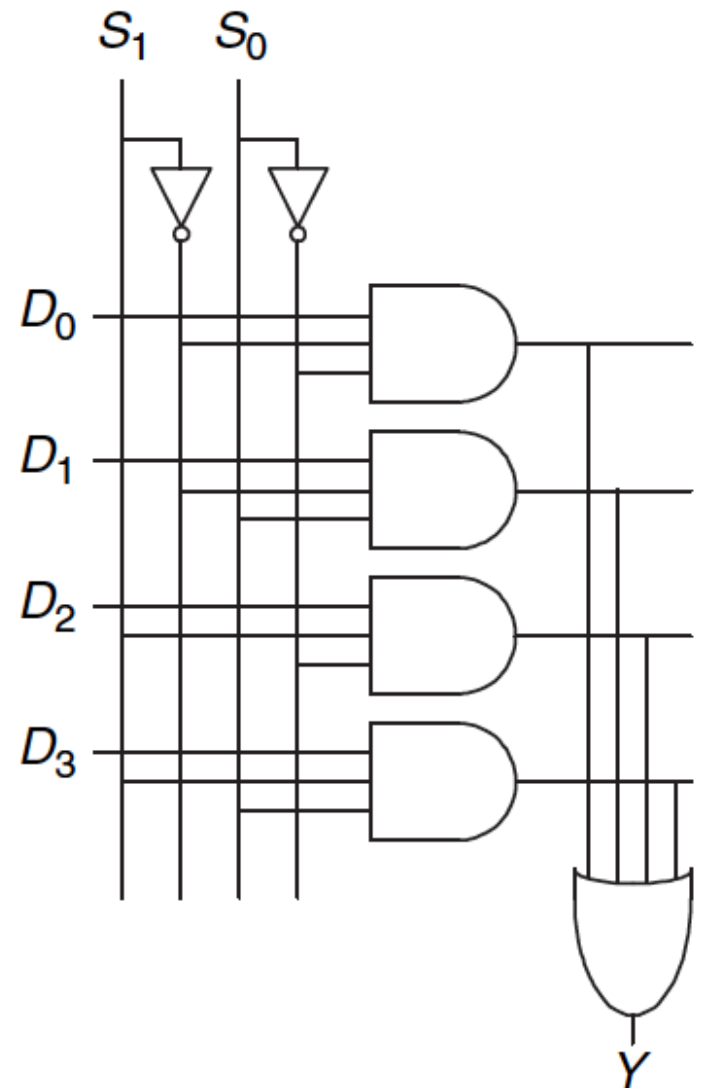
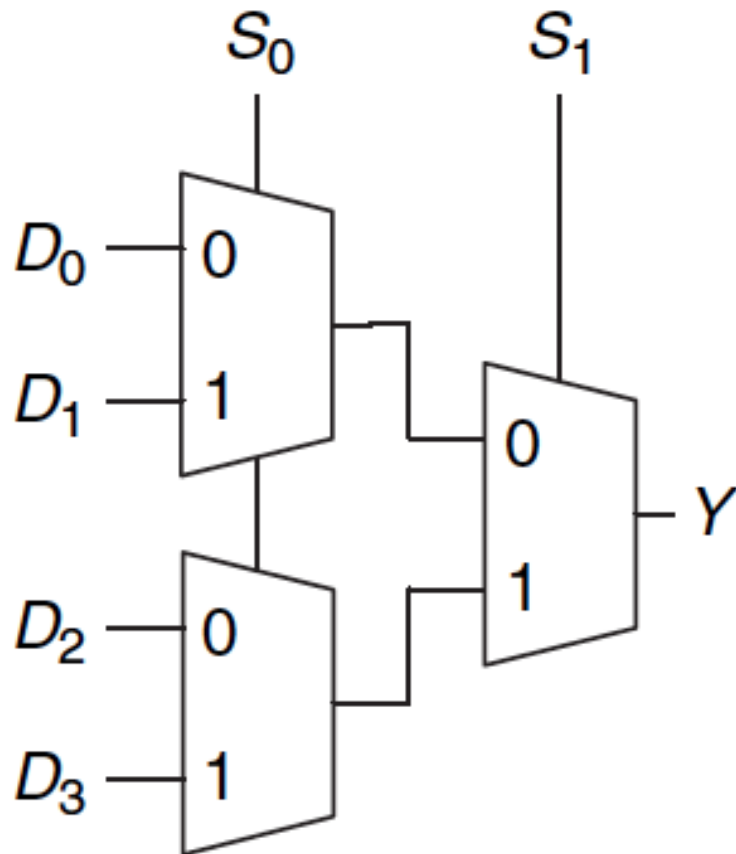
- The output C is always connected to either the input A or the input B
 - Output value depends on the value of the **select line S**

S	C
0	A
1	B



- **Your task:** Draw the schematic for an 4-input (4:1) MUX
 - Gate level: as a combination of basic AND, OR, NOT gates
 - Module level: As a combination of 2-input (2:1) MUXes

Recall: A 4-to-1 Multiplexer



Recall: Aside: Logic Using Multiplexers

- Multiplexers can be used as “lookup tables” to perform logic functions

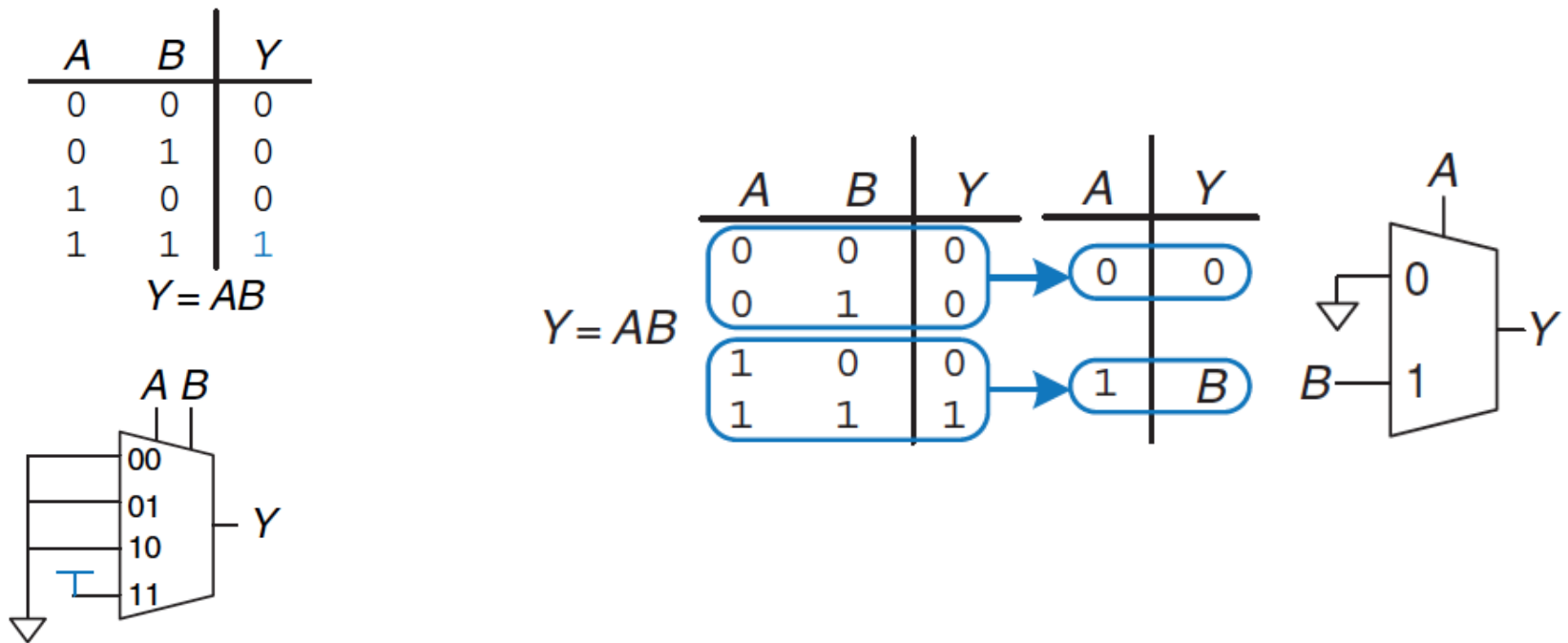
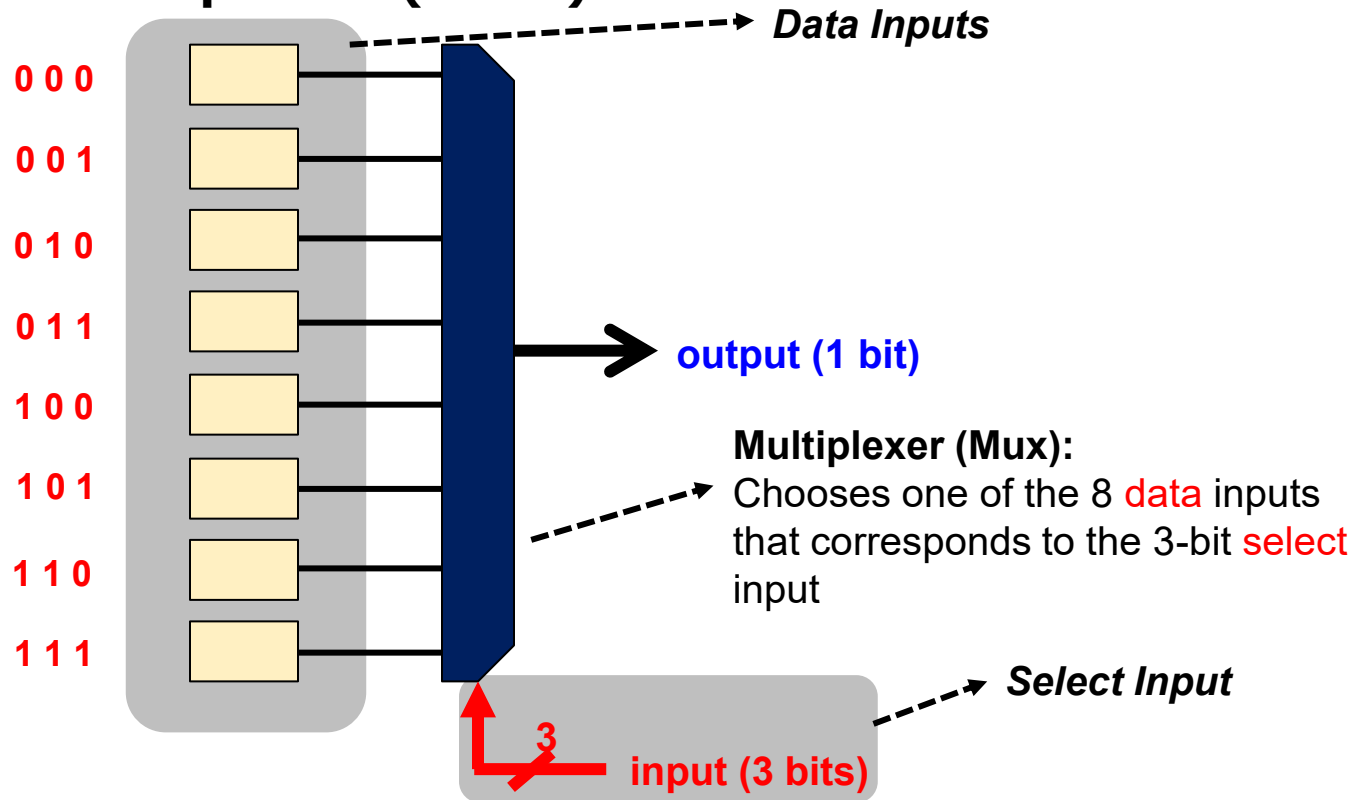


Figure 2.59 4:1 multiplexer implementation of two-input AND function

Recall: 8-Input Lookup Table (LUT)

■ 3-bit input LUT (3-LUT)



3-LUT can implement
any 3-bit input function

Recall: An Example of Programming a LUT

- A function that outputs '1' when there are **at least two '1's in a 3-bit input**

In C:

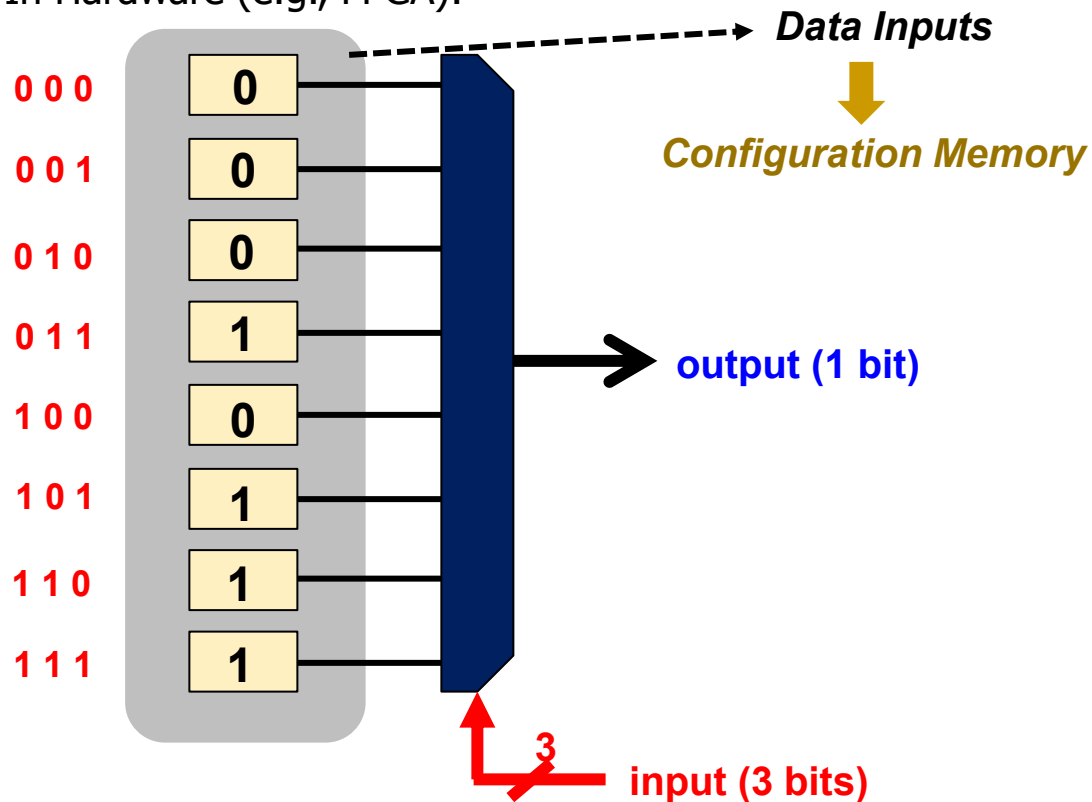
```
int count = 0;
for(int i = 0; i < 3; i++) {
    count += input & 1;
    input = input >> 1;
}

if(count > 1) return 1;

return 0;
```

```
switch(input){
    case 0:
    case 1:
    case 2:
    case 4:
        return 0;
    default:
        return 1;}
```

In Hardware (e.g., FPGA):



Recall: Aside: Logic Using Decoders (I)

- Decoders can be combined with OR gates to build logic functions.

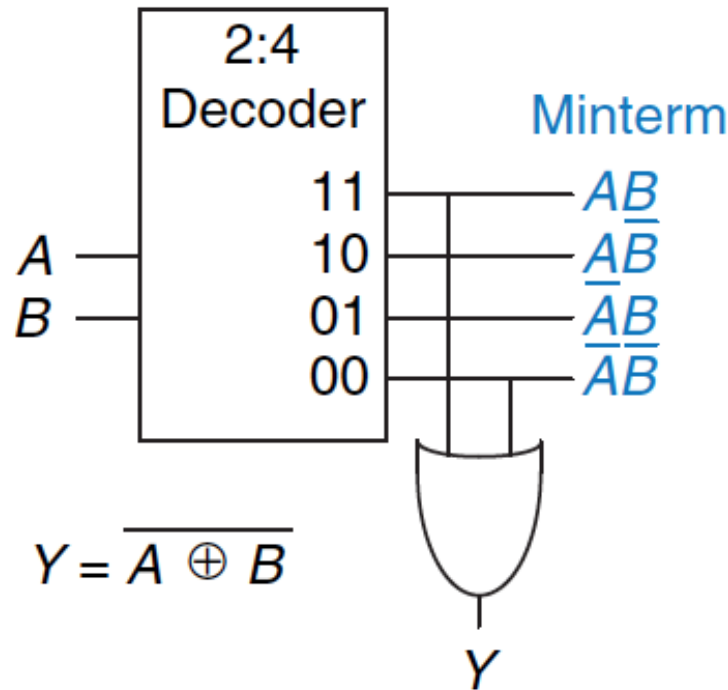


Figure 2.65 Logic function using decoder

We Will Cover Many Building Blocks

- Basic logic gates (AND, OR, NOT, NAND, NOR, XOR)
 - Decoder
 - Multiplexer
 - Full Adder
 - Programmable Logic Array (PLA)
 - Comparator
 - Arithmetic Logic Unit (ALU)
 - Tri-State Buffer
-
- Standard form representations: SOP & POS
 - Logic simplification via Boolean Algebra
 - Logical completeness

Full Adder

Full Adder (I)

■ Binary addition

- Similar to decimal addition
- From right to left
- One column at a time
- One sum and one carry bit

$$\begin{array}{r}
 a_{n-1}a_{n-2} \dots a_1a_0 \\
 b_{n-1}b_{n-2} \dots b_1b_0 \\
 \underline{C_n C_{n-1} \dots C_1} \\
 S_{n-1} \dots S_1S_0
 \end{array}$$

↓

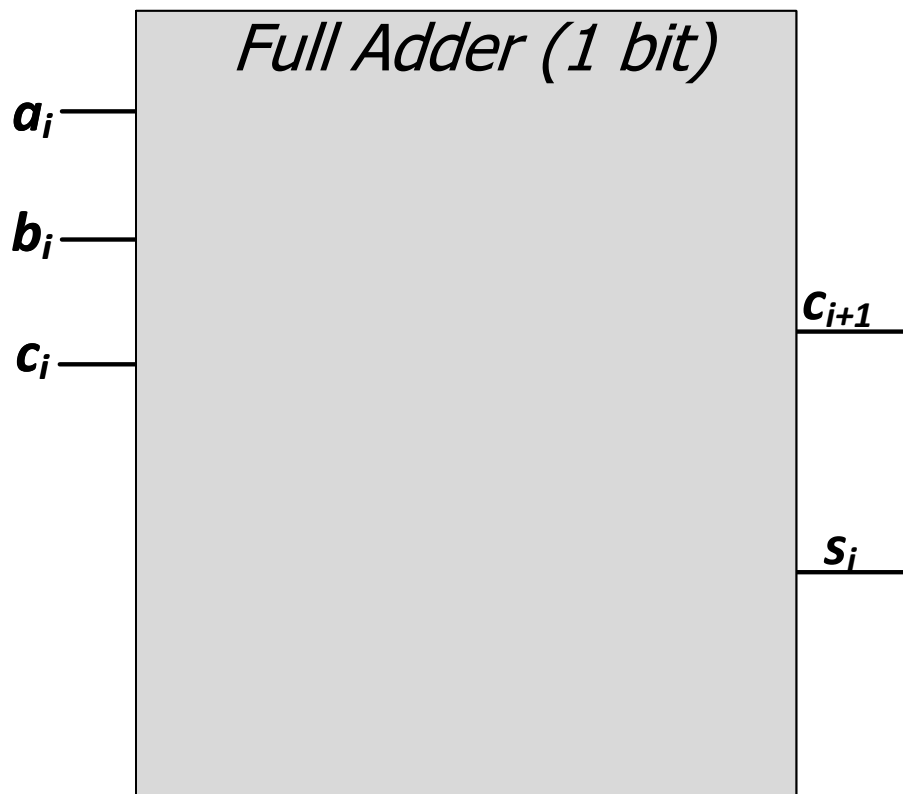
- Truth table of binary addition on **one column** of bits within two n-bit operands

a_i	b_i	$carry_i$	$carry_{i+1}$	S_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Full Adder (II)

■ Binary addition

- N 1-bit additions
- **SOP of 1-bit addition**



$$\begin{array}{r}
 a_{n-1}a_{n-2} \dots a_1a_0 \\
 b_{n-1}b_{n-2} \dots b_1b_0 \\
 \hline
 C_n C_{n-1} \dots C_1 \\
 \hline
 S_{n-1} \dots S_1S_0
 \end{array}$$

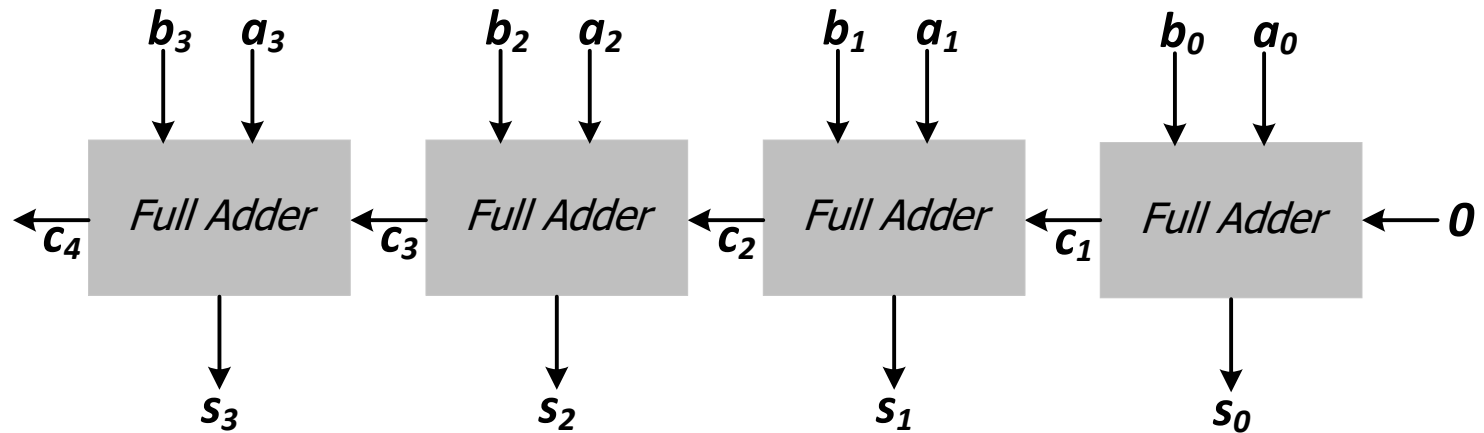
↓

a_i	b_i	$carry_i$	$carry_{i+1}$	S_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

MAJ XOR

4-Bit Adder from Full Adders

- Creating a **4-bit adder** out of 1-bit full adders
 - To add two 4-bit binary numbers A and B



$$\begin{array}{rcccc} & a_3 & a_2 & a_1 & a_0 \\ + & b_3 & b_2 & b_1 & b_0 \\ \hline c_4 & c_3 & c_2 & c_1 & \\ \hline s_3 & s_2 & s_1 & s_0 & \end{array}$$

$$\begin{array}{rcccc} & 1 & 0 & 1 & 1 \\ + & 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 1 & 1 & \\ \hline 0 & 1 & 0 & 0 & \end{array}$$

Adder Design: Ripple Carry Adder

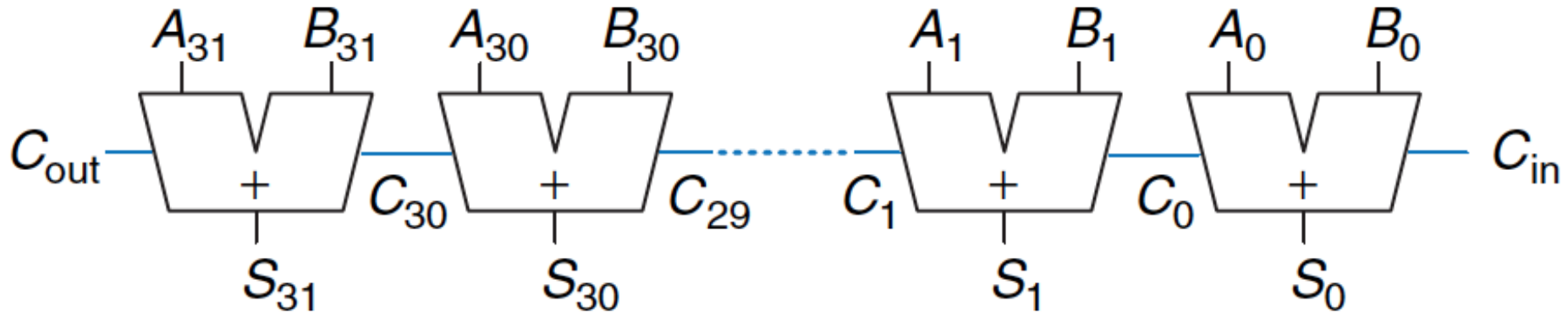
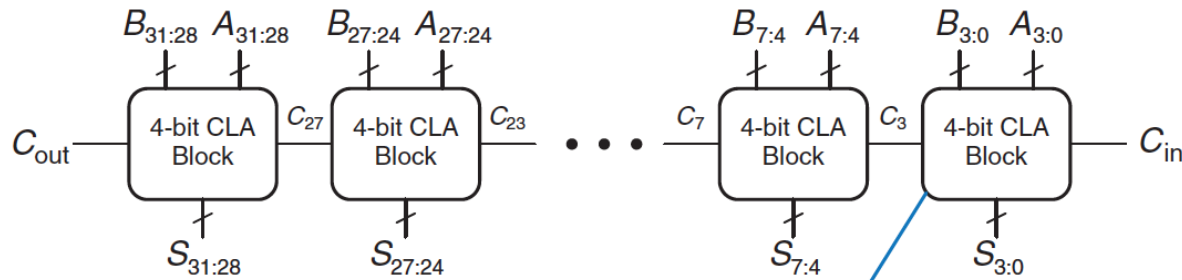
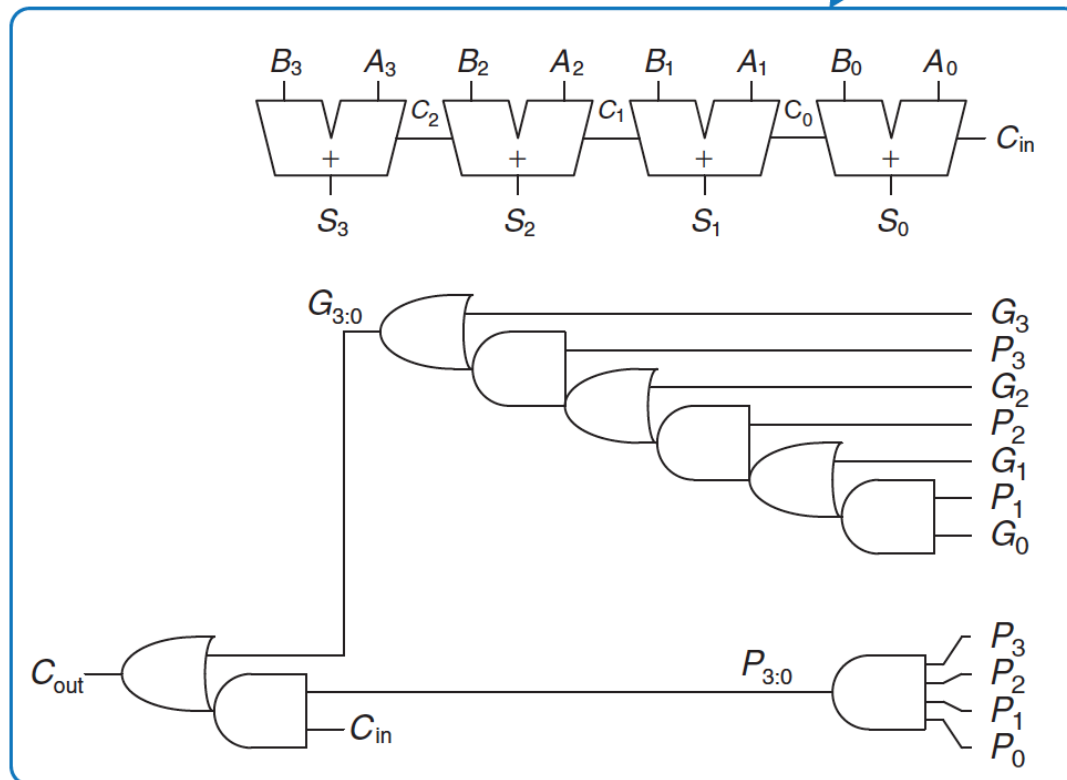


Figure 5.5 32-bit ripple-carry adder

Adder Design: Carry Lookahead Adder



(a)



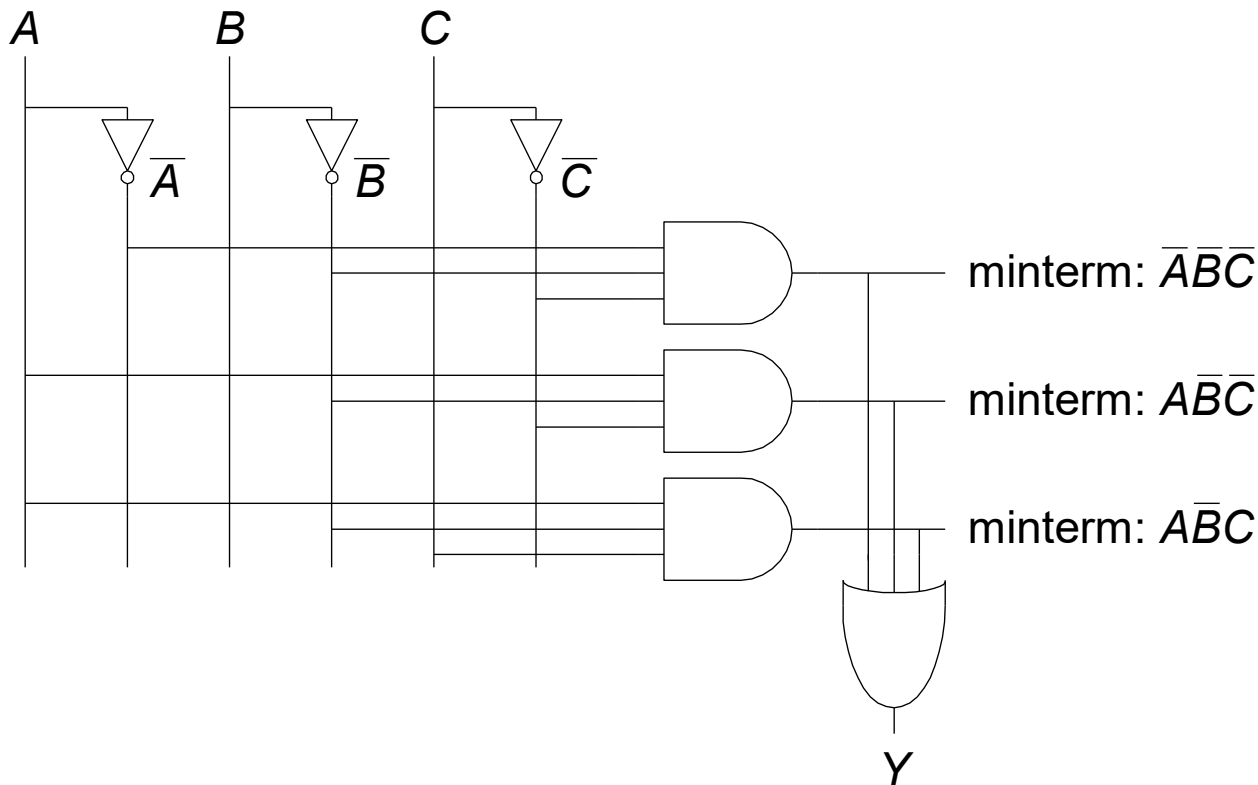
(b)

**Example of
logic specialization:
Specialized logic for
fast carry calculation**

Programmable Logic Array (PLA)

PLA: Recall: SOP Form

- **SOP (sum-of-products) leads to two-level logic**
- Example: $Y = (\bar{A} \cdot \bar{B} \cdot \bar{C}) + (A \cdot \bar{B} \cdot \bar{C}) + (A \cdot \bar{B} \cdot C)$



A PLA enables the two-level SOP implementation of **any** N-input M-output function

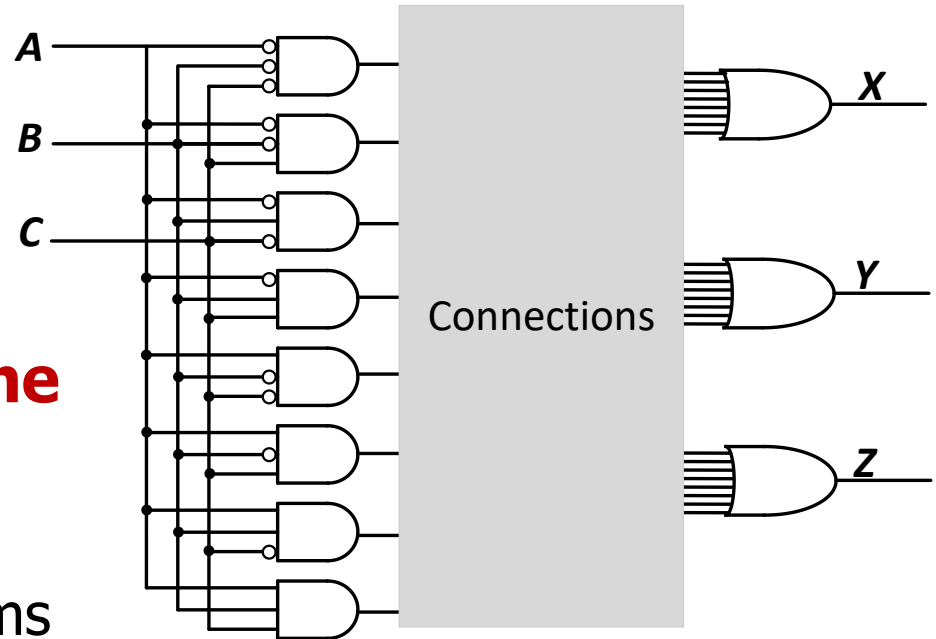
The Programmable Logic Array (PLA)

- The below logic structure is a very **common** building block for implementing any collection of logic functions one wishes to

- An **array** of AND gates followed by an **array** of OR gates
- **How do we determine the number of AND gates?**

- **Remember SOP:** the number of possible minterms
- For an n -input logic function, we need a PLA with 2^n n -input AND gates

- **How do we determine the number of OR gates?** The number of output columns in the truth table

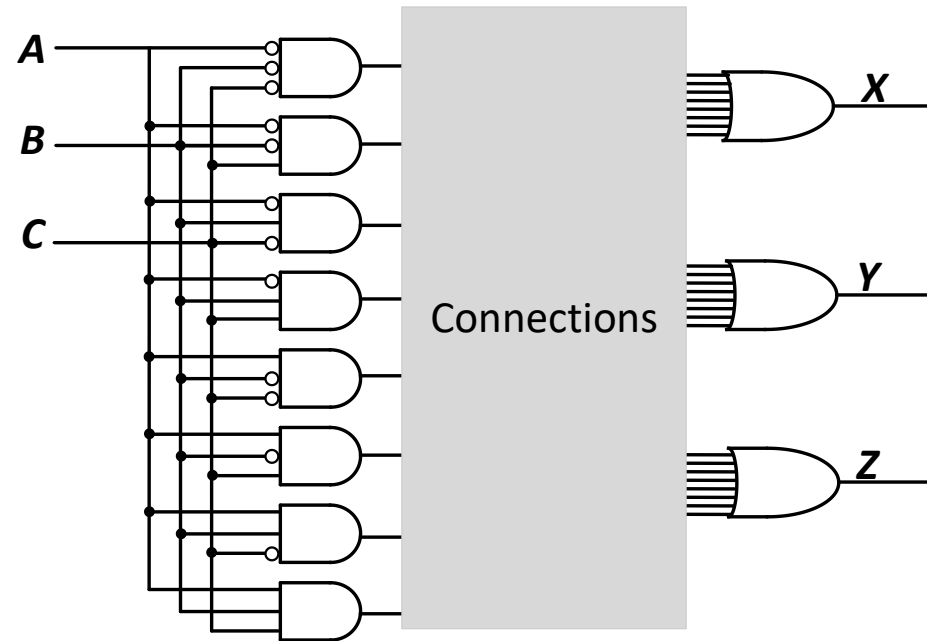


A PLA enables the two-level SOP implementation of **any** N -input M -output function

The Programmable Logic Array (PLA)

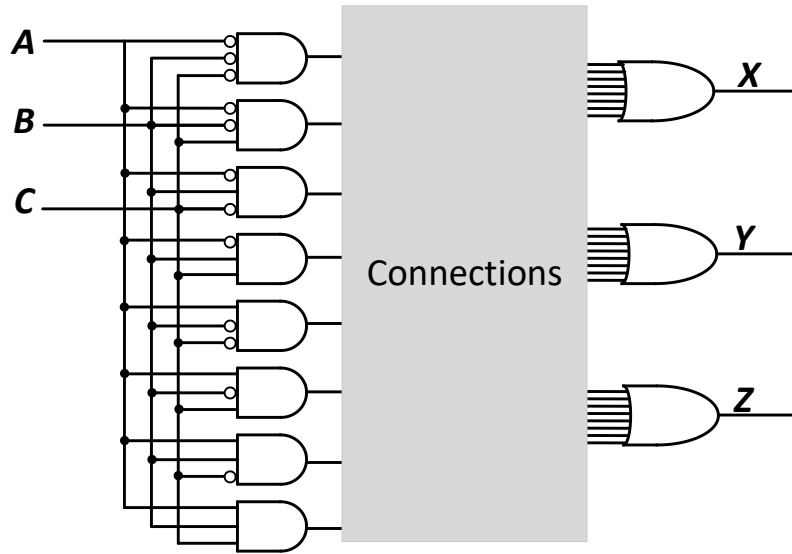
- How do we implement a logic function?
 - Connect the output of an AND gate to the input of an OR gate if the corresponding minterm is included in the SOP
 - This is a simple programmable logic construct

- **Programming a PLA:** we program the connections from AND gate outputs to OR gate inputs to implement a desired logic function



- Is there any other type of programmable logic?
 - Yes! An FPGA...
 - An FPGA uses more advanced structures, as we see in the labs

Implementing a Full Adder Using a PLA

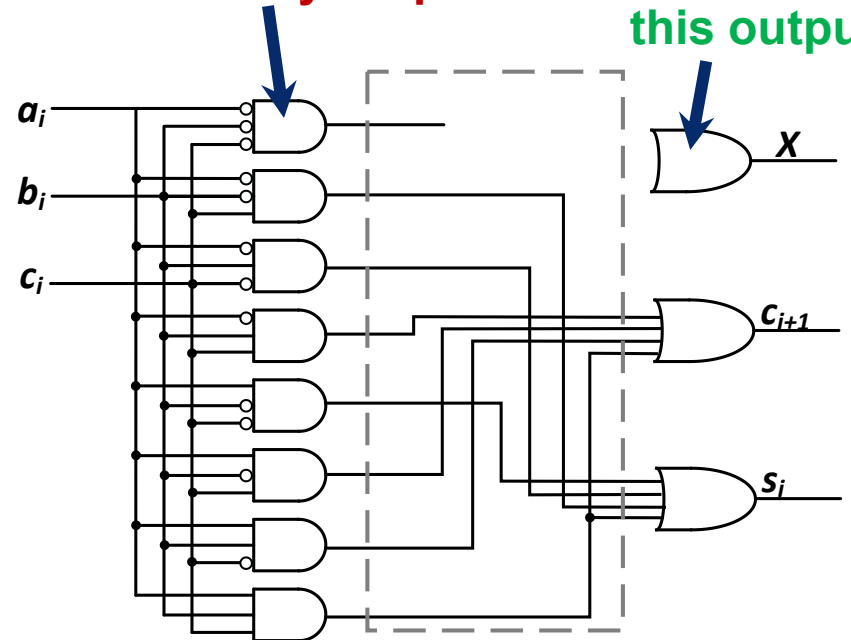


Truth table of a full adder

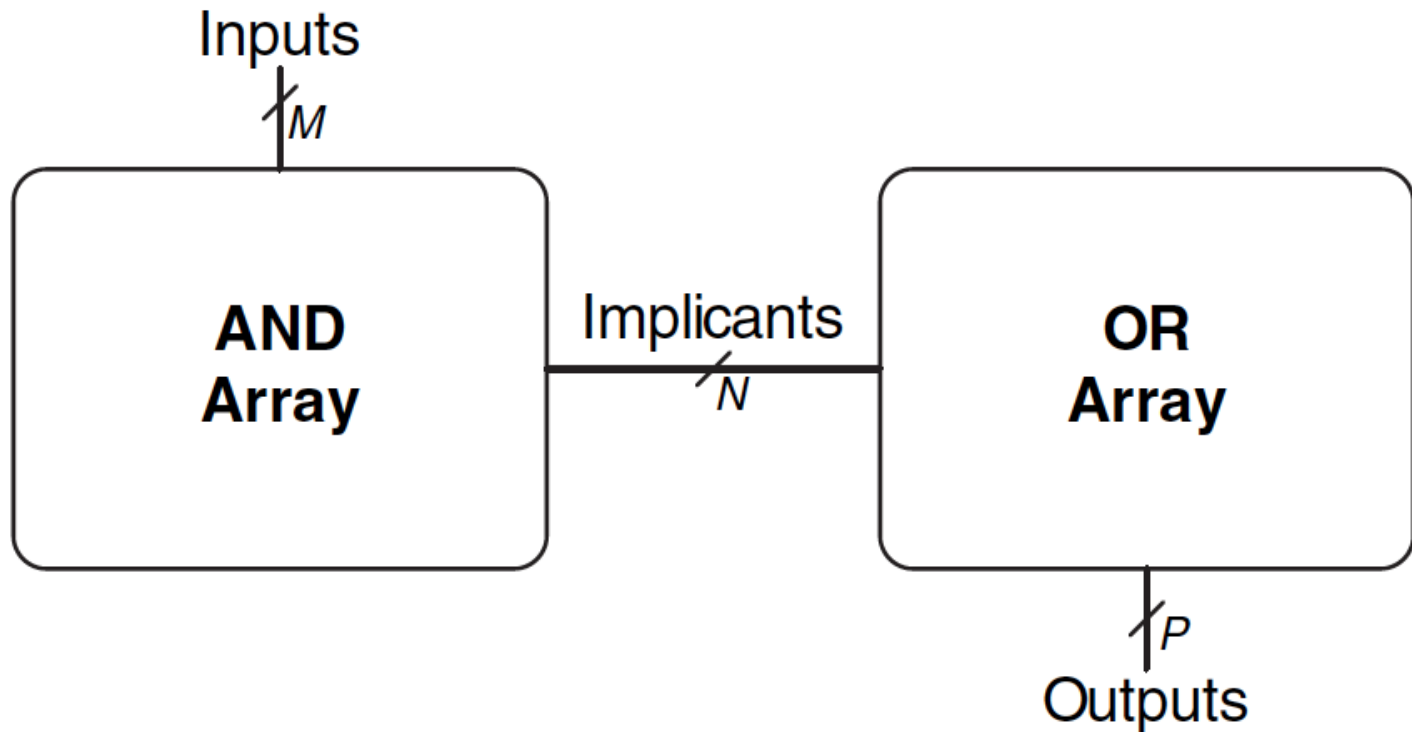
a_i	b_i	$carry_i$	$carry_{i+1}$	S_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

This input should not be connected to any outputs

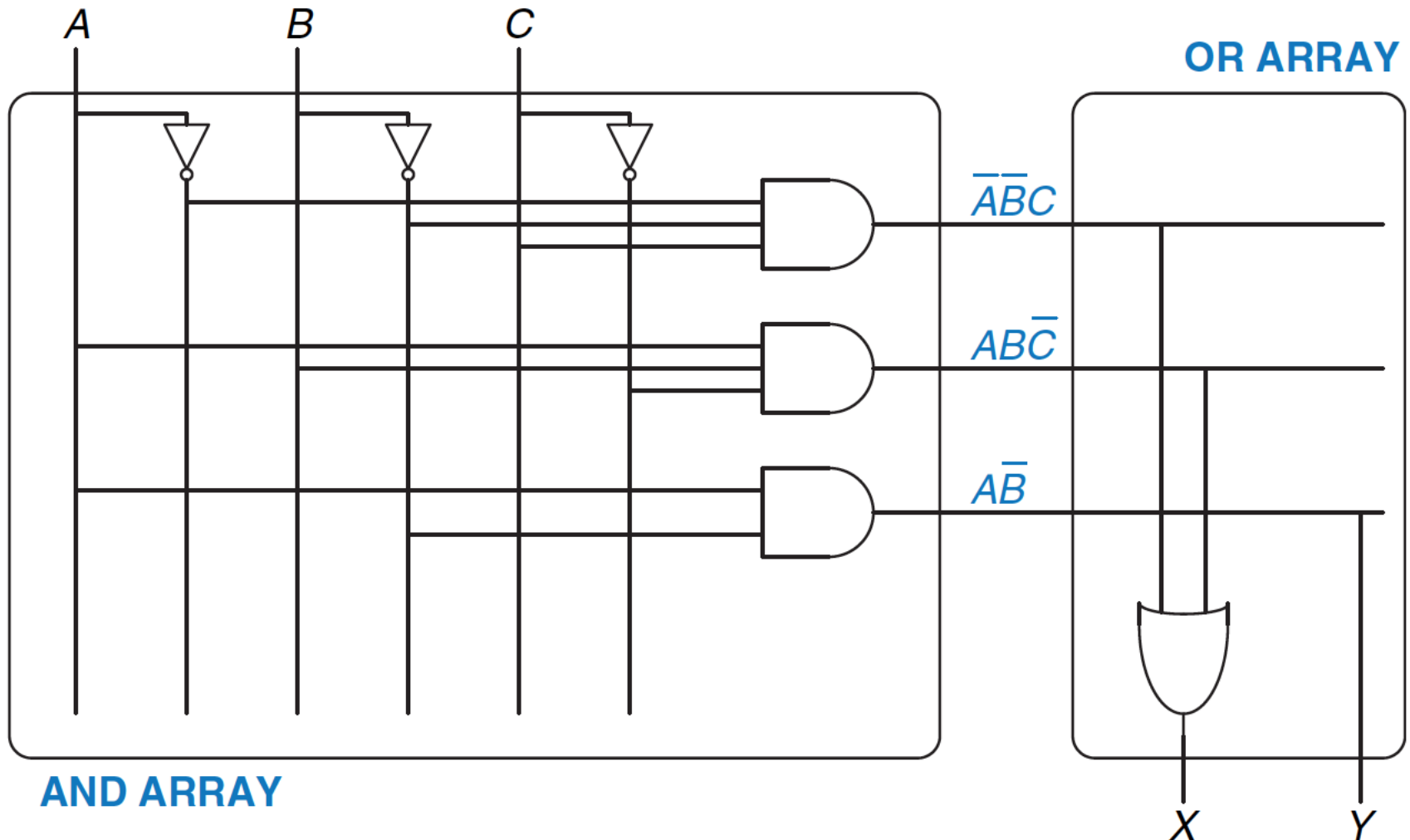
We do not need this output



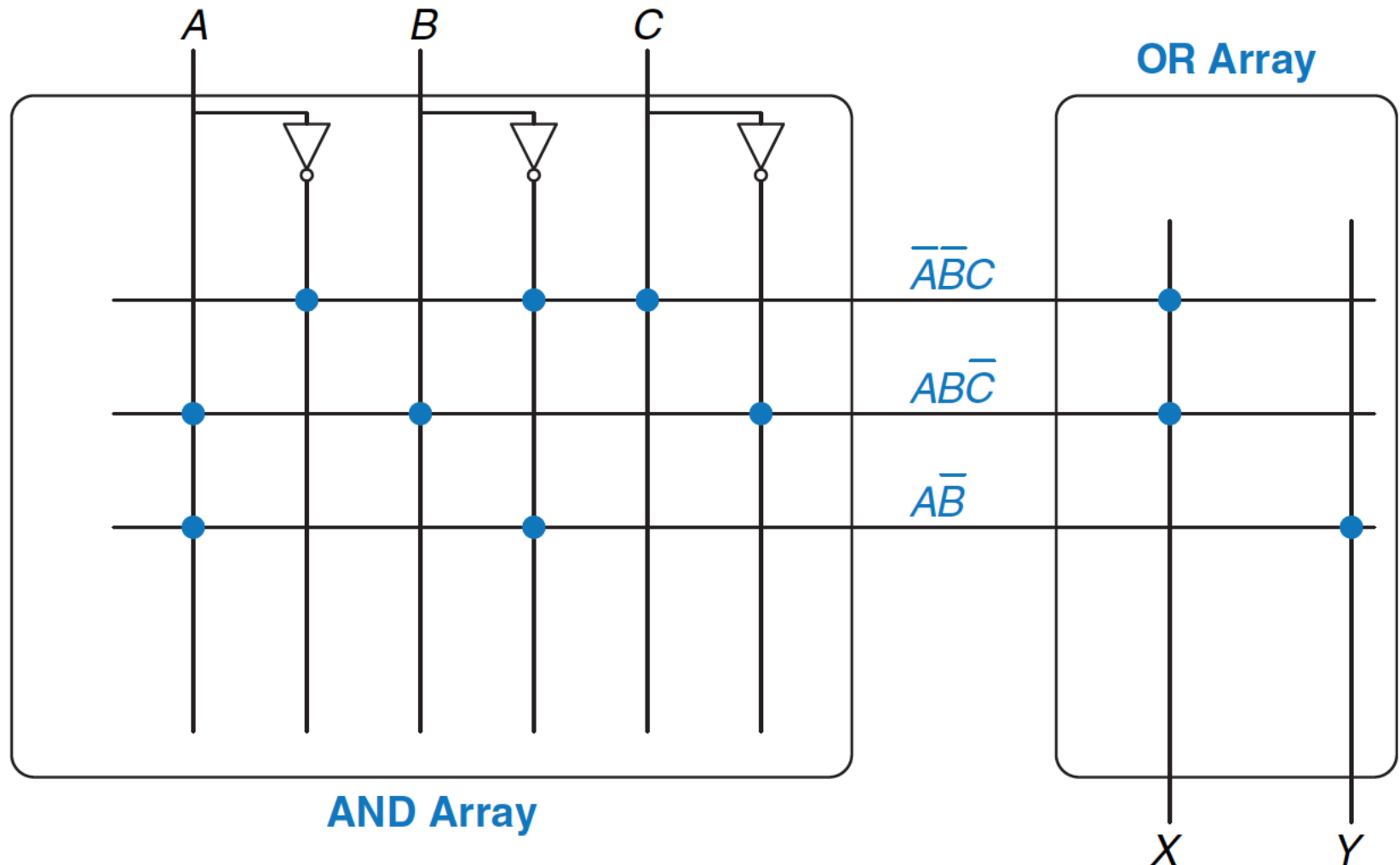
PLA Example (I)



PLA Example Function (II)



PLA Example Function (III)



Logical Completeness

Logical (Functional) Completeness

- **Any logic function** we wish to implement could be accomplished with a PLA
 - PLA consists of **only** AND gates, OR gates, and inverters
 - We just have to program connections based on SOP of the intended logic function
- The set of gates {AND, OR, NOT} is **logically complete** because we can build a circuit to carry out the specification of **any truth table** we wish, without using any other kind of gate
- NAND is also logically complete. So is NOR.
 - **Your task:** Prove this.

More Combinational Blocks

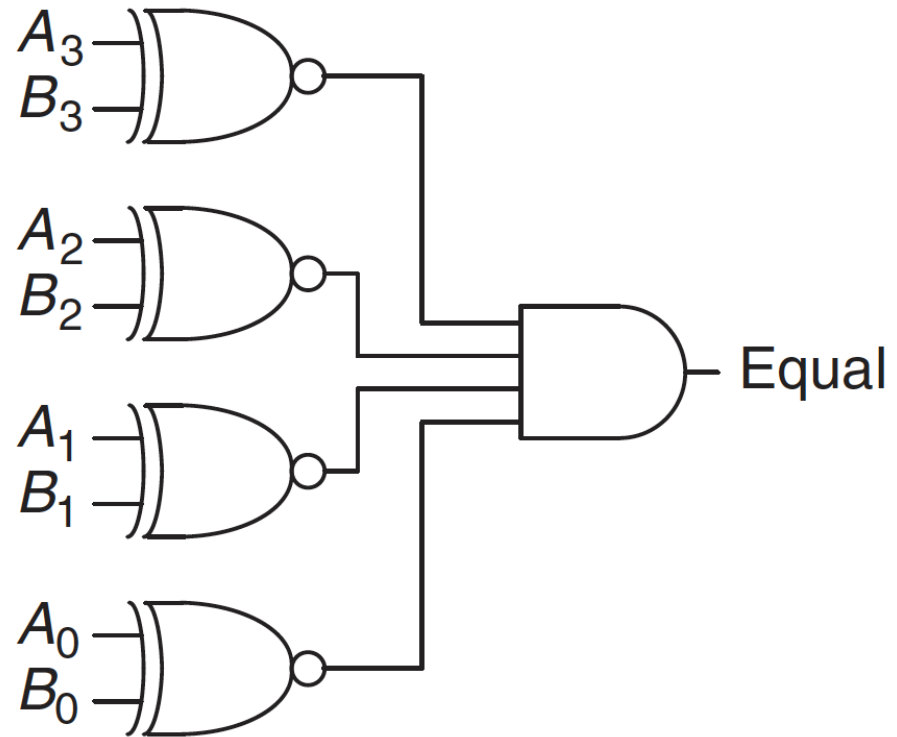
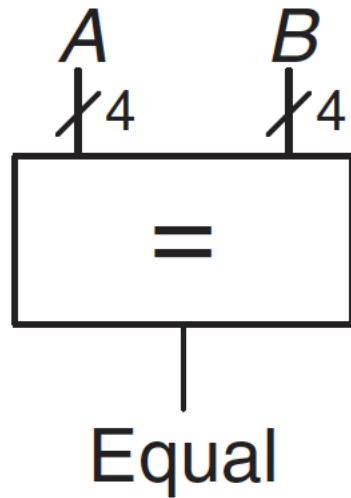
More Combinational Building Blocks

- H&H Chapter 2 in full
 - Required Reading
 - E.g., see Tri-state Buffer and Z values in Section 2.6
- H&H Chapter 5
 - Will be required reading soon
- You will benefit greatly by reading the “combinational” parts of Chapter 5 soon.
 - Sections 5.1 and 5.2
 - E.g., Adder, Subtractor, Comparator, Shifter/Rotator, Multiplier, Divider

Comparator

Equality Checker (Compare if Equal)

- Checks if two N-input values are exactly the same
- Example: 4-bit Comparator



ALU (Arithmetic Logic Unit)

ALU (Arithmetic Logic Unit)

- Combines a variety of arithmetic and logical operations into a single unit (that performs only one function at a time)
- Usually denoted with this symbol:

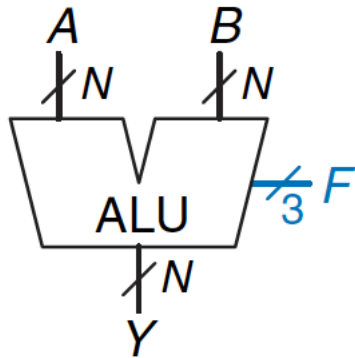


Figure 5.14 ALU symbol

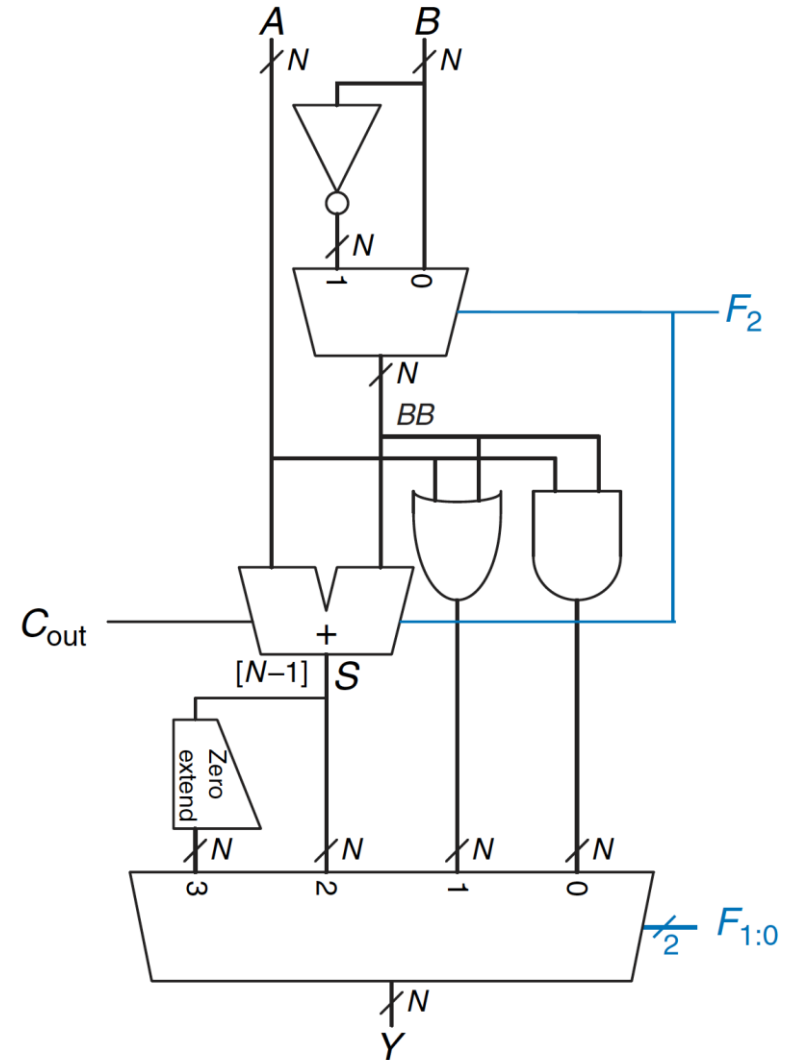
Table 5.1 ALU operations

$F_{2:0}$	Function
000	A AND B
001	A OR B
010	A + B
011	not used
100	A AND $\bar{B}$
101	A OR $\bar{B}$
110	A - B
111	SLT

Example ALU (Arithmetic Logic Unit)

Table 5.1 ALU operations

$F_{2:0}$	Function
000	A AND B
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010	A + B
011	not used
100	A AND $\overline{B}$
101	A OR $\overline{B}$
110	A – B
111	SLT



More Combinational Building Blocks

- See H&H Chapter 5.2 for
 - Subtractor (using 2's Complement Representation)
 - Shifter and Rotator
 - Multiplier
 - Divider
 - ...

More Combinational Building Blocks

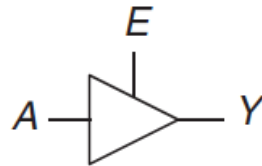
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Tri-State Buffer

Tri-State Buffer

- A tri-state buffer enables gating of different signals onto a wire

Tristate
Buffer



<i>E</i>	<i>A</i>	<i>Y</i>
0	0	Z
0	1	Z
1	0	0
1	1	1

**A tri-state buffer
acts like a switch**

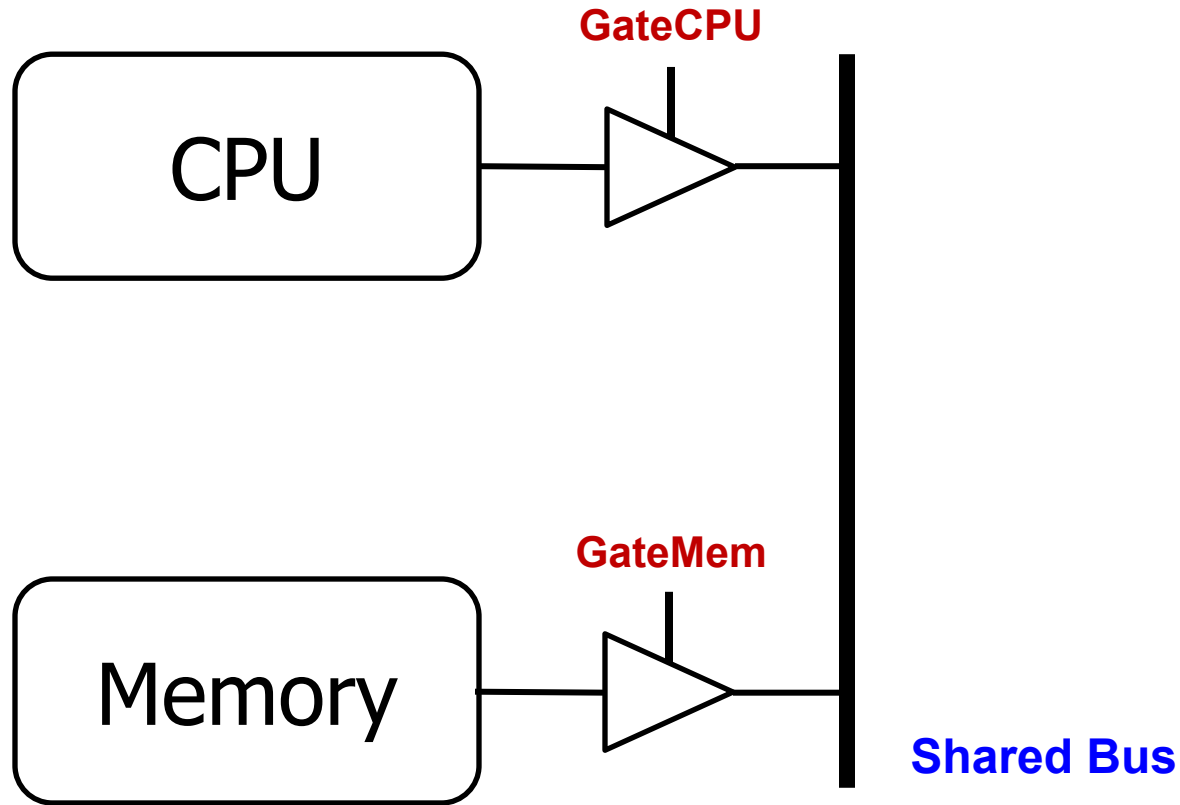
Figure 2.40 Tristate buffer

- **Floating signal (Z):** Signal that is not driven by any circuit
 - Open circuit, floating wire

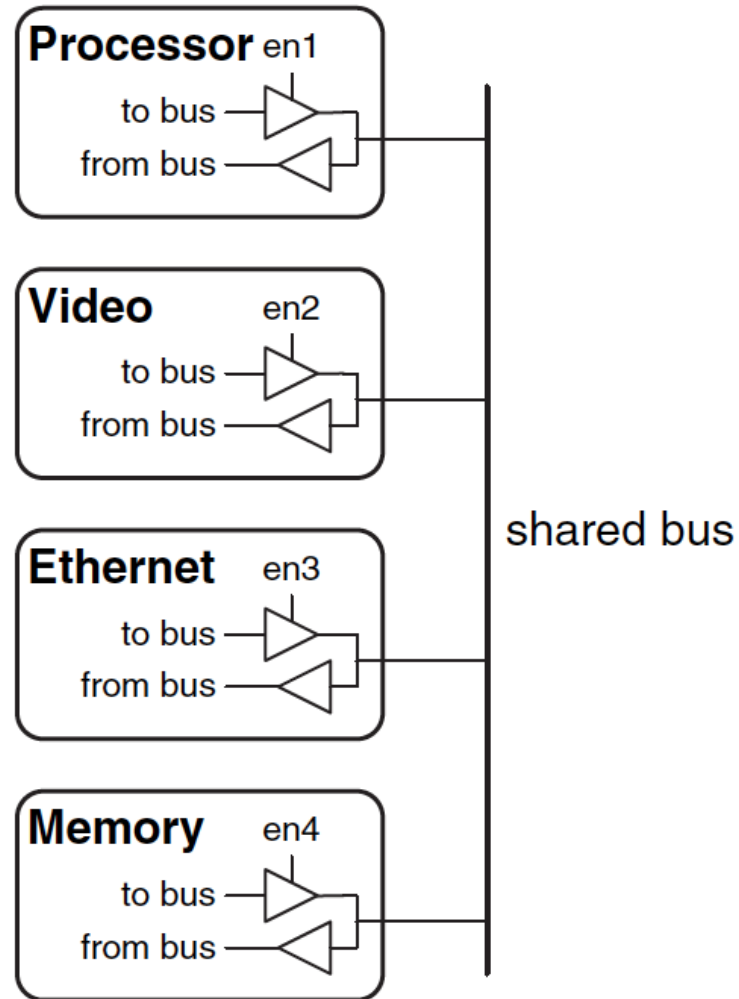
Example: Use of Tri-State Buffers

- Imagine a wire connecting the CPU and memory
 - At any time only the CPU or the memory can place a value on the wire, both not both
 - You can have two tri-state buffers: one driven by CPU, the other memory; and ensure at most one is enabled at any time

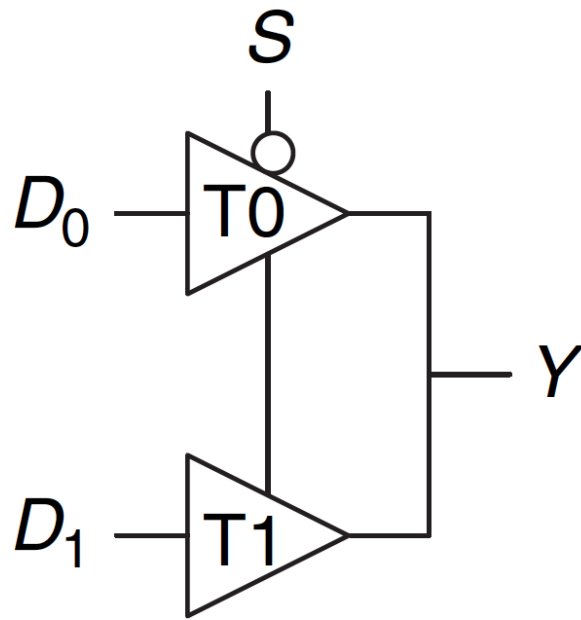
Example Design with Tri-State Buffers



Another Example

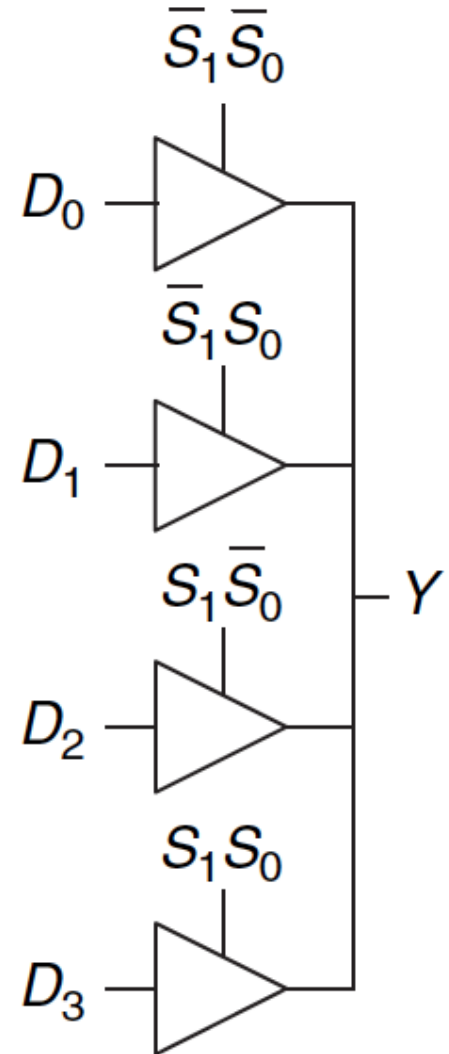


Multiplexer Using Tri-State Buffers

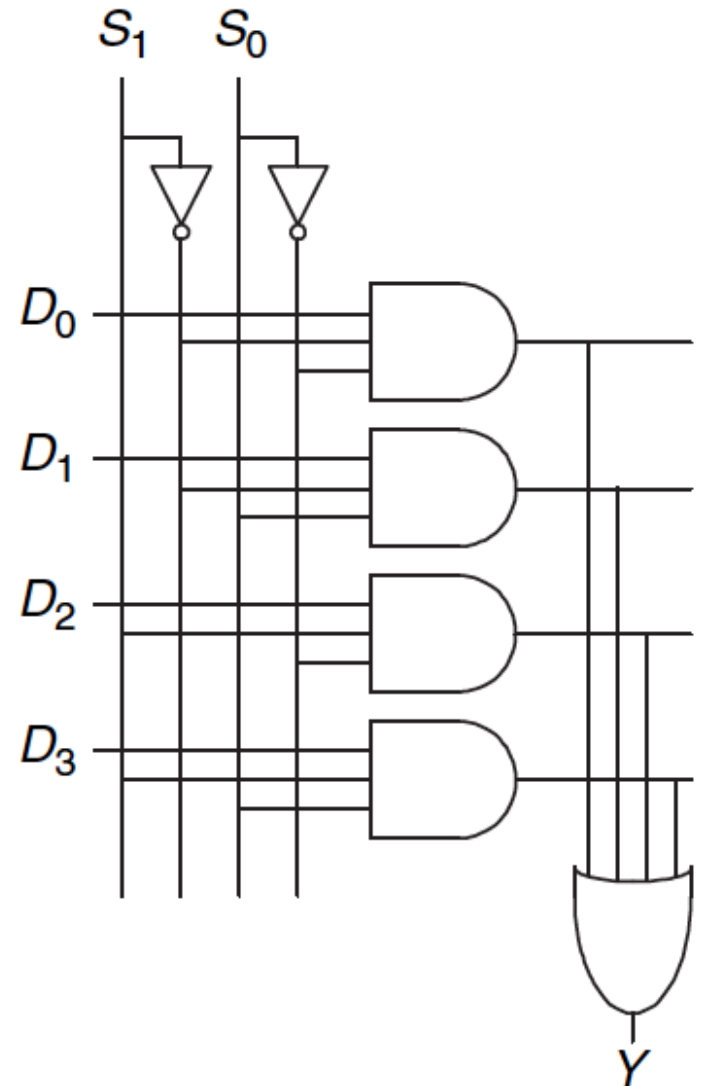
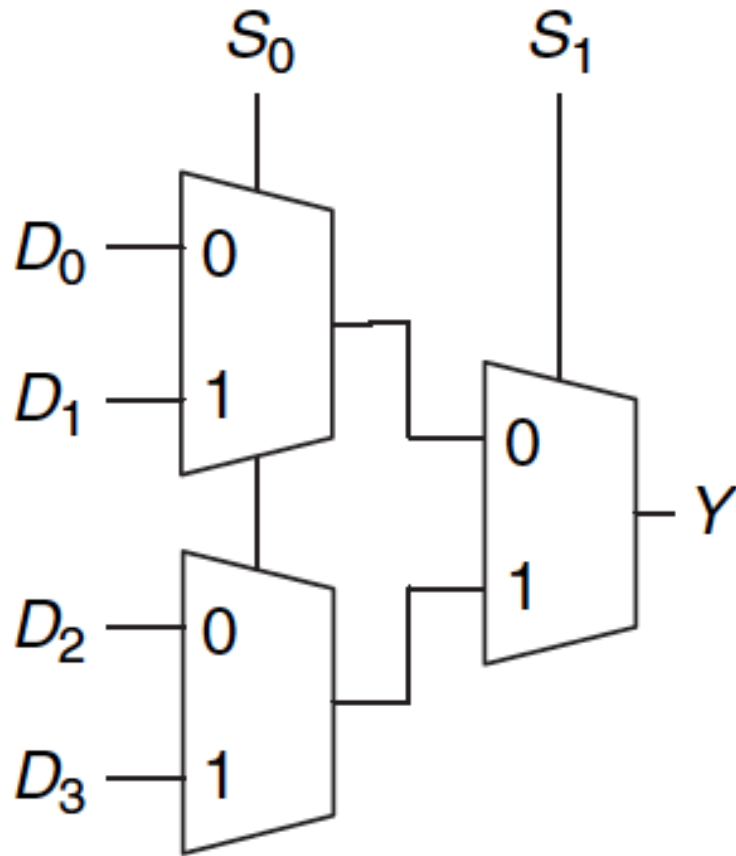


$$Y = D_0 \bar{S} + D_1 S$$

Figure 2.56 Multiplexer using tristate buffers

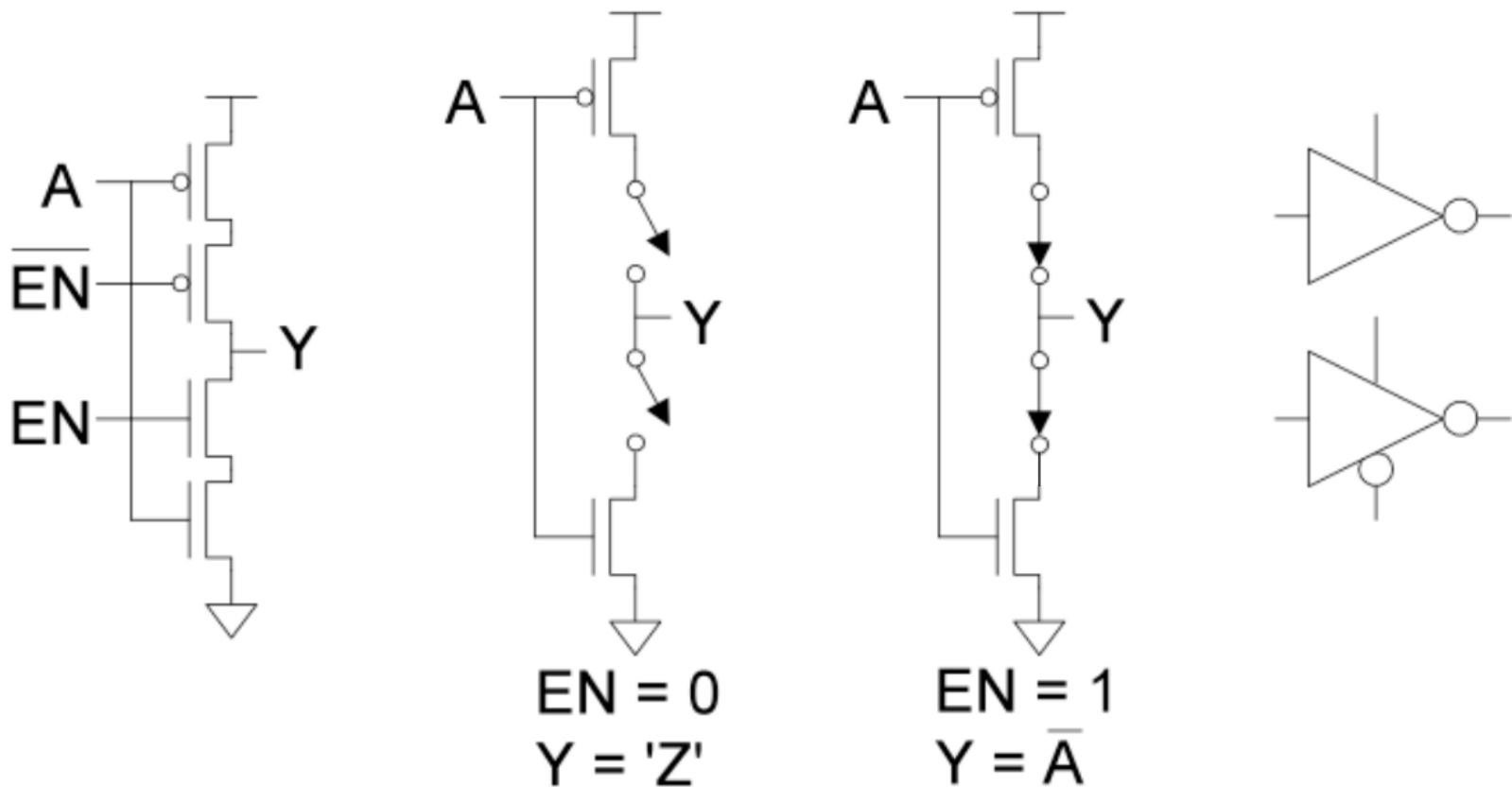


Recall: A 4-to-1 Multiplexer



Digging Deeper: Tri-State Buffer in CMOS

- How do you implement a Tri-State Buffer using transistors?

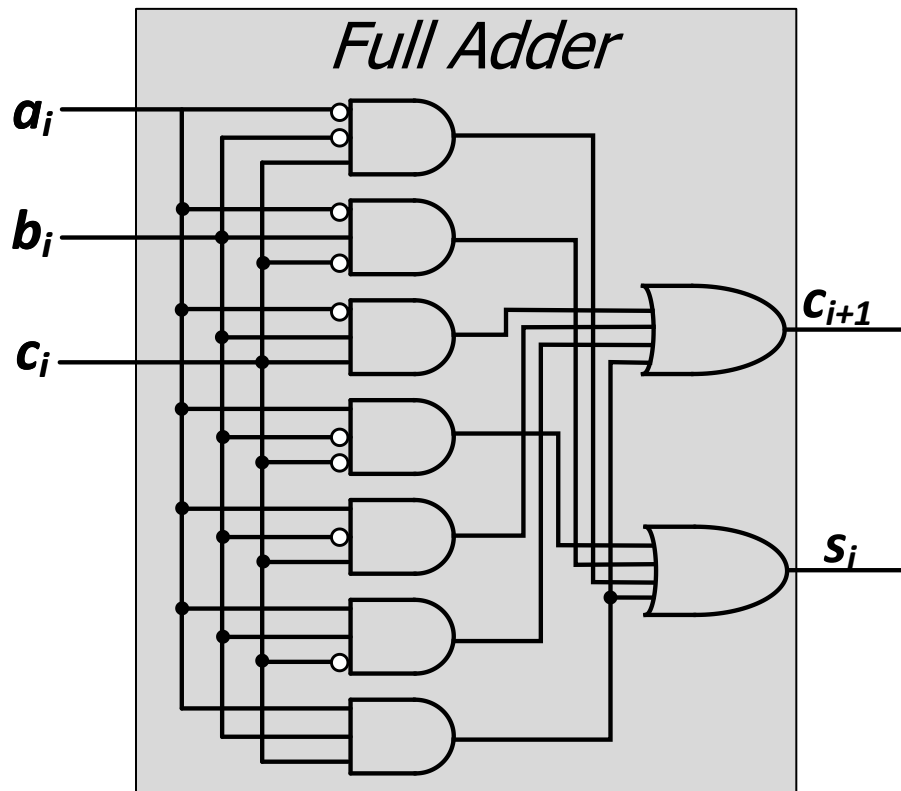


We Covered Combinational Logic Blocks

- Basic logic gates (AND, OR, NOT, NAND, NOR, XOR)
 - Decoder
 - Multiplexer
 - Full Adder
 - Programmable Logic Array (PLA)
 - Comparator
 - Arithmetic Logic Unit (ALU)
 - Tri-State Buffer
-
- Standard form representations: SOP & POS
 - Logic simplification via Boolean Algebra
 - Logical completeness

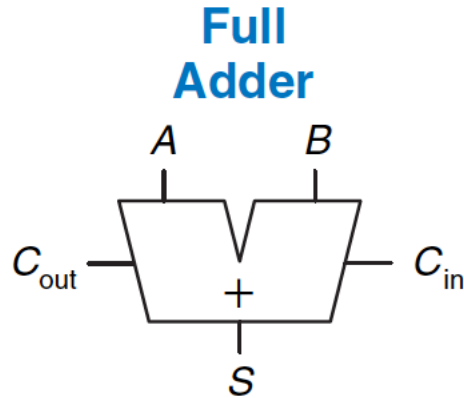
Logic Simplification using Boolean Algebra Rules

Recall: Full Adder in SOP Form Logic



a_i	b_i	$carry_i$	$carry_{i+1}$	S_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Goal: Simplified Full Adder



$$S = A \oplus B \oplus C_{\text{in}} \quad \text{3-input XOR}$$

$$C_{\text{out}} = AB + AC_{\text{in}} + BC_{\text{in}} \quad \text{3-input majority}$$

C_{in}	A	B	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

How do we simplify Boolean logic?

How do we automate simplification?

Quick Recap on Logic Simplification

- The original Boolean expression (i.e., logic circuit) may not be optimal (e.g., in number of terms or logic gates)

$$F = \sim A(A + B) + (B + AA)(A + \sim B)$$

- Can we reduce a given Boolean expression to an equivalent expression **with fewer terms**?

$$F = A + B$$

- The **goal** of logic simplification:
 - **Reduce** the number of gates/inputs
 - **Reduce** implementation cost (and potentially latency & power)

A basis for what the automated design tools are doing today

Logic Simplification

■ Systematic techniques for simplifications

- amenable to automation

Key Tool: The Uniting Theorem — $F = A\bar{B} + AB$

A	B	F
0	0	0
0	1	0
1	0	1
1	1	1

$$F = A\bar{B} + AB = A(\bar{B} + B) = A(1) = A$$

B's value changes within the rows where $F=1$ ("ON set")

A's value does NOT change within the ON-set rows

If an input (B) can change without changing the output, that input value is not needed

→ *B is eliminated, A remains*

A	B	G
0	0	1
0	1	0
1	0	1
1	1	0

$$G = \bar{A}\bar{B} + A\bar{B} = (\bar{A} + A)\bar{B} = \bar{B}$$

B's value stays the same within the ON-set rows

A's value changes within the ON-set rows

→ *A is eliminated, B remains*

Logic Simplification

■ Systematic techniques for simplifications

- amenable to automation

Key Tool: The Uniting Theorem — $F = A\bar{B} + AB$

A	B	F
0	0	0
0	1	0
1	0	1
1	1	1

$$F = A\bar{B} + AB = A(\bar{B} + B) = A(1) = A$$

Essence of Simplification:

Find two-element subsets of the ON-set where only one variable changes its value. This single varying variable *can be eliminated!*

value is not needed

→ *B is eliminated, A remains*

A	B	G
0	0	1
0	1	0
1	0	1
1	1	0

$$G = \bar{A}\bar{B} + A\bar{B} = (\bar{A} + A)\bar{B} = \bar{B}$$

B's value stays the same within the ON-set rows

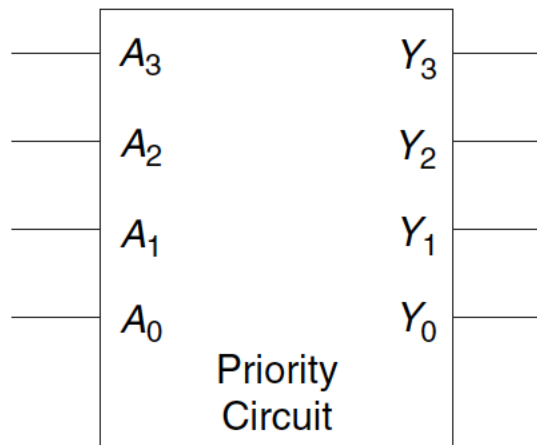
A's value changes within the ON-set rows

→ *A is eliminated, B remains*

Logic Simplification Example: Priority Circuit

■ Priority Circuit

- Inputs: "Requestors" with priority levels
- Outputs: "Grant" signal for each requestor
- Example 4-bit priority circuit
- Real life example: Imagine a bus requested by 4 processors



A_3	A_2	A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	0
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	0
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	0
1	1	1	1	1	0	0	0

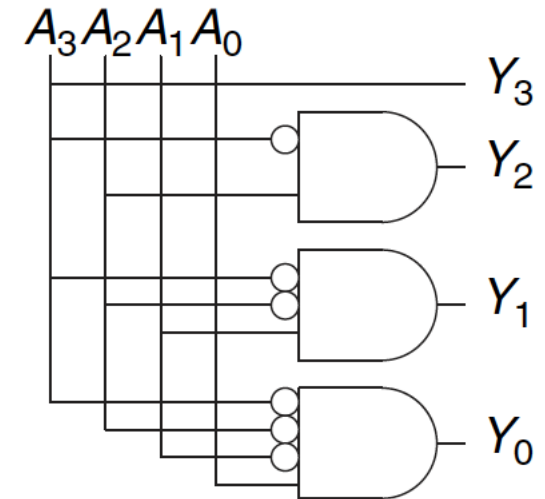
Simplified Priority Circuit

- Priority Circuit
 - Inputs: "Requestors" with priority levels
 - Outputs: "Grant" signal for each requestor
 - Example 4-bit priority circuit

A_3	A_2	A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	0
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	0
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	0
1	1	1	1	1	0	0	0

A_3	A_2	A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	X	0	0	1	0
0	1	X	X	0	1	0	0
1	X	X	X	1	0	0	0

Figure 2.29 Priority circuit truth table with don't cares (X's)



X (Don't Care) means *I don't care what the value of this input is*

Logic Simplification: Karnaugh Maps (K-Maps)

Karnaugh Maps are Fun...

- A pictorial way of minimizing circuits by visualizing opportunities for simplification
- They are for you to **study on your own...**
 - We may cover them later if time permits (very unlikely)
- See backup slides
- Read H&H Section 2.7
- Watch videos of Lectures 5 and 6 from 2019 DDCA course:
 - <https://youtu.be/0ks0PeaOUjE?list=PL5Q2soXY2Zi8J58xLKBNFQFHRO3GrXxA9&t=4570>
 - <https://youtu.be/ozs18ARNG6s?list=PL5Q2soXY2Zi8J58xLKBNFQFHRO3GrXxA9&t=220>

We Are Done with Combinational Logic

- Building blocks of modern computers
 - Transistors
 - Logic gates
- Combinational logic circuits
- Boolean algebra
- Using Boolean algebra to represent combinational circuits
- Basic combinational logic blocks
- Simplifying combinational logic circuits

Sequential Logic

Agenda for Today and Tomorrow

■ Today

- Start (and finish) Sequential Logic

■ Tomorrow & Next week

- Hardware Description Languages and Verilog
 - Combinational Logic
 - Sequential Logic
- Timing and Verification

Sequential Logic Circuits and Design

What We Will Learn Today

■ Circuits that can store information

- ❑ Cross-coupled inverter
- ❑ R-S Latch
- ❑ Gated D Latch
- ❑ D Flip-Flop
- ❑ Register
- ❑ Memory

■ Sequential logic circuits

- ❑ State & Clock
- ❑ Asynchronous vs. Synchronous

■ Finite State Machines (FSM)

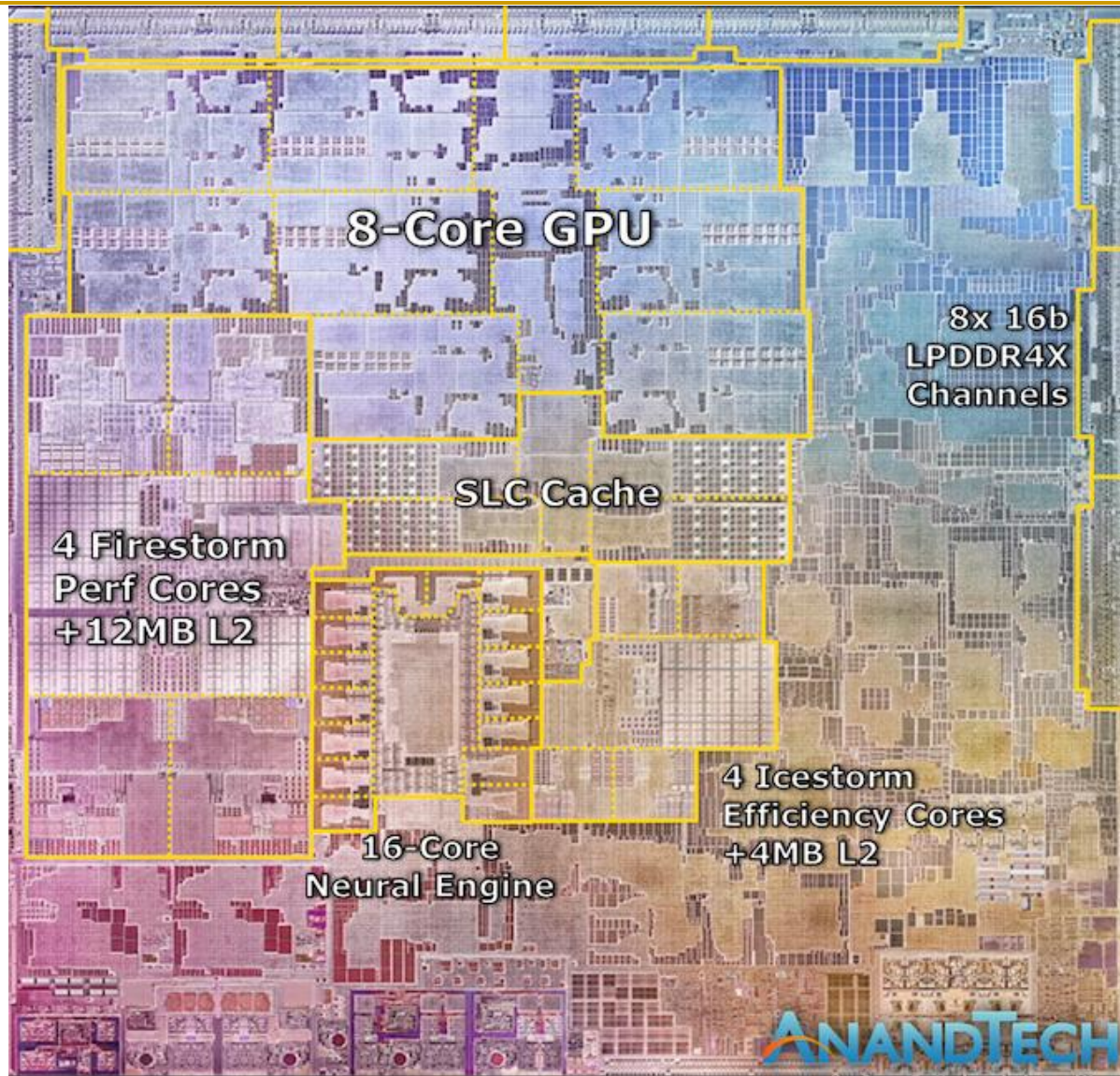
- ❑ How to design FSMs

Readings

- Combinational Logic
 - P&P Chapter 3 until 3.3 + H&H Chapter 2
- Sequential Logic
 - P&P Chapter 3.4 until end + H&H Chapter 3 in full
- Hardware Description Languages and Verilog
 - H&H Chapter 4 in full
- Timing and Verification
 - H&H Chapters 2.9 and 3.5 + (start Chapter 5)

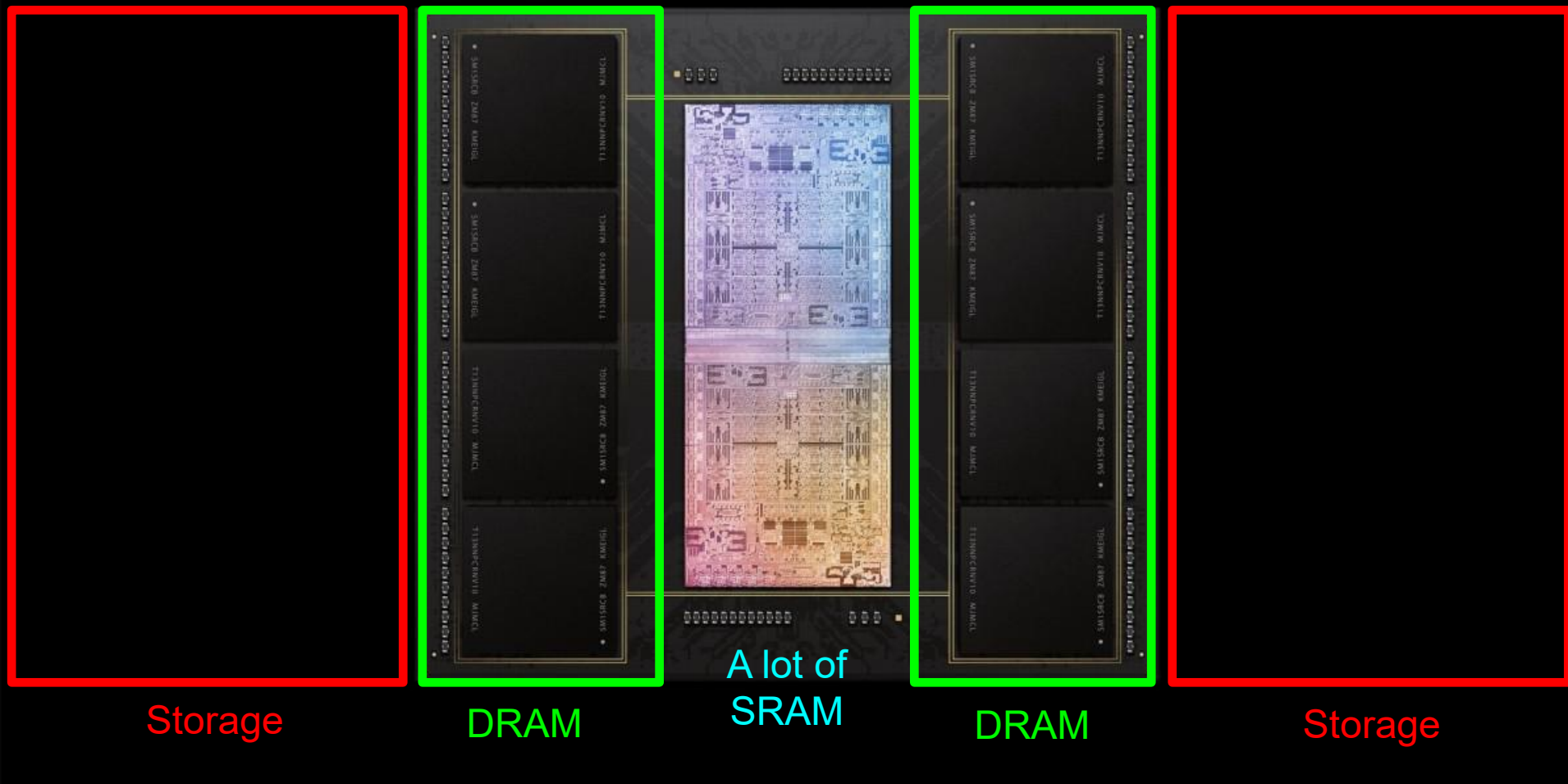
- By the end of next week, make sure you are done with
 - **P&P Chapters 1-3 + H&H Chapters 1-4**

No Real Computer Can Work w/o Memory



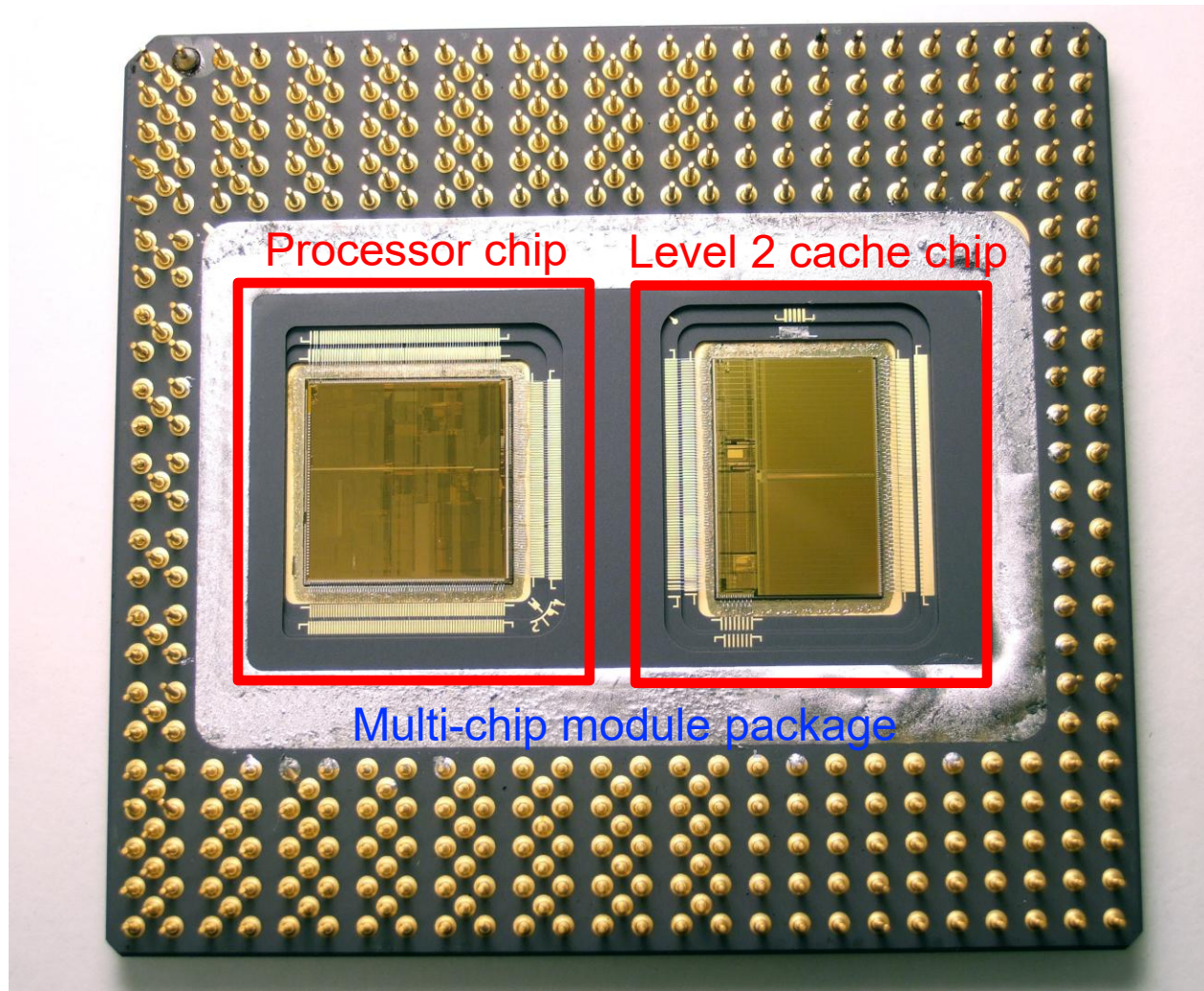
Apple M1,
2021

A Large Fraction of Modern Systems is Memory



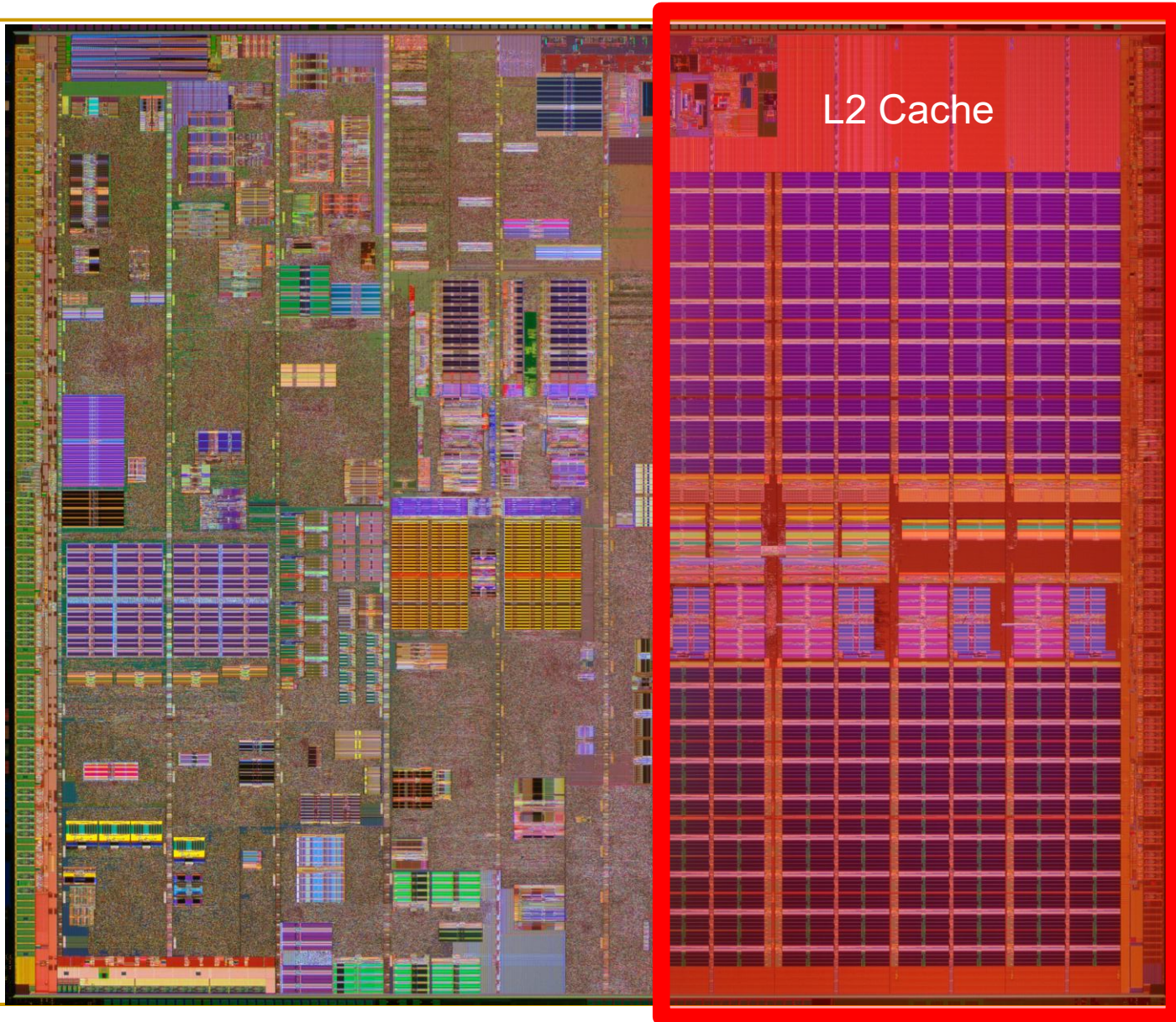
Apple M1 Ultra System (2022)

A Large Fraction of Modern Systems is Memory

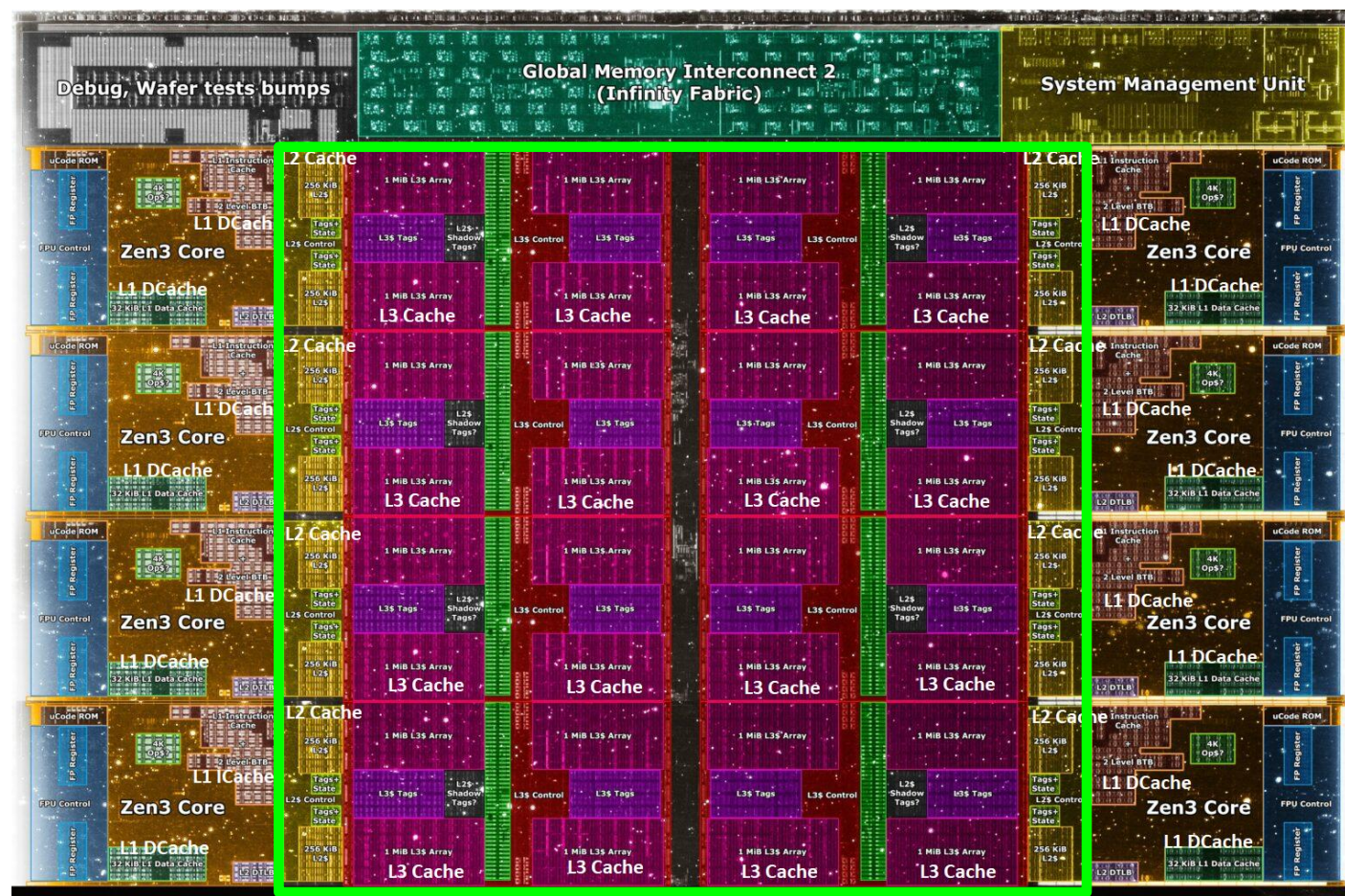


Intel Pentium Pro, 1995

A Large Fraction of Modern Systems is Memory



A Large Fraction of Modern Systems is Memory



Core Count:
8 cores/16 threads

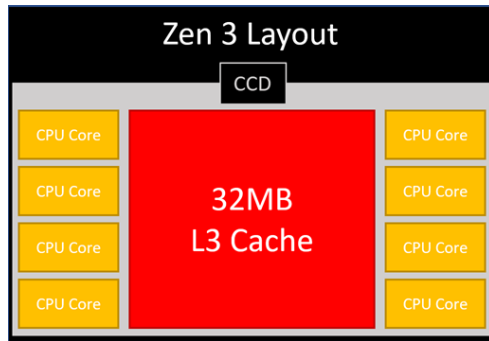
L1 Caches:
32 KB per core

L2 Caches:
512 KB per core

L3 Cache:
32 MB shared

AMD Ryzen 5000, 2020

Adding Even More Memory in 3D (2021)

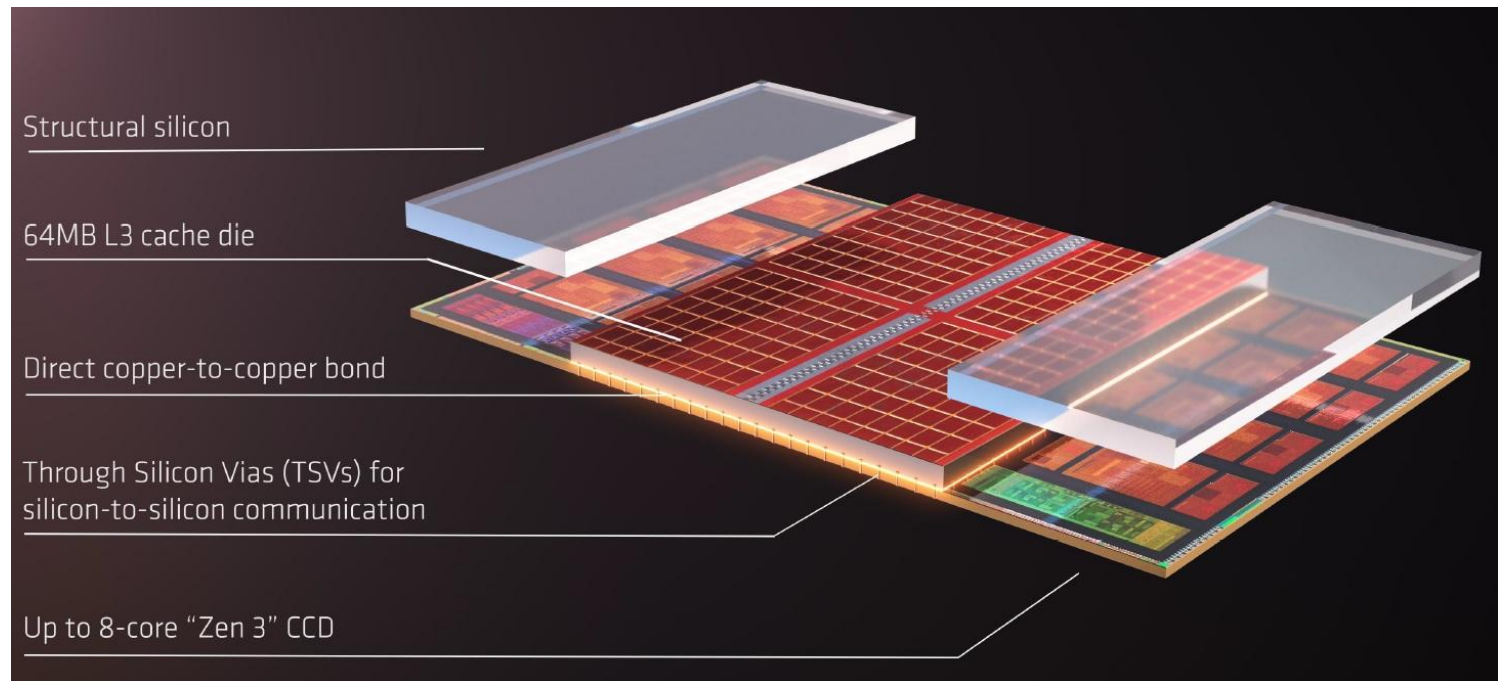


<https://community.microcenter.com/discussion/5134/comparing-zen-3-to-zen-2>

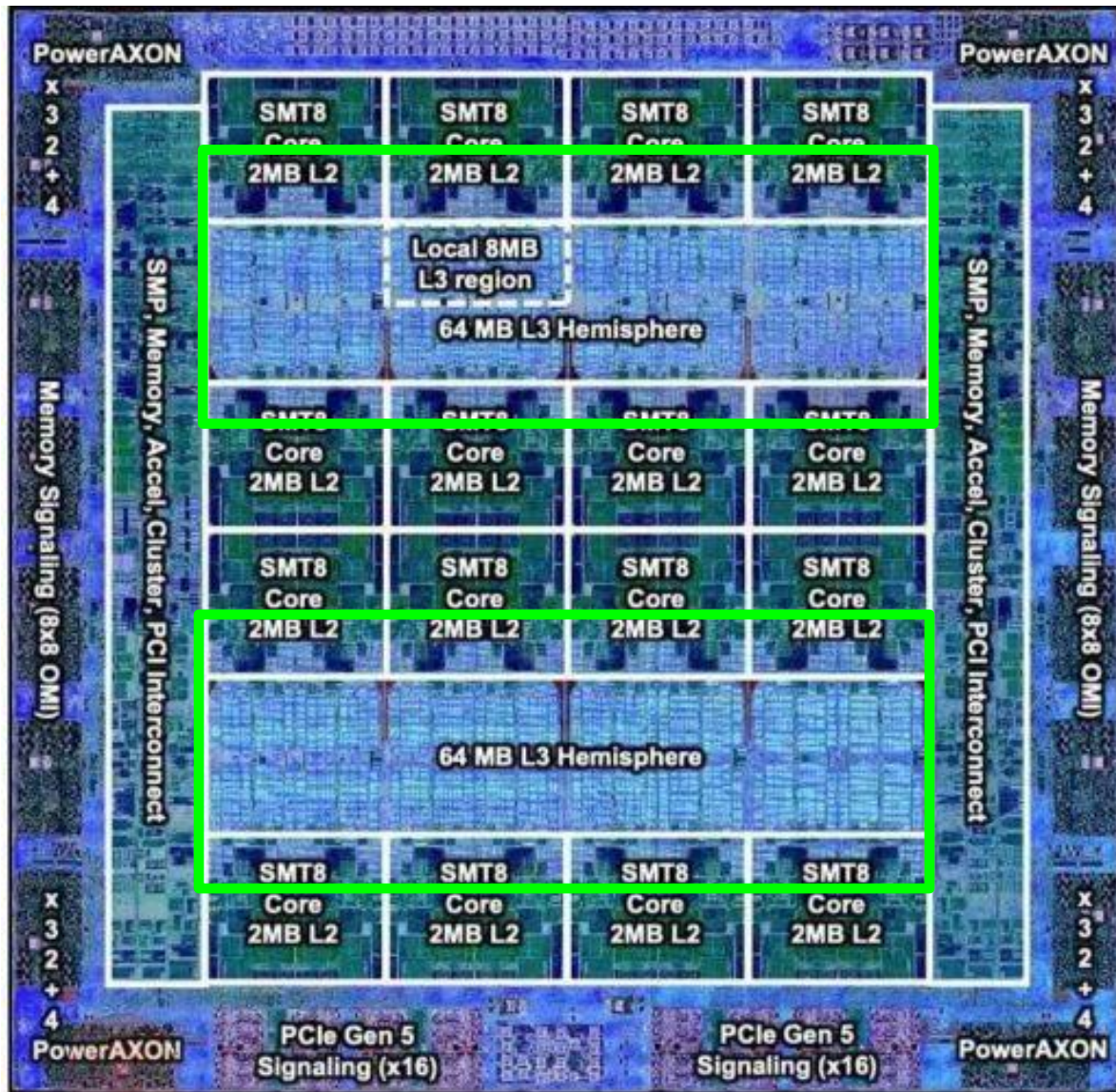
AMD increases the L3 size of their 8-core Zen 3 processors from 32 MB to 96 MB

Additional 64 MB L3 cache die
stacked on top of the processor die

- Connected using Through Silicon Vias (TSVs)
- Total of 96 MB L3 cache



A Large Fraction of Modern Systems is Memory



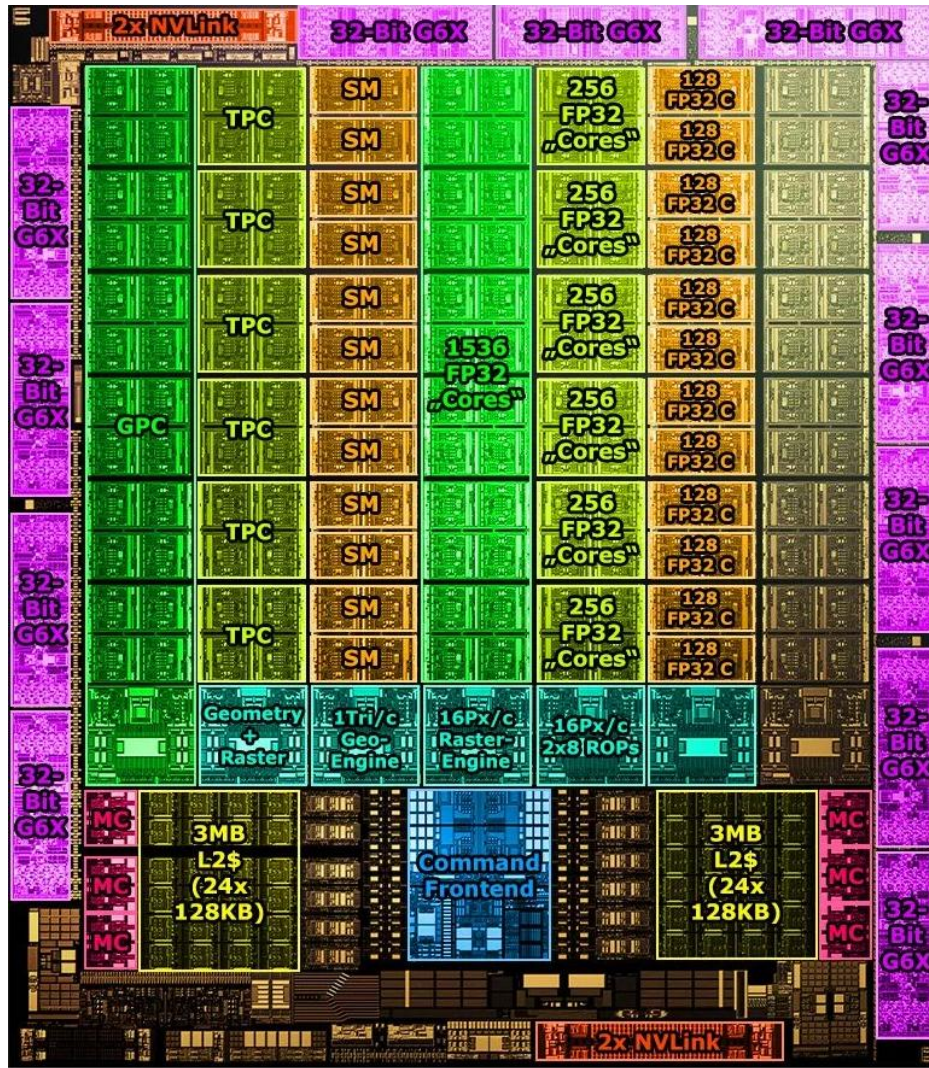
IBM POWER10,
2020

Cores:
15-16 cores,
8 threads/core

L2 Caches:
2 MB per core

L3 Cache:
120 MB shared

A Large Fraction of Modern Systems is Memory



Nvidia Ampere, 2020

Cores:

128 Streaming Multiprocessors

L1 Cache or Scratchpad:

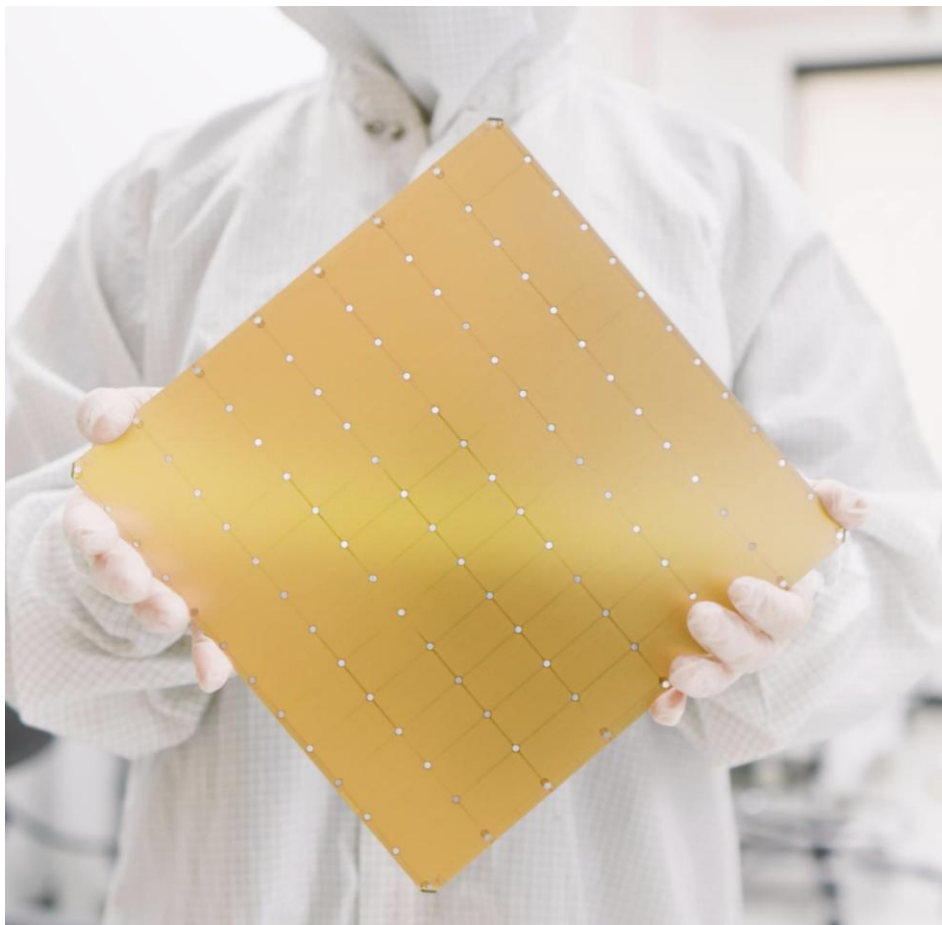
192KB per SM

Can be used as L1 Cache and/or Scratchpad

L2 Cache:

40 MB shared

Cerebras's Wafer Scale Engine-3 (2023)



Cerebras Wafer-Scale Engine

The largest chip ever produced

46,225 mm² silicon

4 trillion transistors

900,000 AI cores

125 Petaflops of AI compute

44 Gigabytes of on-chip memory

21 PByte/s memory bandwidth

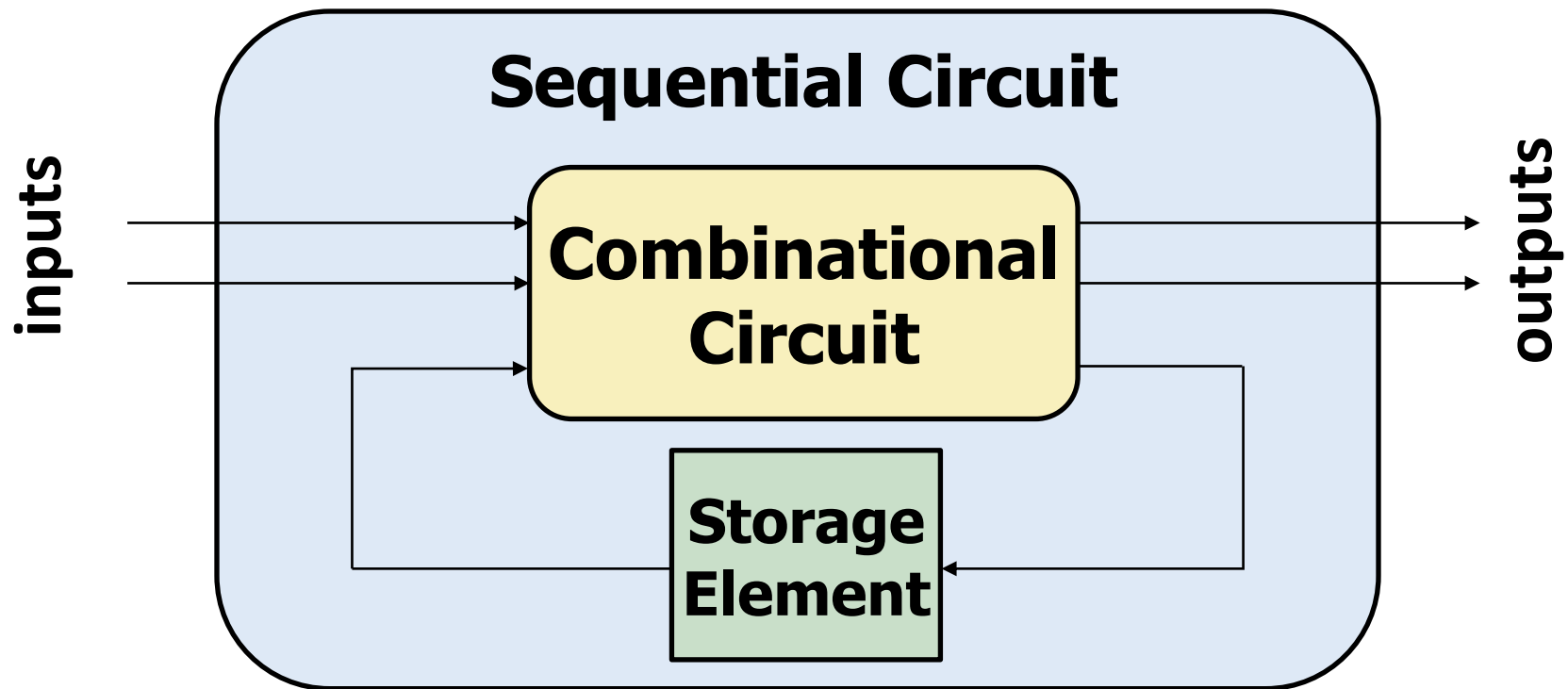
214 Pbit/s fabric bandwidth

5nm TSMC process

Circuits That Can Store Information

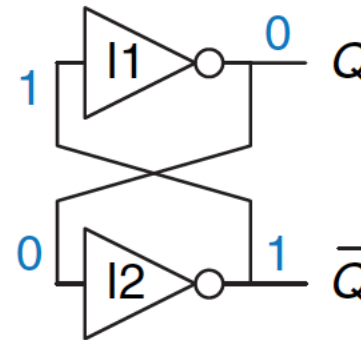
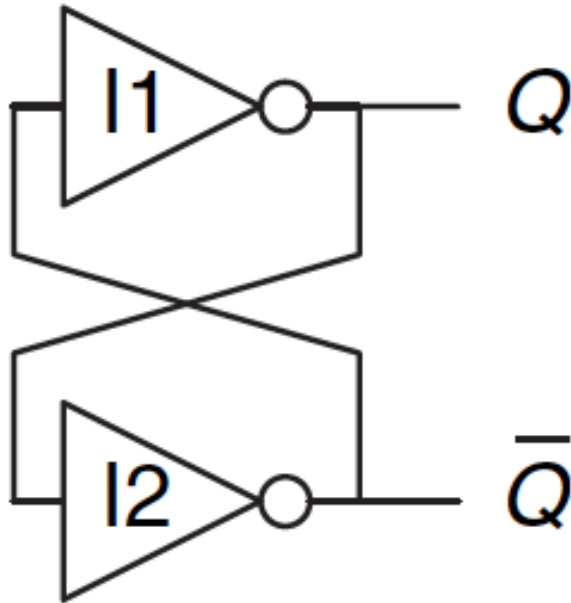
Introduction

- Combinational circuit output depends **only** on **current** input
- We want circuits that produce output depending on **current** and **past** input values – circuits with **memory**
- How can we design a circuit that **stores information**?

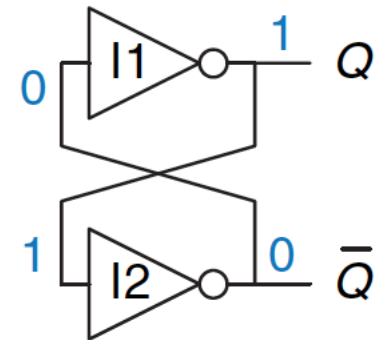


Capturing Data

Basic Element: Cross-Coupled Inverters



(a)



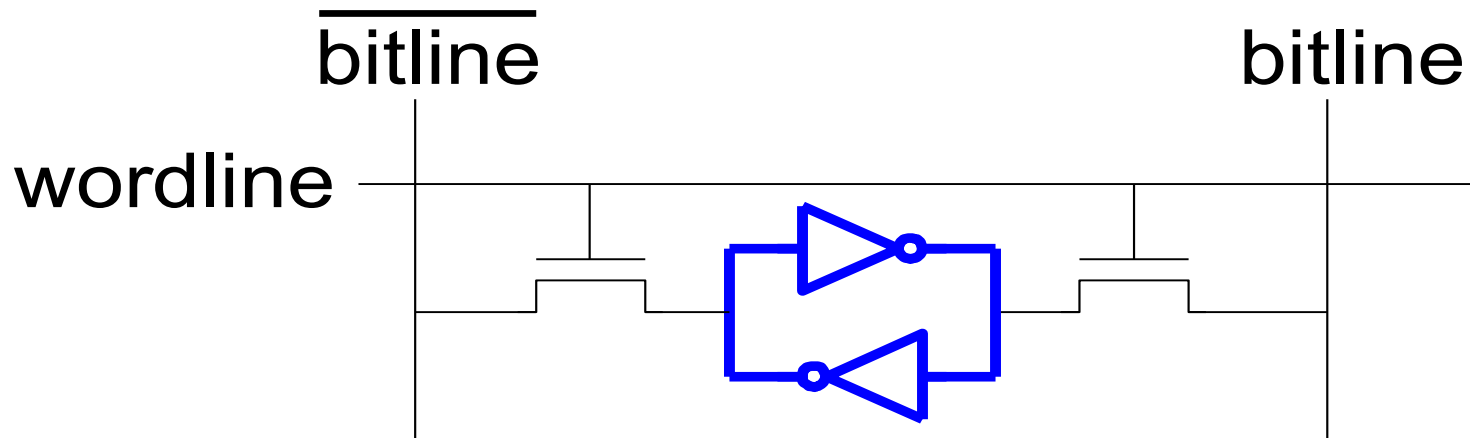
(b)

- Has two stable states: $Q=1$ or $Q=0$.
- Has a third possible "metastable" state with both outputs oscillating between 0 and 1 (we will see this later)
- Not useful without a *control mechanism* for setting Q

More Realistic Storage Elements

- **Have a control mechanism for setting Q**

- We will see the R-S latch soon
- Let's look at an SRAM (static random access memory) cell first



SRAM cell

- We will get back to SRAM (and DRAM) later

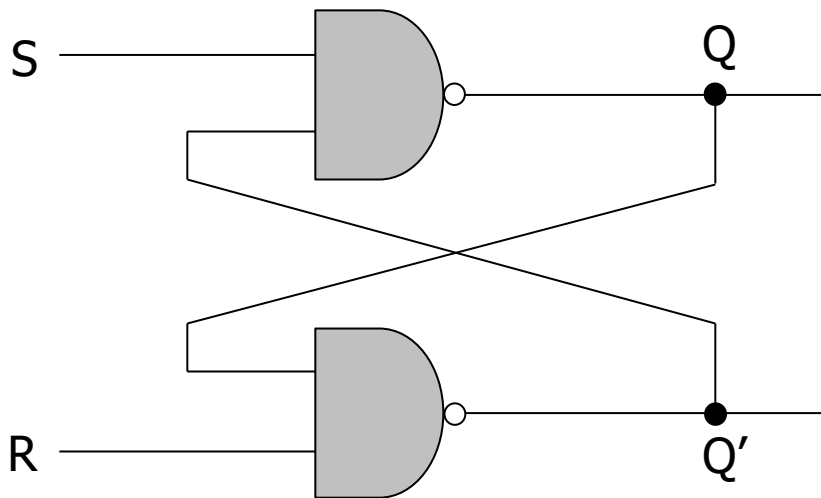
The Big Picture: Storage Elements

- Latches and Flip-Flops
 - Very fast, parallel access
 - Very expensive (one bit costs tens of transistors)
 - Static RAM (SRAM)
 - Relatively fast
 - Expensive (one bit costs 6+ transistors)
 - Dynamic RAM (DRAM)
 - Slower, reading destroys content (refresh), needs special process for manufacturing
 - Cheap (one bit costs only one transistor plus one capacitor)
 - Other storage technology (flash memory, hard disk, tape)
 - Much slower, access takes a long time, non-volatile
 - Very cheap
-

Basic Storage Element: The R-S Latch

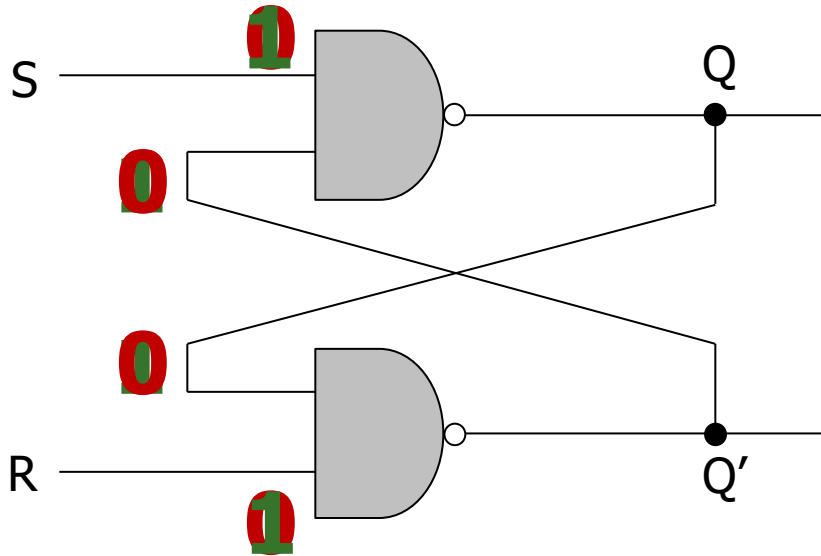
The R-S (Reset-Set) Latch

- Cross-coupled **NAND gates**
 - Data is stored at **Q** (inverse at **Q'**)
 - **S** and **R** are control inputs
 - In *quiescent (idle) state*, **both S and R are held at 1**
 - **S (set)**: drive **S** to 0 (keeping **R** at 1) to change **Q** to 1
 - **R (reset)**: drive **R** to 0 (keeping **S** at 1) to change **Q** to 0
- **S** and **R** should not **both** be 0 at the same time



Input		Output
R	S	Q
1	1	Q_{prev}
1	0	1
0	1	0
0	0	Forbidden

Why not $R=S=0$?



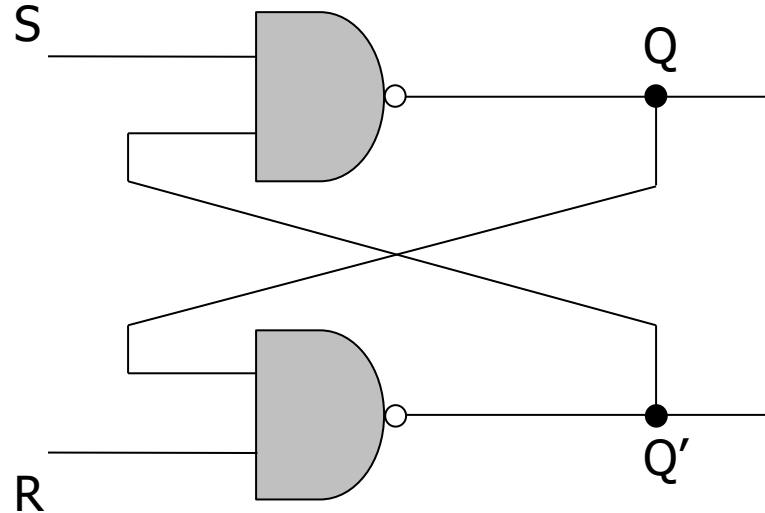
Input		Output
R	S	Q
1	1	Q_{prev}
1	0	1
0	1	0
0	0	Forbidden

1. If $R=S=0$, Q and Q' will both settle to 1, which **breaks** our invariant that $Q = !Q'$
2. If S and R transition back to 1 at the same time, Q and Q' begin to oscillate between 1 and 0 because their final values depend on each other (**metastability**)
 - This eventually settles depending on **variation in the circuits** (more on this in the **Timing Lecture**)

Gated D Latch

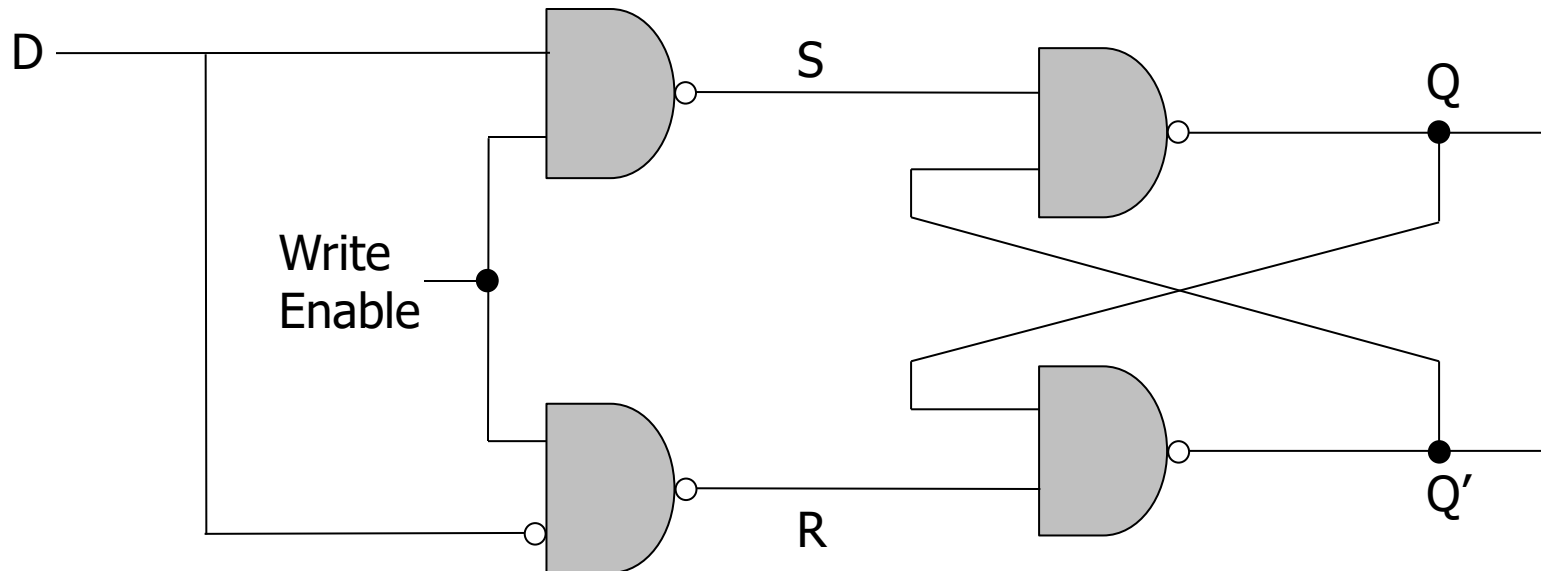
The Gated D Latch

- How do we **guarantee** correct operation of an R-S Latch?



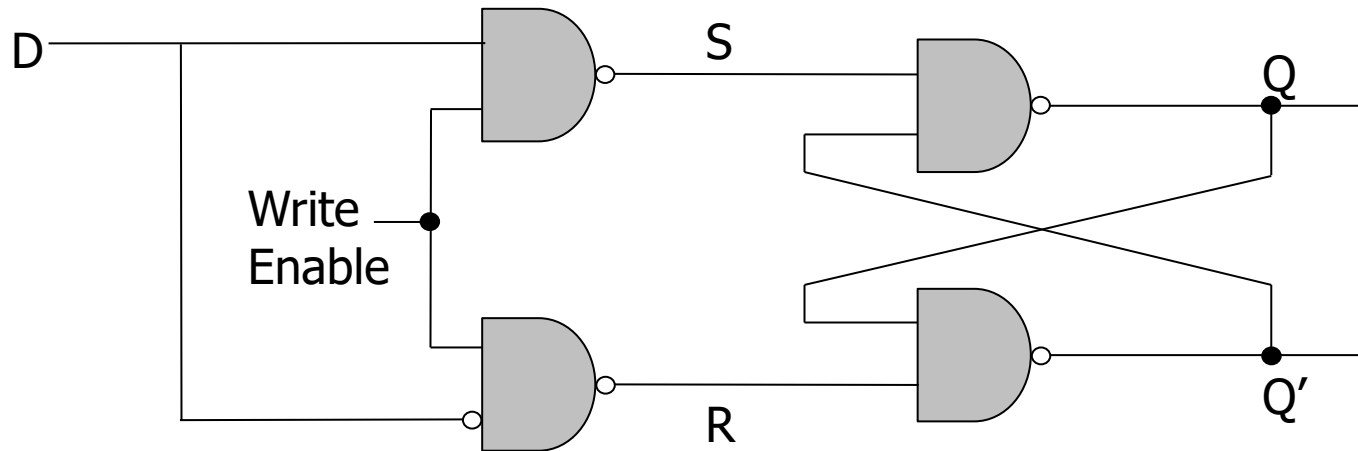
The Gated D Latch

- How do we **guarantee** correct operation of an R-S Latch?
 - Add two more NAND gates!



- **Q** takes the value of **D**, when **write enable (WE)** is set to 1
- **S** and **R** can never be 0 at the same time!

The Gated D Latch



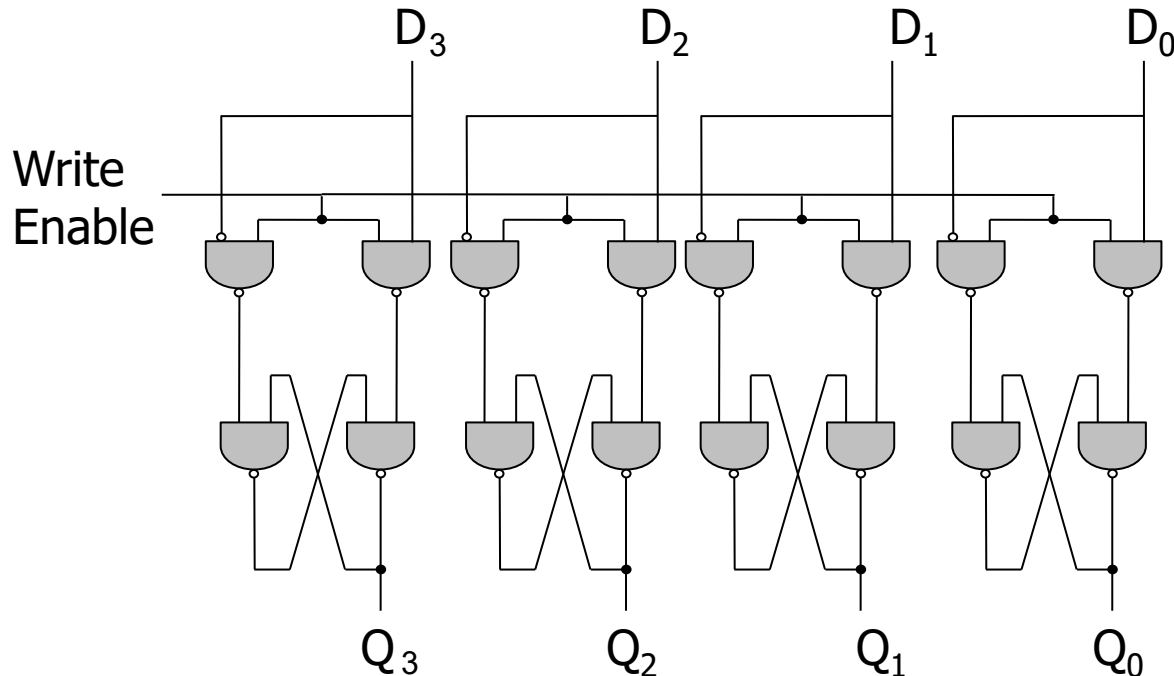
Input		Output
WE	D	Q
0	0	Q_{prev}
0	1	Q_{prev}
1	0	0
1	1	1

Register

The Register

How can we use D latches to store **more** data?

- Use **more** D latches!
- A single WE signal for all latches for simultaneous writes



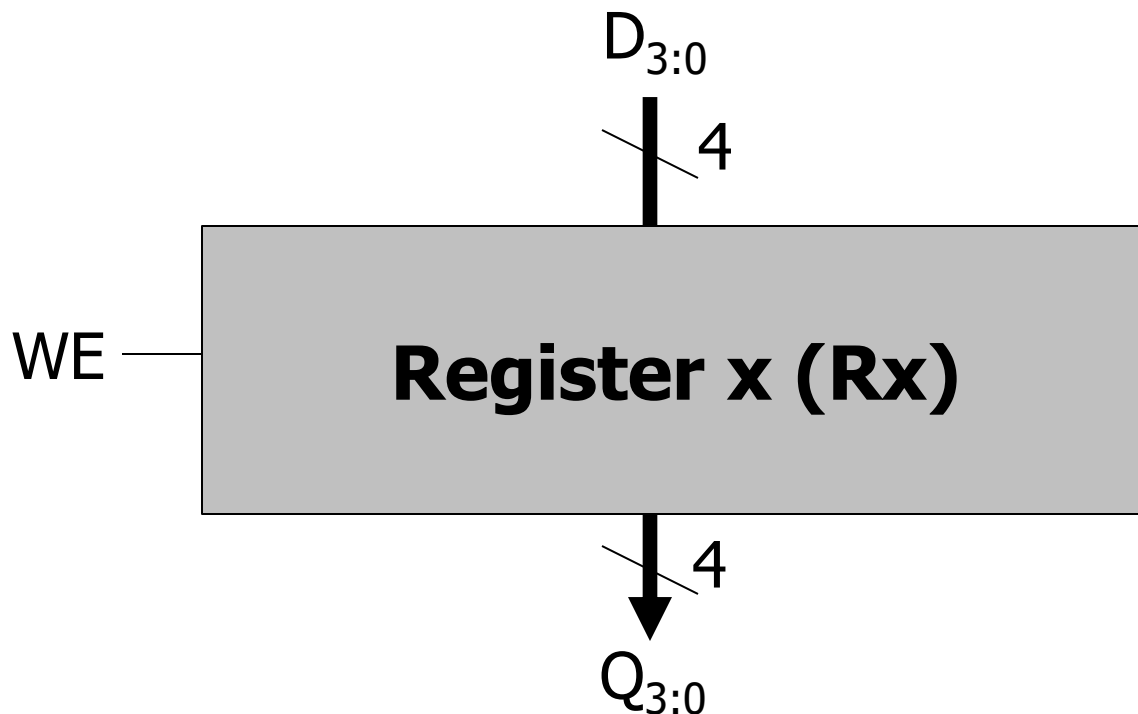
Here we have a **register**, or a structure that stores more than one bit and can be read from and written to

This **register** holds 4 bits, and its data is referenced as Q[3:0]

The Register

How can we use D latches to store **more** data?

- Use **more** D latches!
- A single WE signal for all latches for simultaneous writes



Here we have a **register**, or a structure that stores more than one bit and can be read from and written to

This **register** holds 4 bits, and its data is referenced as Q[3:0]

Memory

Memory

- **Memory** is comprised of locations that can be written to or read from. An example memory array with 4 locations:

Addr(00): 0100 1001	Addr(01): 0100 1011
Addr(10): 0010 0010	Addr(11): 1100 1001

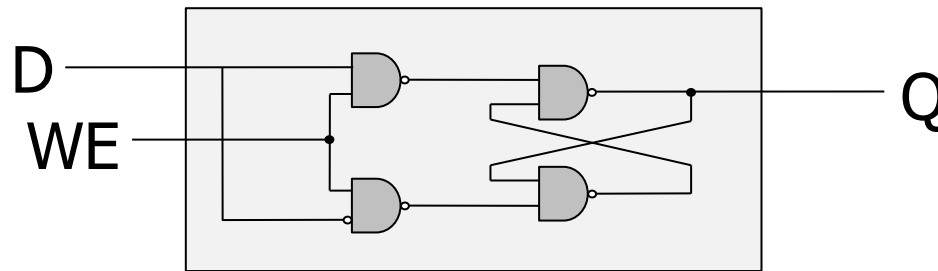
- Every unique location in memory is indexed with a unique **address**. 4 locations require 2 address bits ($\log[\text{\#locations}]$).
- **Addressability**: the number of **bits** of information stored **in each location**. This example: addressability is 8 bits.
- The entire set of **unique locations** in memory is referred to as the **address space**.
- Typical memory is **MUCH** larger (e.g., billions of locations)

Addressing Memory

Let's implement a simple memory array with:

- 3-bit addressability & address space size of 2 (total of 6 bits)

1 Bit



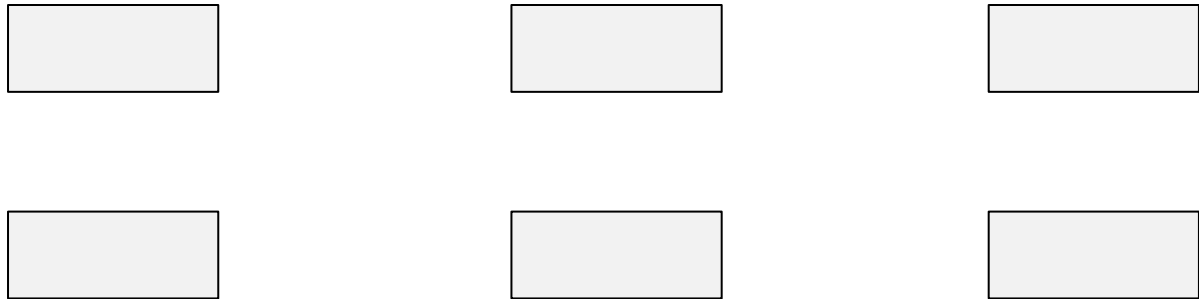
6-Bit Memory Array

Addr(0)	Bit ₂	Bit ₁	Bit ₀
Addr(1)	Bit ₂	Bit ₁	Bit ₀

Reading from Memory

How can we select an address to read?

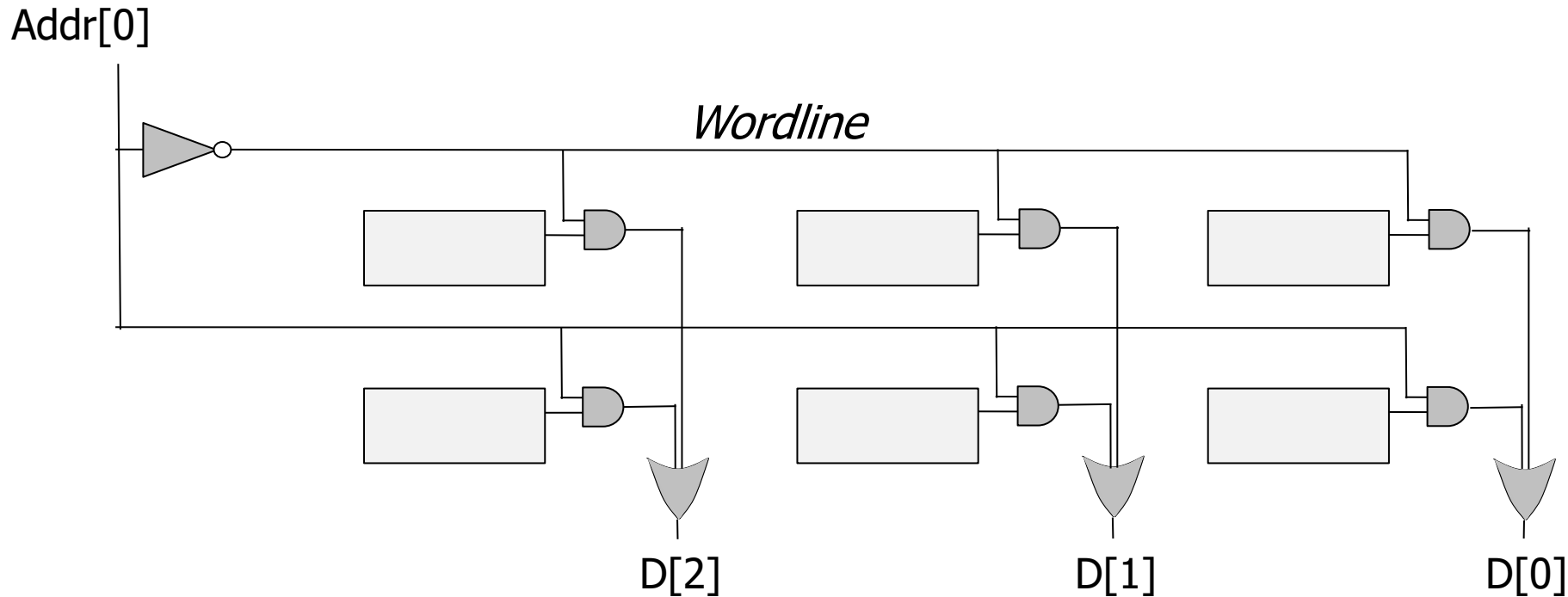
- Because there are 2 addresses, address size is $\log(2)=1$ bit



Reading from Memory

How can we select an address to read?

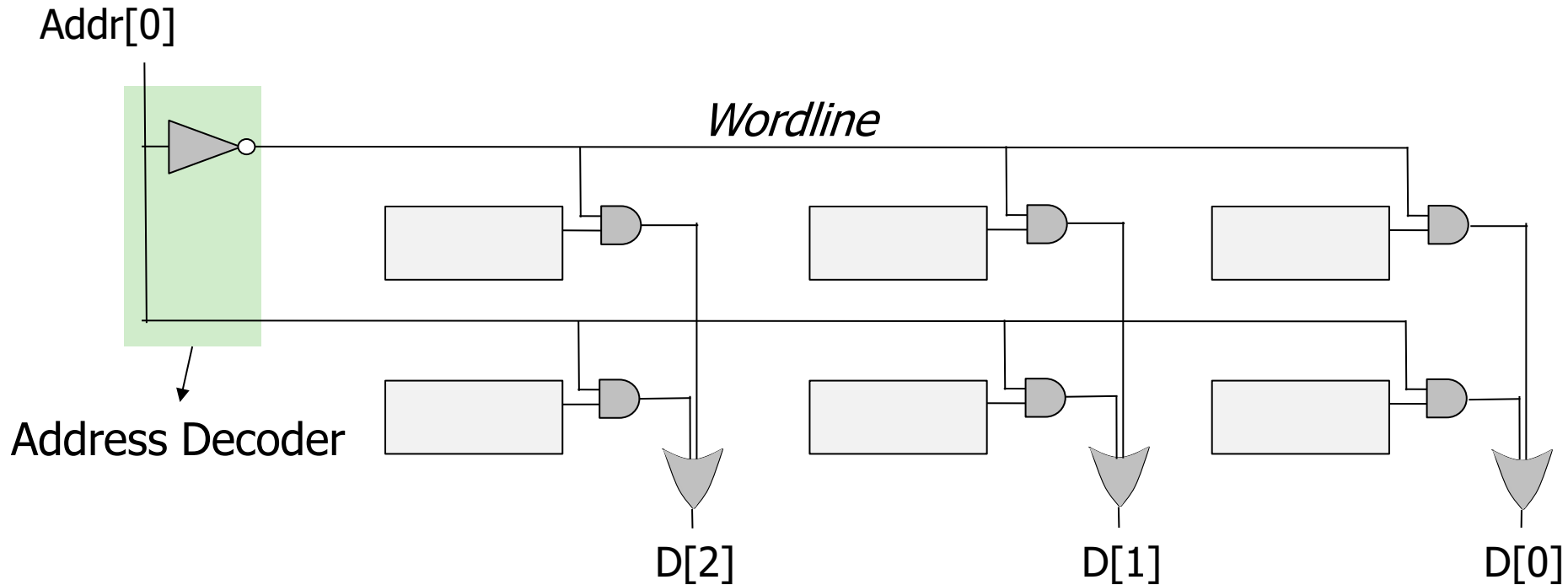
- Because there are 2 addresses, address size is $\log(2)=1$ bit



Reading from Memory

How can we select an address to read?

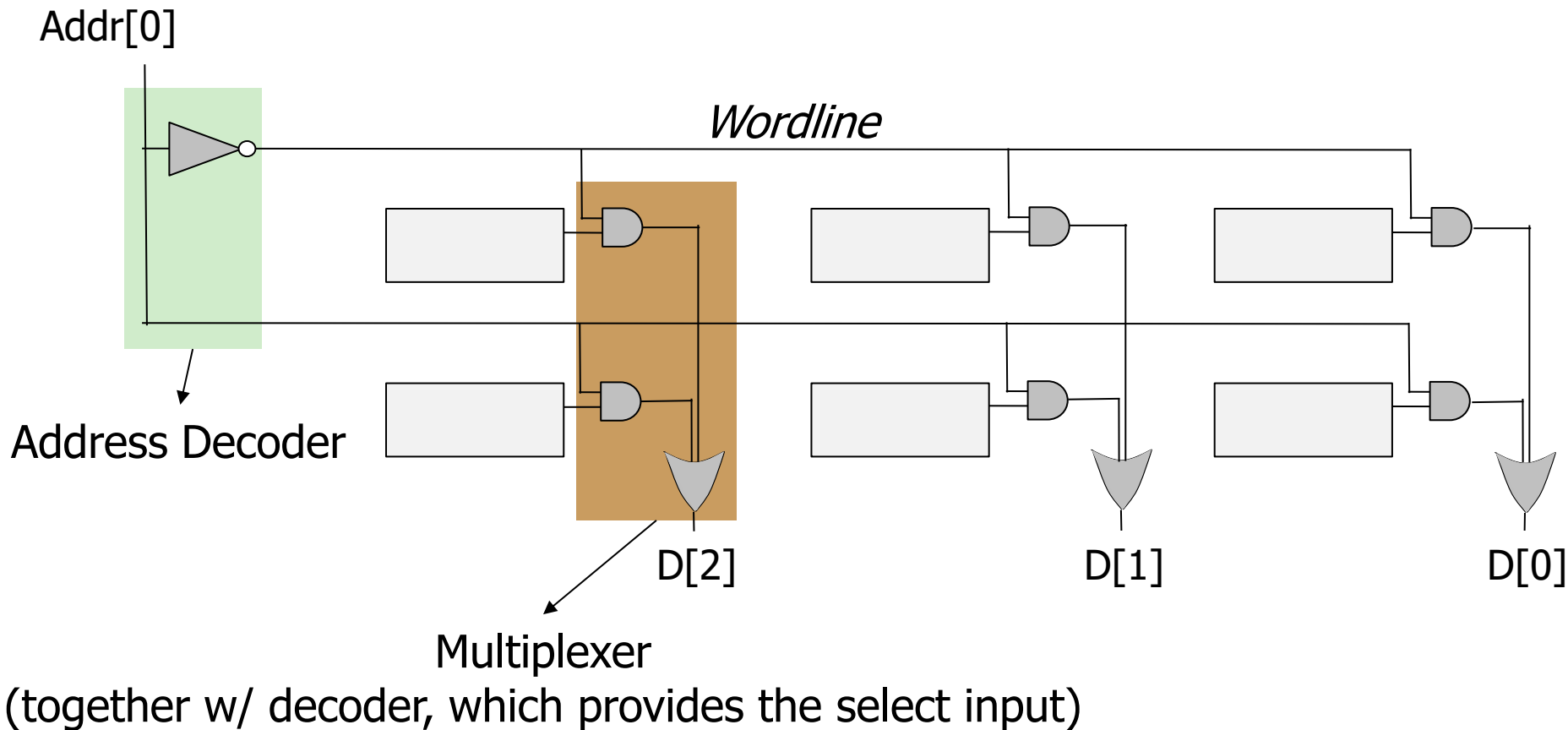
- Because there are 2 addresses, address size is $\log(2)=1$ bit



Reading from Memory

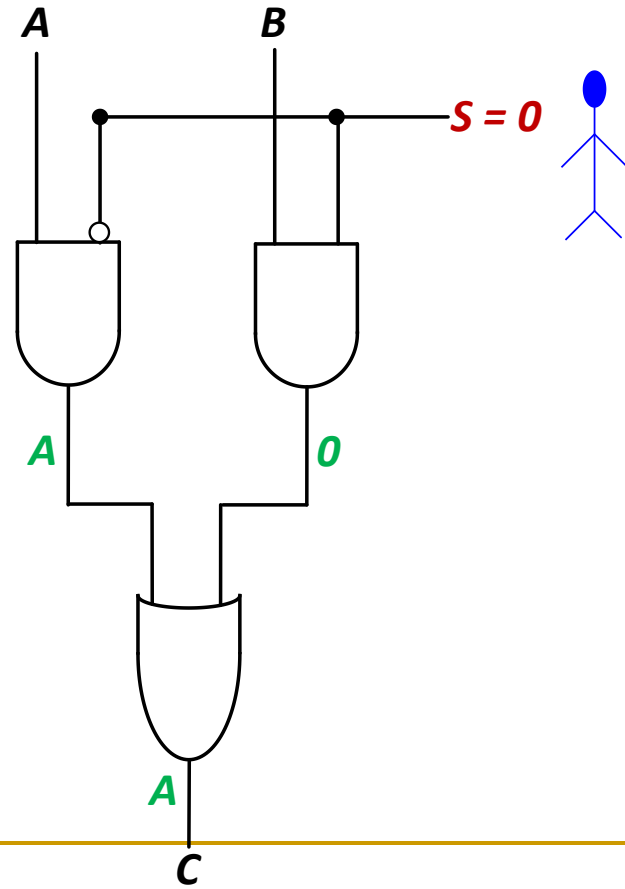
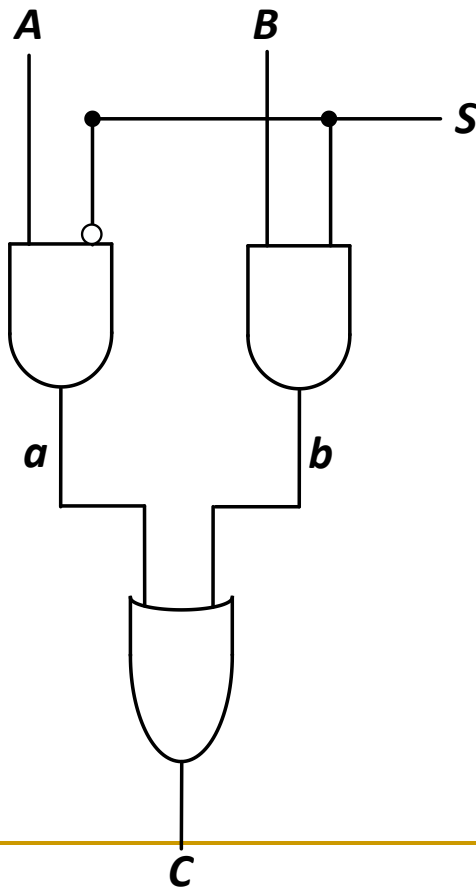
How can we select an address to read?

- Because there are 2 addresses, address size is $\log(2)=1$ bit



Recall: Multiplexer (MUX), or Selector

- **Selects** one of the N inputs to connect it to the output
 - based on the value of a $\log_2 N$ -bit control input called **select**
- Example: 2-to-1 MUX



Writing to Memory

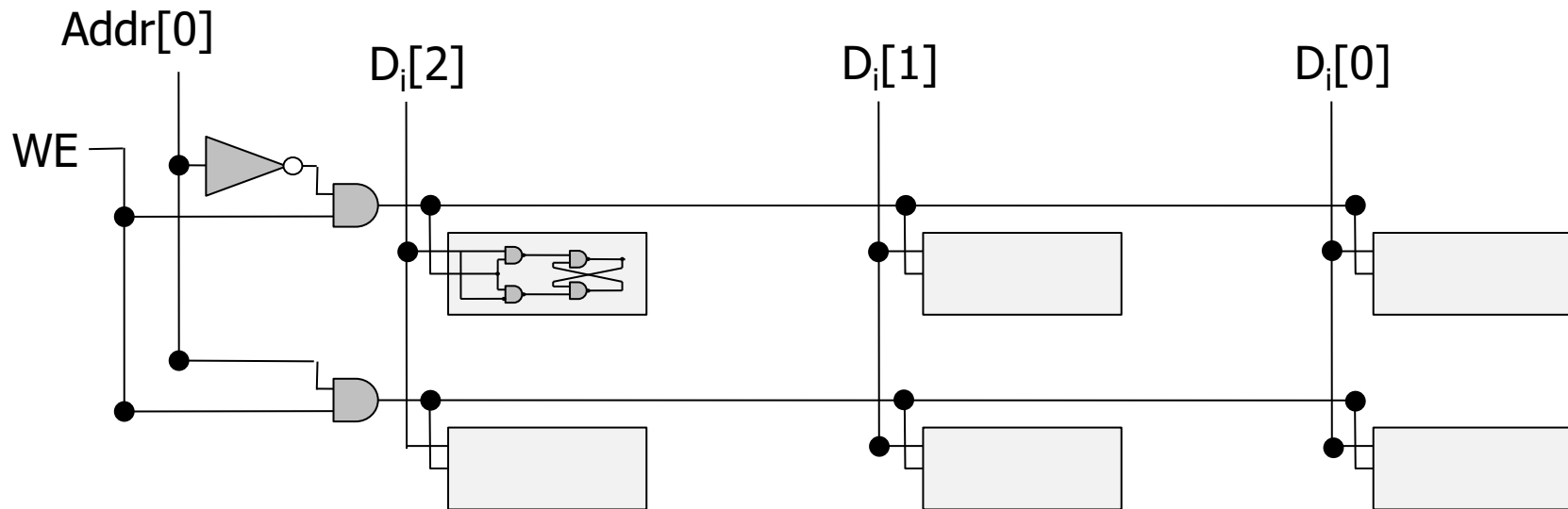
How can we select an address and write to it?



Writing to Memory

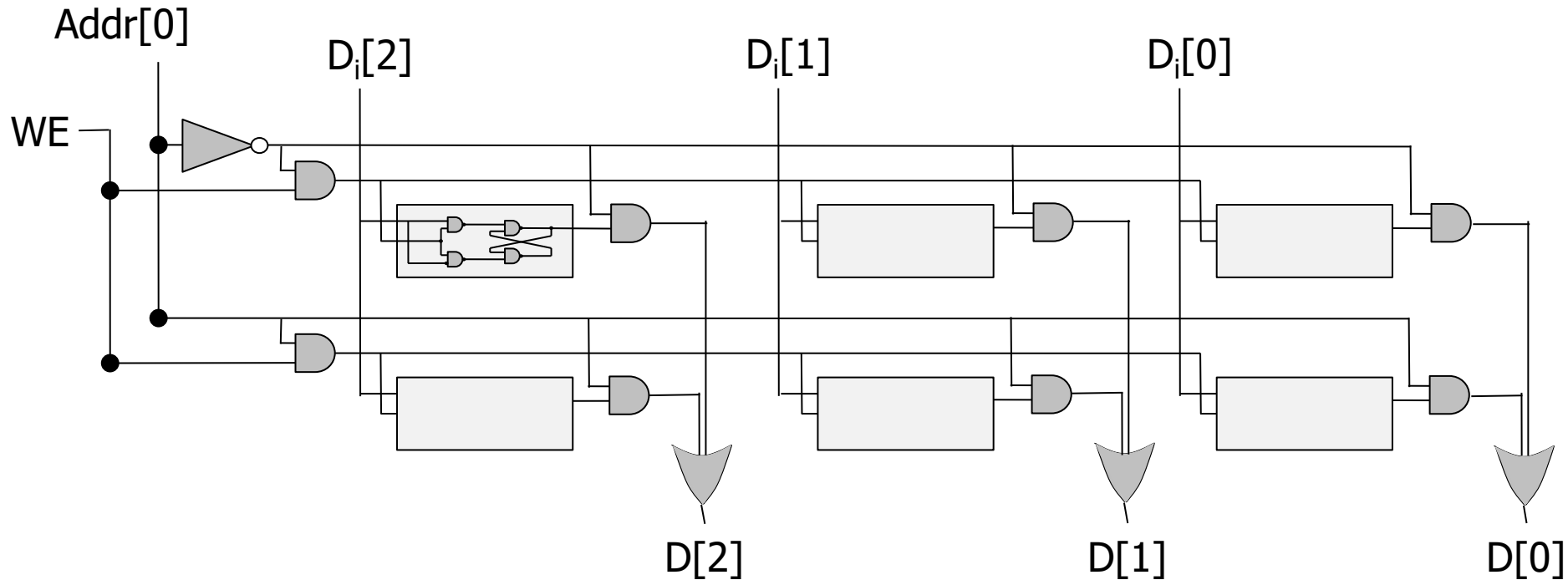
How can we select an address and write to it?

- Input is indicated with D_i

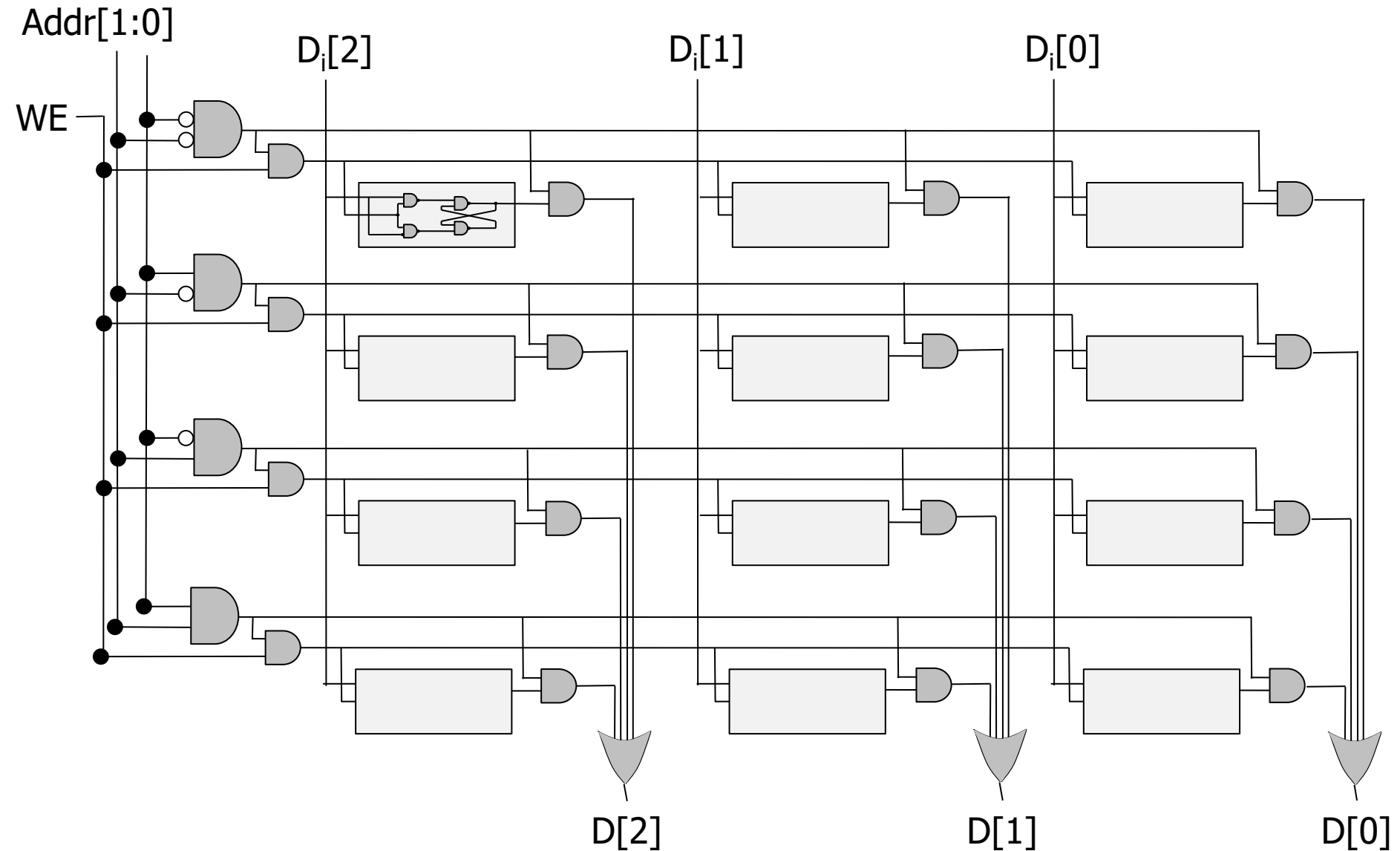


Putting it all Together

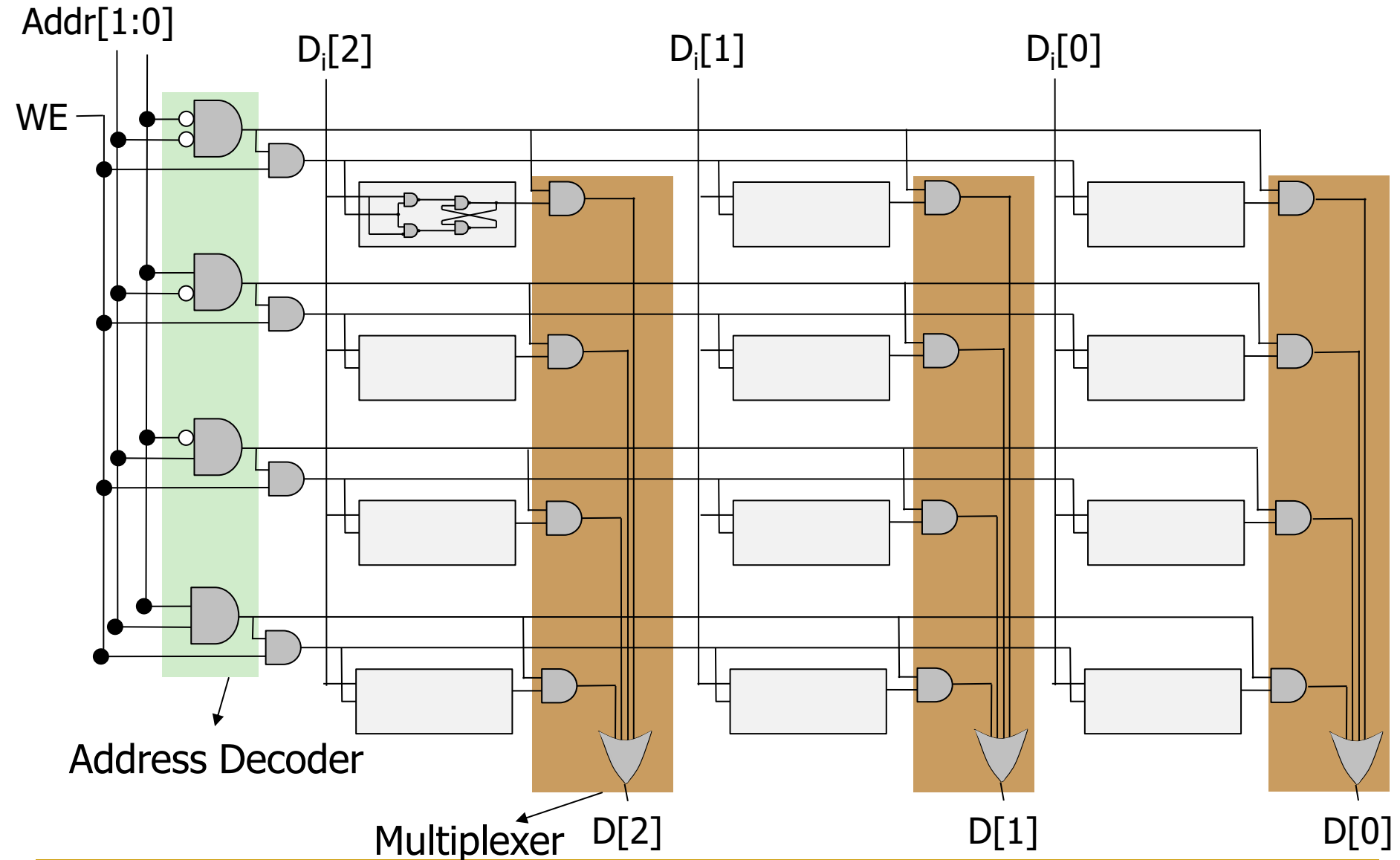
Let's enable reading from and writing to a memory array



A Bigger Memory Array (4 locations X 3 bits)



A Bigger Memory Array (4 locations X 3 bits)



(together w/ decoder, which provides the select input)

Example: Reading Location 3

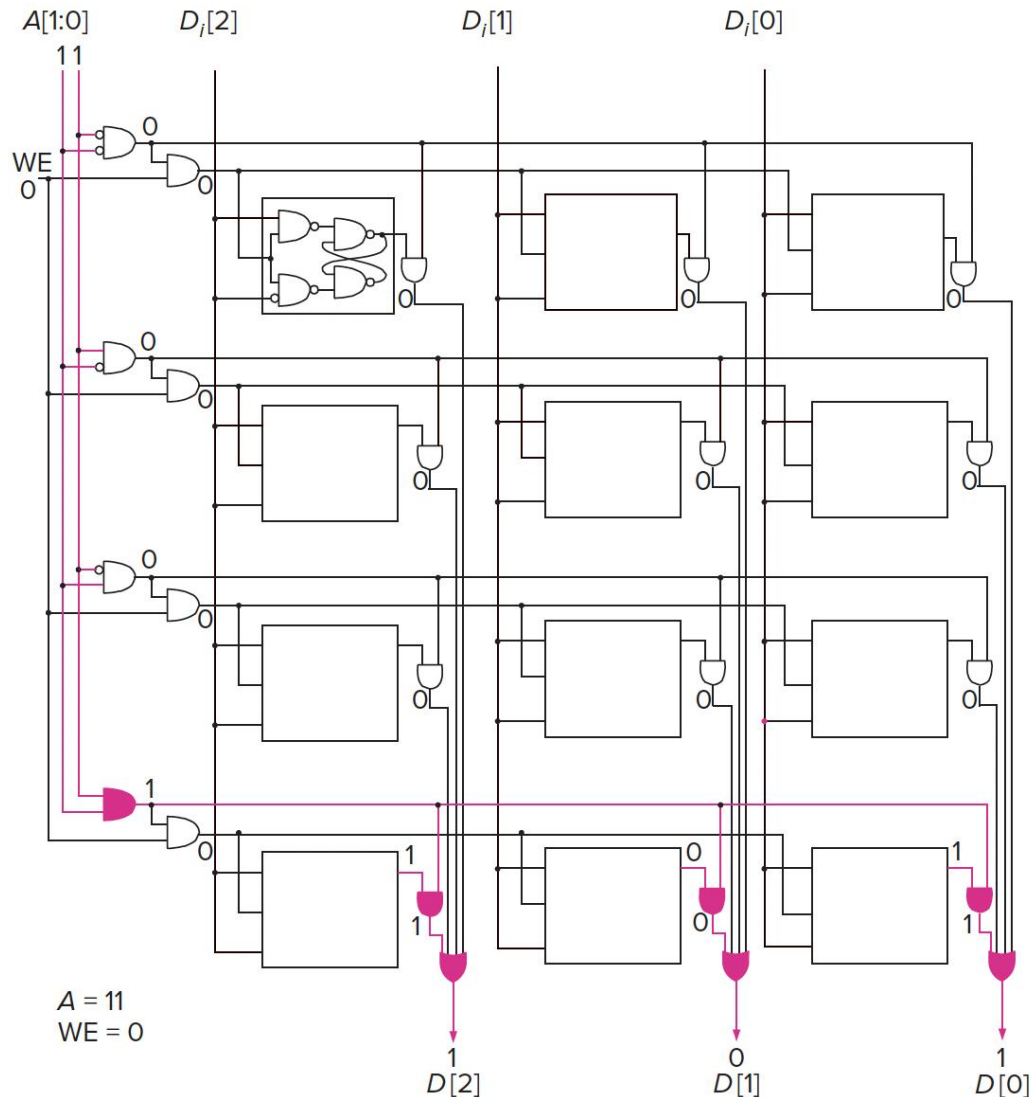
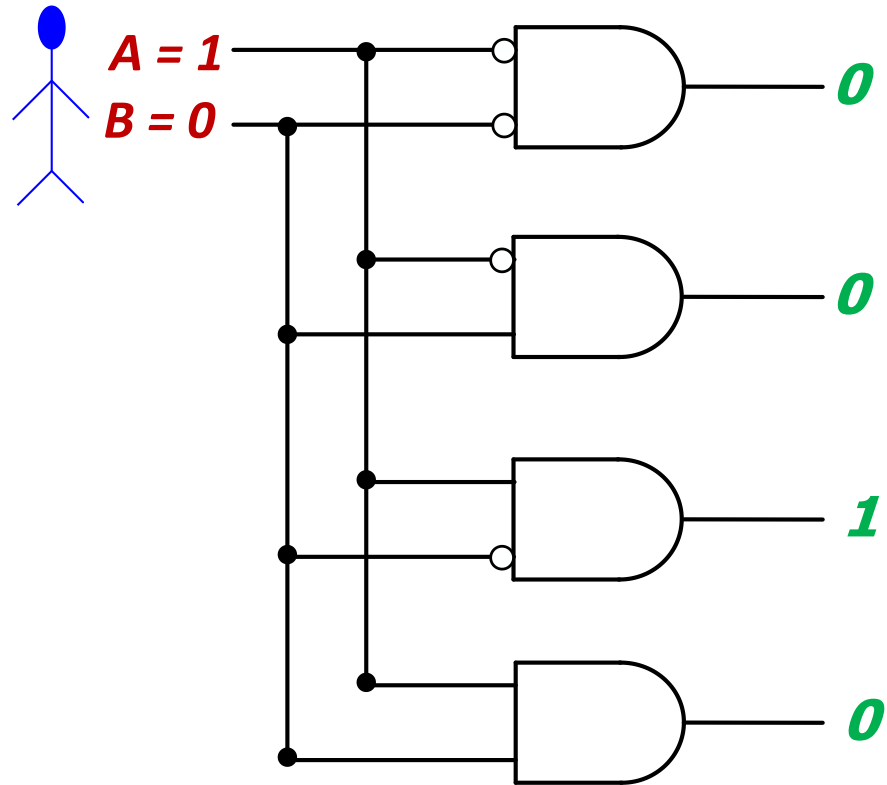


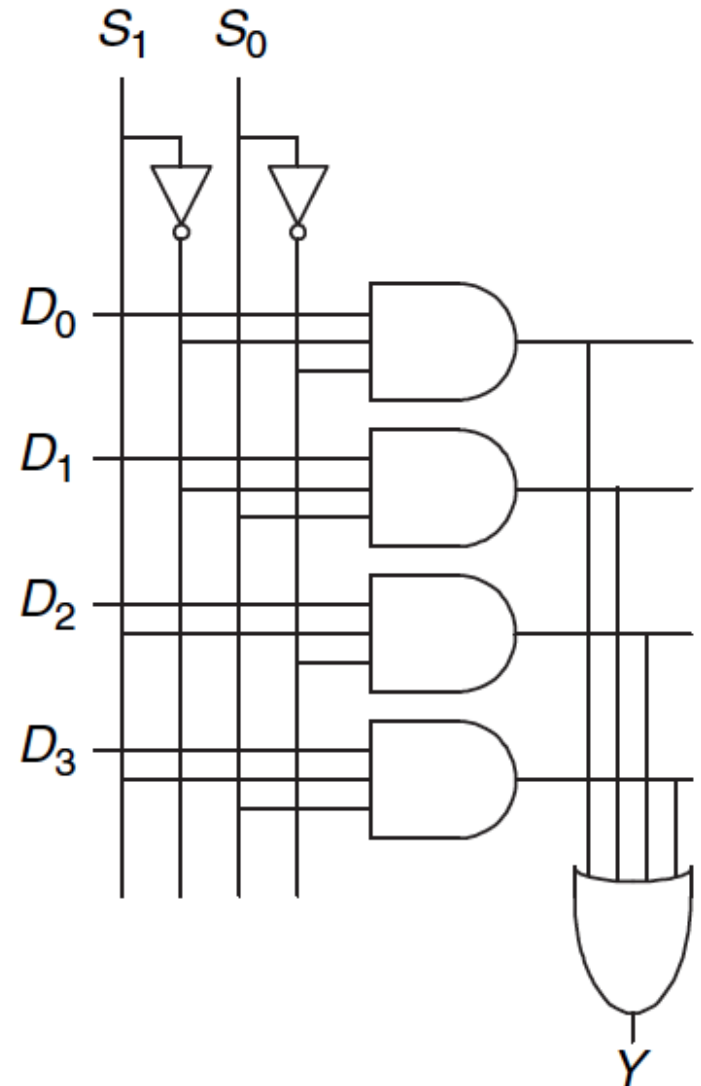
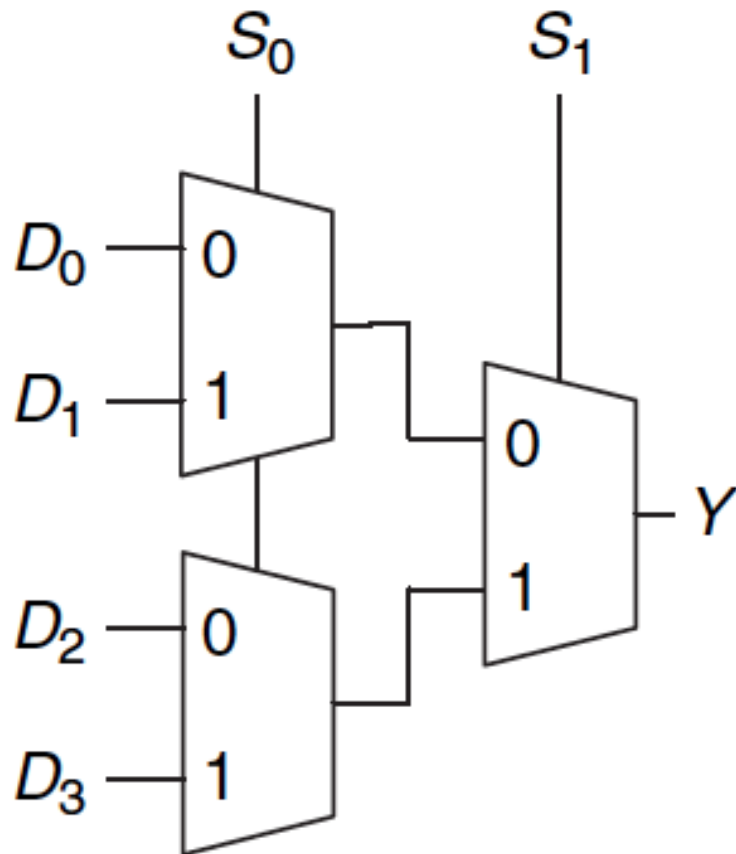
Figure 3.21 Reading location 3 in our 2²-by-3-bit memory.

Recall: Decoder (II)

- The decoder is useful in determining how to interpret a bit pattern
 - **It could be the address of a location in memory, that the processor intends to read from**
 - **It could be an instruction in the program and the processor needs to decide what action to take (based on *instruction opcode*)**

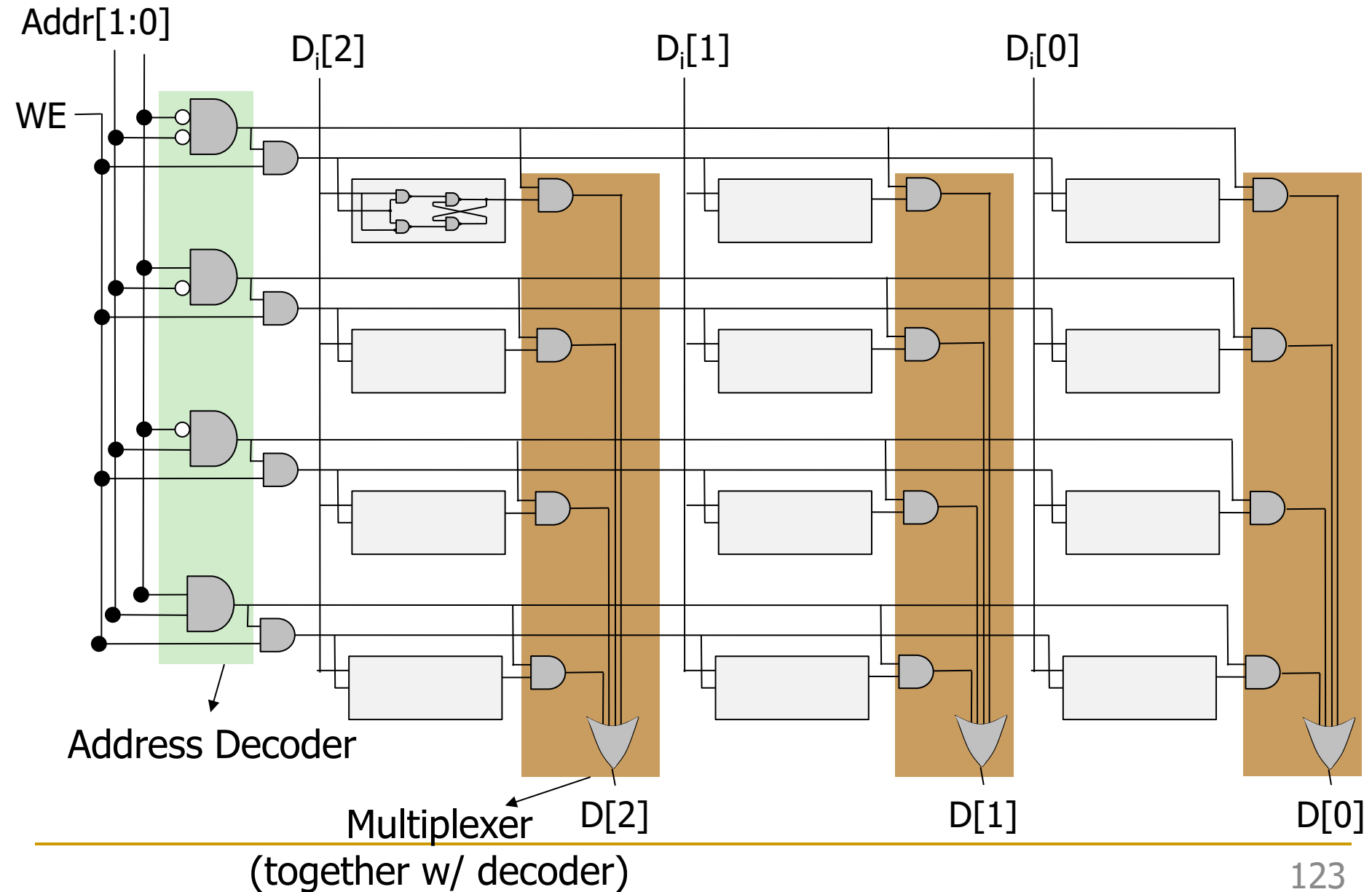


Recall: A 4-to-1 Multiplexer



Aside: Implementing Logic Functions Using Memory

Recall: A Bigger Memory Array (4 locations X 3 bits)



Memory-Based Lookup Table Example

- Memory arrays can also perform Boolean Logic functions
 - ❑ 2^N -location M -bit memory can perform any N -input, M -output function
 - ❑ Lookup Table (LUT): Memory array used to perform logic functions
 - ❑ Each address: row in truth table; each data bit: corresponding output value

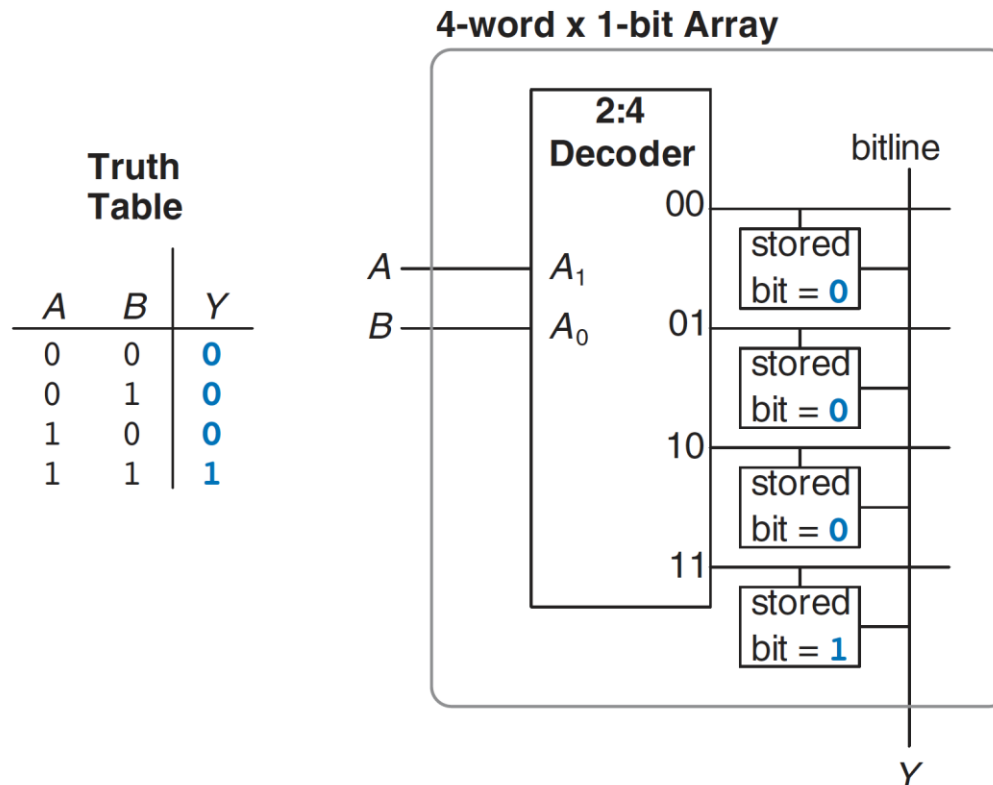


Figure 5.52 4-word $\times$ 1-bit memory array used as a lookup table

Aside: Lookup Tables (LUTs) in FPGAs

- LUTs are commonly used in FPGAs
 - To enable programmable/reconfigurable logic functions
 - To enable easy integration of combinational and sequential logic

(A) data 1	(B) data 2	(C) data 3	data 4	(X) LUT output
0	0	0	X	0
0	0	1	X	1
0	1	0	X	0
0	1	1	X	0
1	0	0	X	0
1	0	1	X	0
1	1	0	X	1
1	1	1	X	0

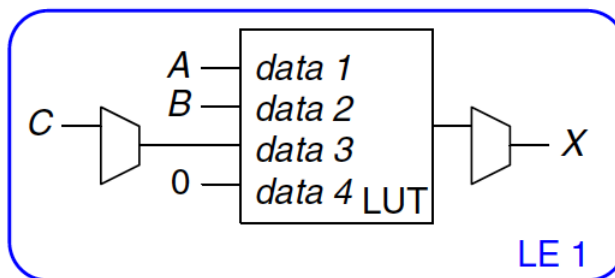
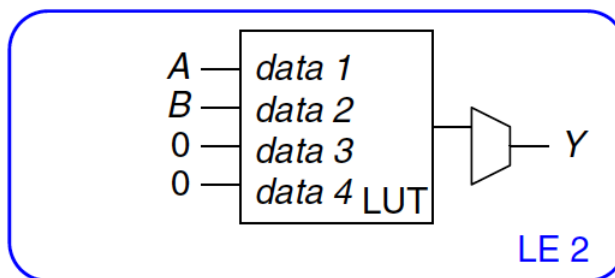


Figure 5.59 LE configuration for two functions of up to four inputs each

(A) data 1	(B) data 2	data 3	data 4	(Y) LUT output
0	0	X	X	0
0	1	X	X	0
1	0	X	X	1
1	1	X	X	0



Recall: A Multiplexer-Based LUT

- Let's implement a function that outputs '1' when there are at least two '1's in a 3-bit input

In C:

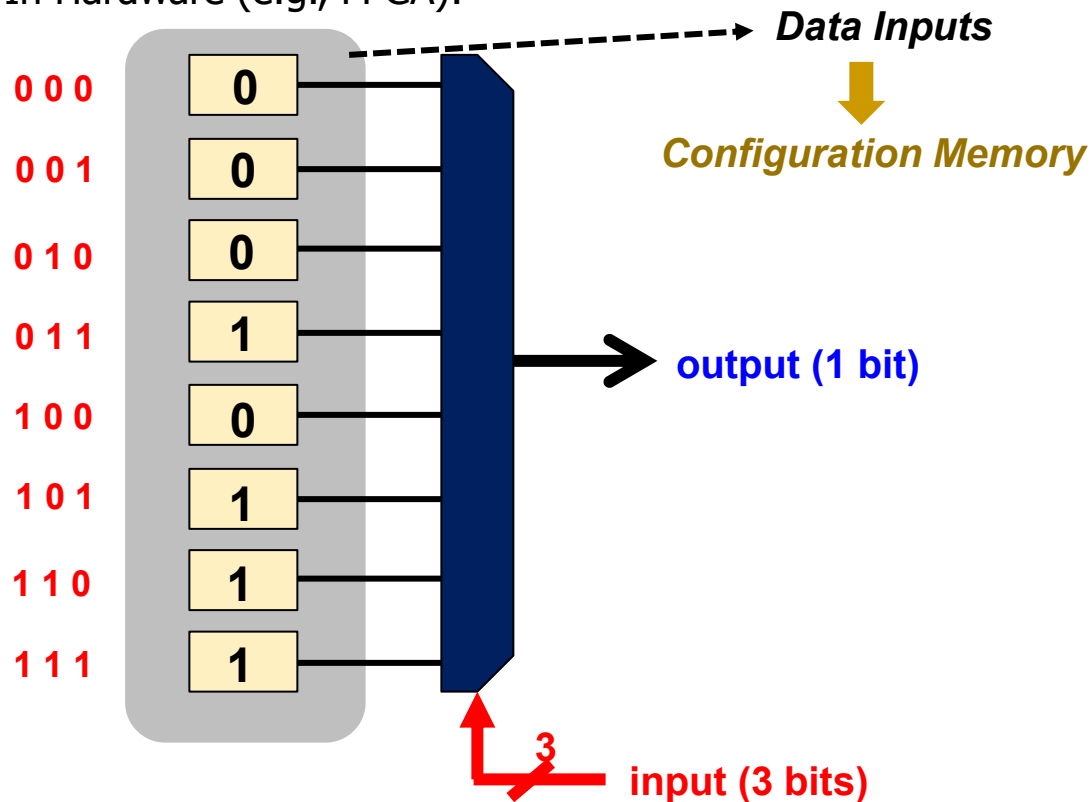
```
int count = 0;
for(int i = 0; i < 3; i++) {
    count += input & 1;
    input = input >> 1;
}

if(count > 1) return 1;

return 0;
```

```
switch(input){
    case 0:
    case 1:
    case 2:
    case 4:
        return 0;
    default:
        return 1;}
```

In Hardware (e.g., FPGA):



Sequential Logic Circuits

Sequential Logic Circuits

- We have examined designs of circuit elements that can **store information**
- Now, we will use these elements to build circuits that **remember** past inputs



Combinational

Only depends on current inputs



Sequential

Opens depending on past inputs

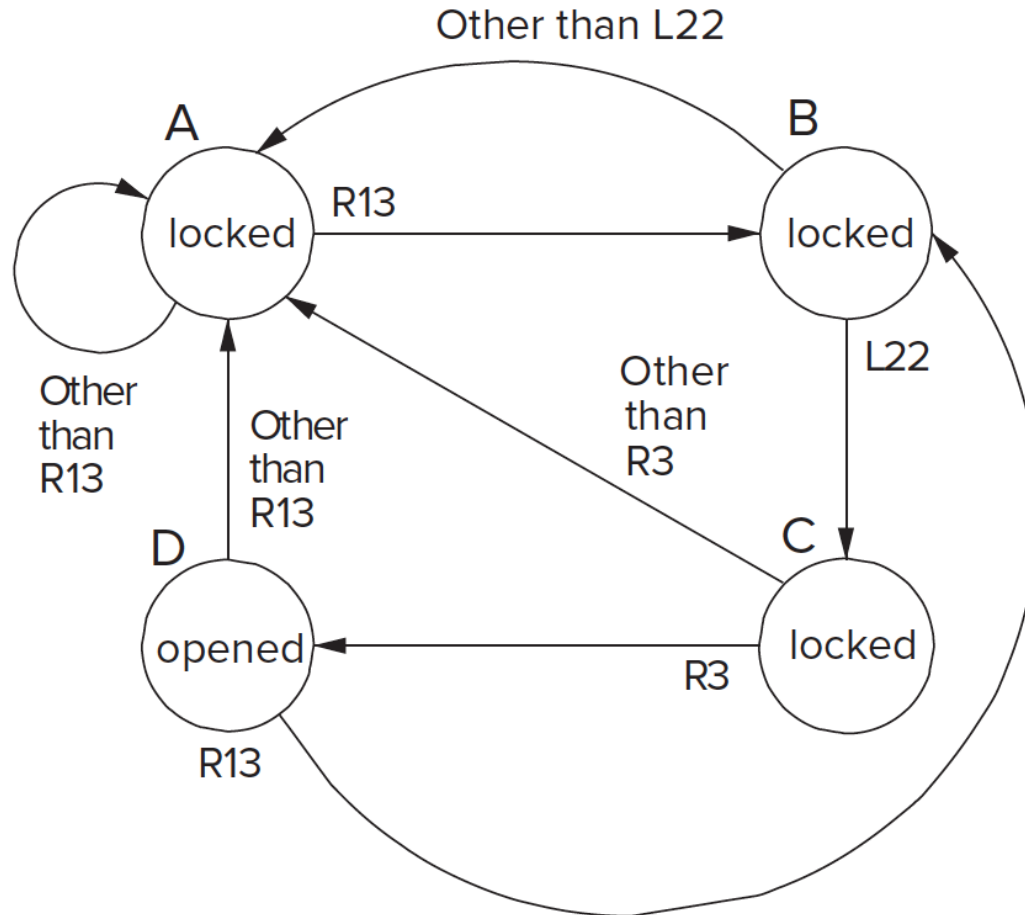
State

- In order for this lock to work, it has to keep track (**remember**) of the past events!
- If passcode is **R13-L22-R3**, sequence of **states** to unlock:
 - A. The lock is not open (locked), and no relevant operations have been performed
 - B. Locked but user has completed R13
 - C. Locked but user has completed R13-L22
 - D. Unlocked: user has completed R13-L22-R3
- The **state** of a system is a snapshot of all relevant elements of the system at the moment of the snapshot
 - ❑ To open the lock, **states A-D must be completed in order**
 - ❑ If anything else happens (e.g., L5), lock **returns** to state A



State Diagram of Our Sequential Lock

- Completely describes the operation of the sequential lock



- We will understand "state diagrams" fully later today

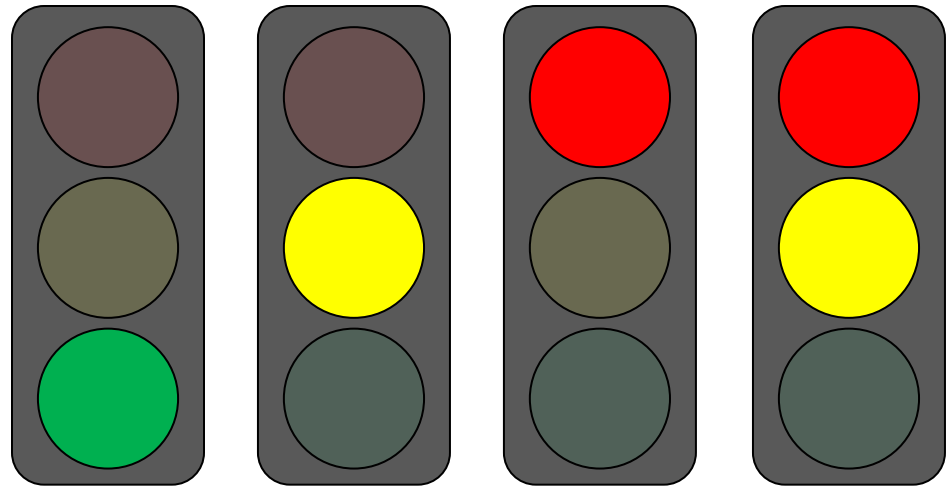
Asynchronous vs. Synchronous State Changes

- Sequential lock we saw is an **asynchronous** “machine”
 - State transitions can take place immediately in response to input
 - There is nothing that synchronizes when each state transition must occur
- Most modern computers are **synchronous** “machines”
 - State transitions take place at fixed units of time (i.e., potentially delayed response to input, synchronized to an external signal)
 - Controlled in part by a clock, as we will see soon
- These are two different design paradigms, with tradeoffs

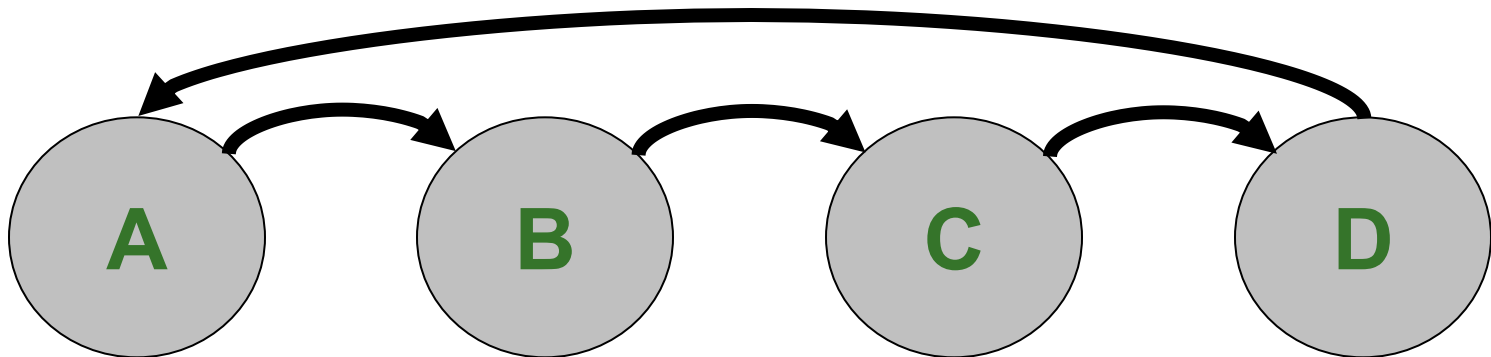
Another Simple Example of State

- A standard Swiss traffic light has **4 states**

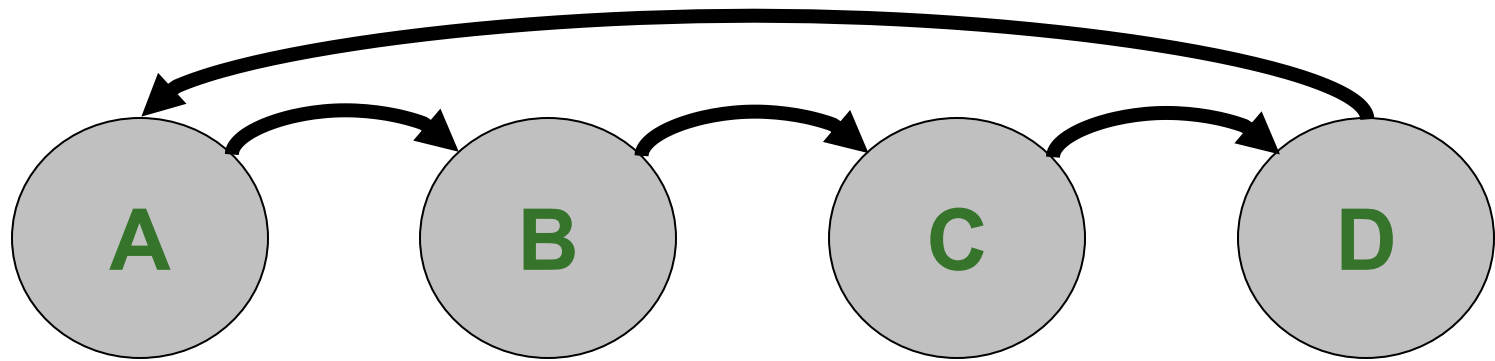
- A. Green
- B. Yellow
- C. Red
- D. Red and Yellow



- The sequence of these states are always as follows



Changing State: The Notion of Clock (I)



- When should the light change from one state to another?
- We need a **clock** to dictate when to change state
 - Clock signal alternates between 0 & 1

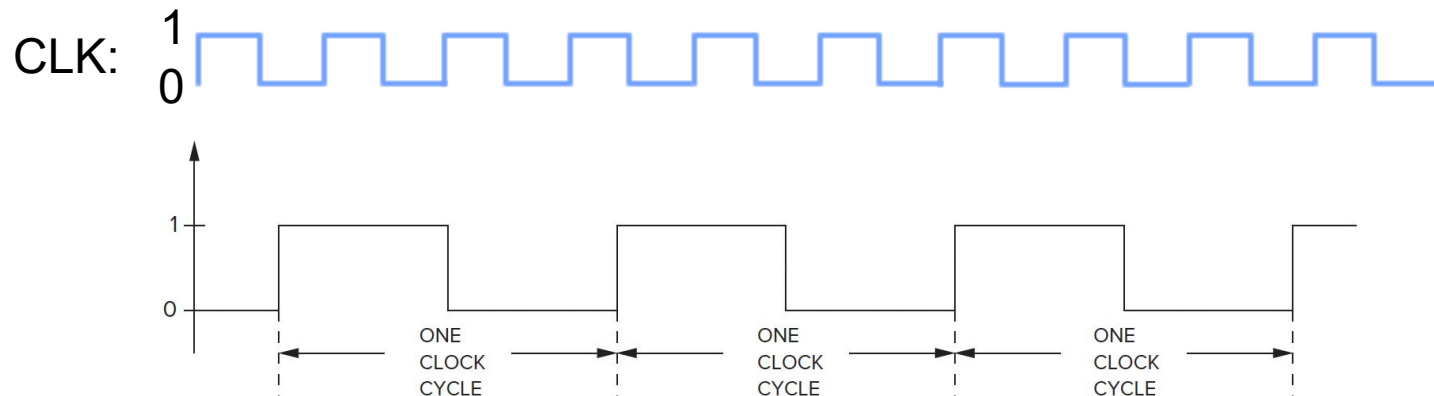
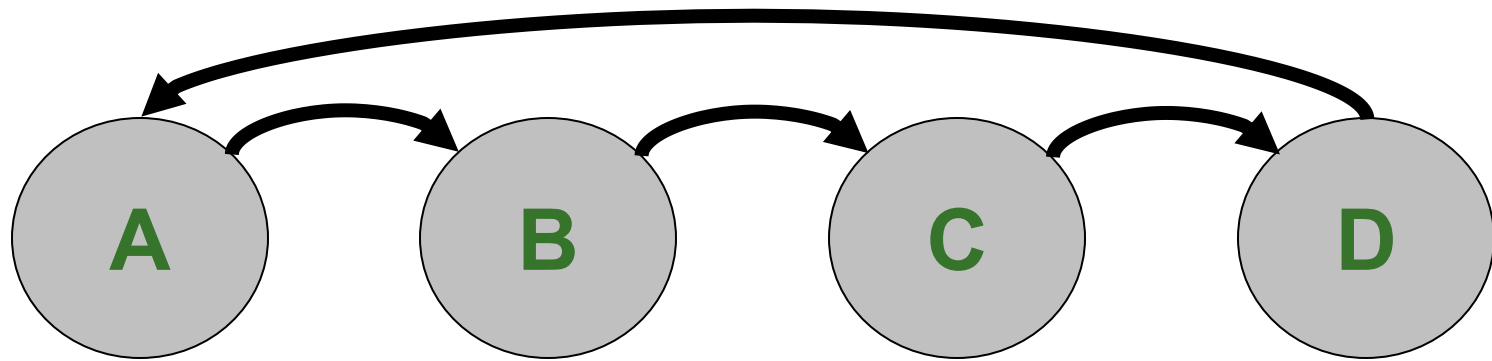


Figure 3.28 A clock signal.

Changing State: The Notion of Clock (I)



- When should the light change from one state to another?
- We need a **clock** to dictate when to change state
 - Clock signal alternates between 0 & 1



- At the start of a clock cycle (), system state changes
 - During a clock cycle, the state stays constant
 - In this traffic light example, we are assuming the traffic light stays in each state an equal amount of time

Changing State: The Notion of Clock (II)

- **Clock** is a general mechanism that **triggers transition from one state to another** in a (synchronous) sequential circuit
- Clock **synchronizes state changes** across many sequential circuit elements
- Combinational logic evaluates for the **length of the clock cycle**
- Clock cycle should be chosen to accommodate maximum combinational circuit delay
 - More on this later, when we discuss timing

Asynchronous vs. Synchronous State Changes

- Sequential lock we saw is an **asynchronous** “machine”
 - State transitions can take place immediately in response to input
 - There is nothing that synchronizes when each state transition must occur
- Most modern computers are **synchronous** “machines”
 - State transitions take place at fixed units of time (i.e., synch.)
 - Controlled in part by a clock, as we will see soon
- **These are two different design paradigms, with tradeoffs**
 - Synchronous control can be easier to get correct when the system consists of many components and many states
 - Asynchronous control can be more efficient (no clock overheads)

Finite State Machines

Finite State Machines

- What is a **Finite State Machine** (FSM)?
 - **A discrete-time model** of a stateful system
 - Each state represents a snapshot of the system at a given time
- An FSM pictorially shows
 1. the set of all possible **states** that a system can be in
 2. how the system transitions from one state to another
- An FSM can model
 - A traffic light, an elevator, fan speed, a microprocessor, etc.
- **An FSM enables us to pictorially think of a stateful system using simple diagrams**

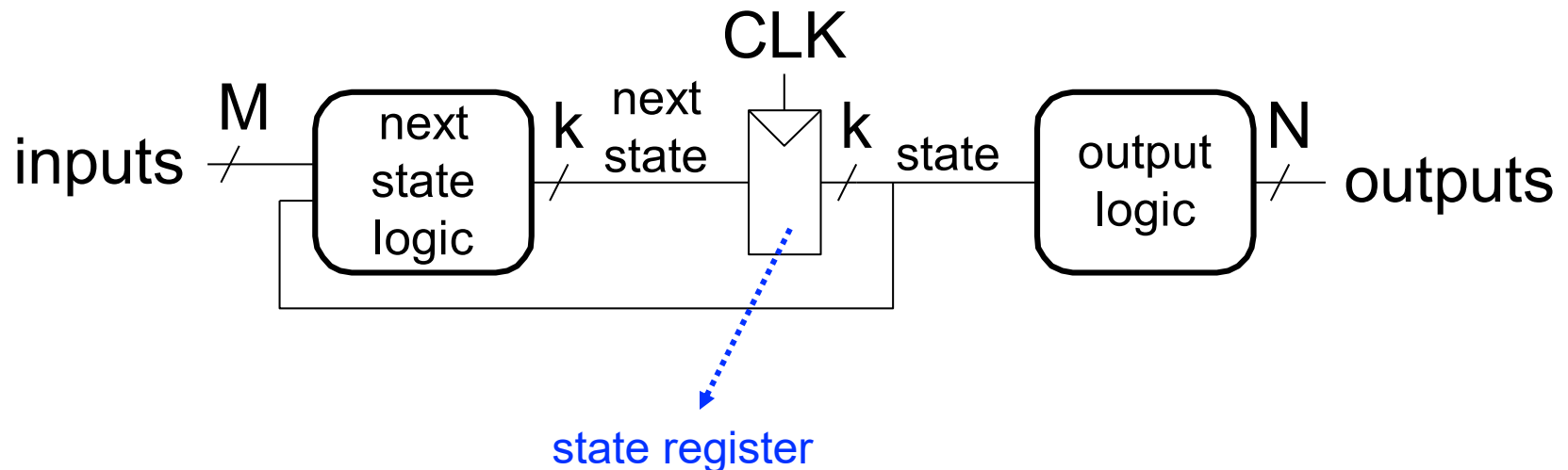
Finite State Machines (FSMs) Consist of:

■ Five elements:

1. A **finite** number of **states**
 - **State**: snapshot of all relevant elements of the system at the time of the snapshot
2. A **finite** number of external **inputs**
3. A **finite** number of external **outputs**
4. An explicit **specification of all state transitions**
 - How to get from one state to another
5. An explicit **specification of what determines each external output value**

Finite State Machines (FSMs)

- Each FSM consists of three separate parts:
 - next state logic
 - state register
 - output logic

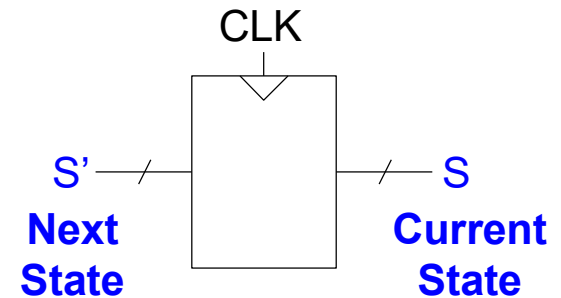


At the beginning of the clock cycle, next state is latched into the state register

Finite State Machines (FSMs) Consist of:

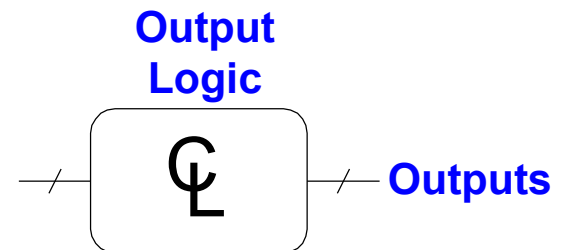
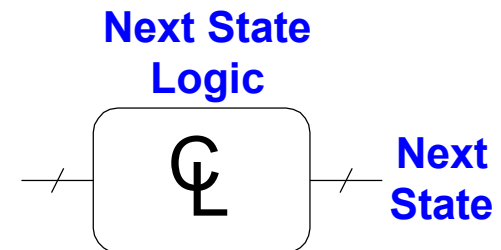
■ Sequential Circuits

- State register(s)
 - Store the current state and
 - Load the next state at the clock edge



■ Combinational Circuits

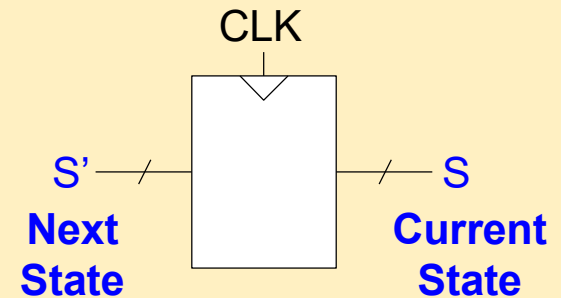
- Next state logic
 - Determines what the next state will be
- Output logic
 - Generates the outputs



Finite State Machines (FSMs) Consist of:

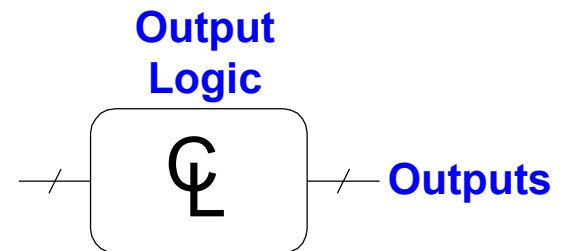
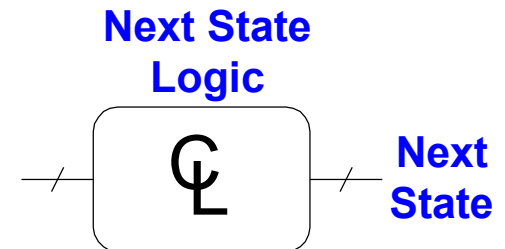
■ Sequential Circuits

- State register(s)
 - Store the current state and
 - Provide the next state at the clock edge



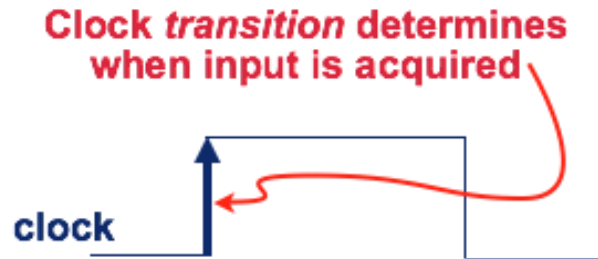
■ Combinational Circuits

- Next state logic
 - Determines what the next state will be
- Output logic
 - Generates the outputs

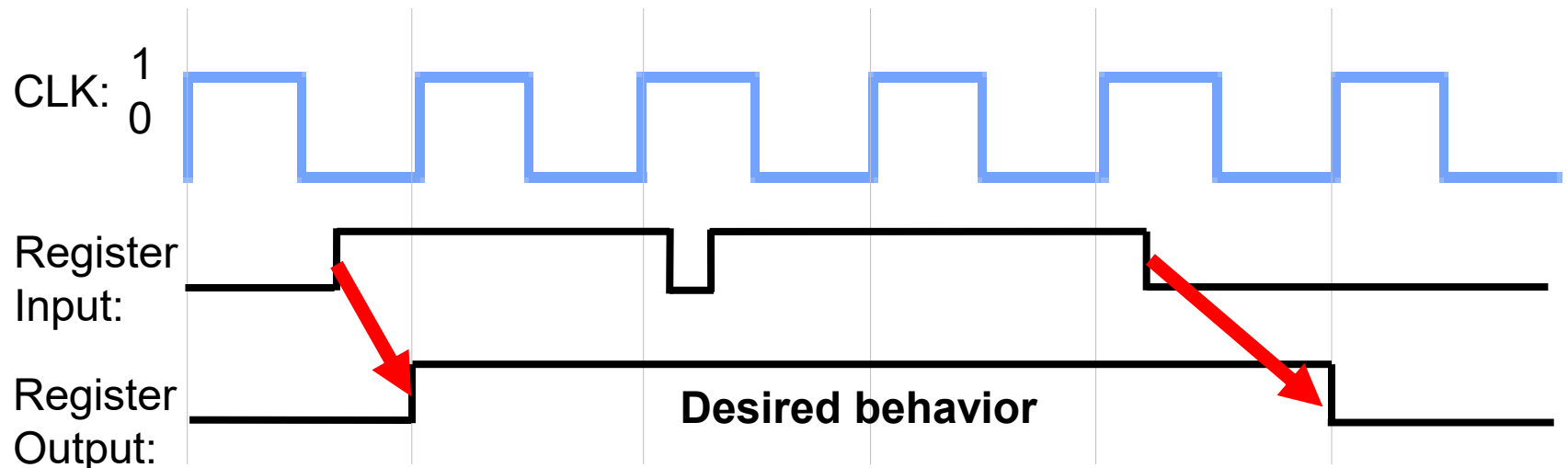


State Register Implementation

- How can we implement a **state register**? Two properties:
 1. We need to store data at the **beginning** of every clock cycle

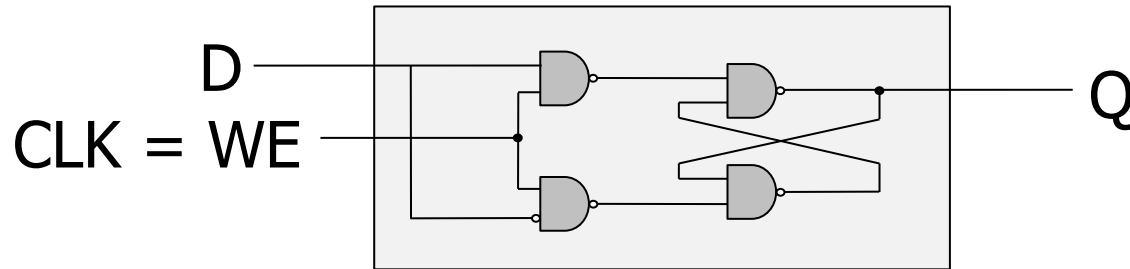


2. The data must be **available** during the **entire clock cycle**

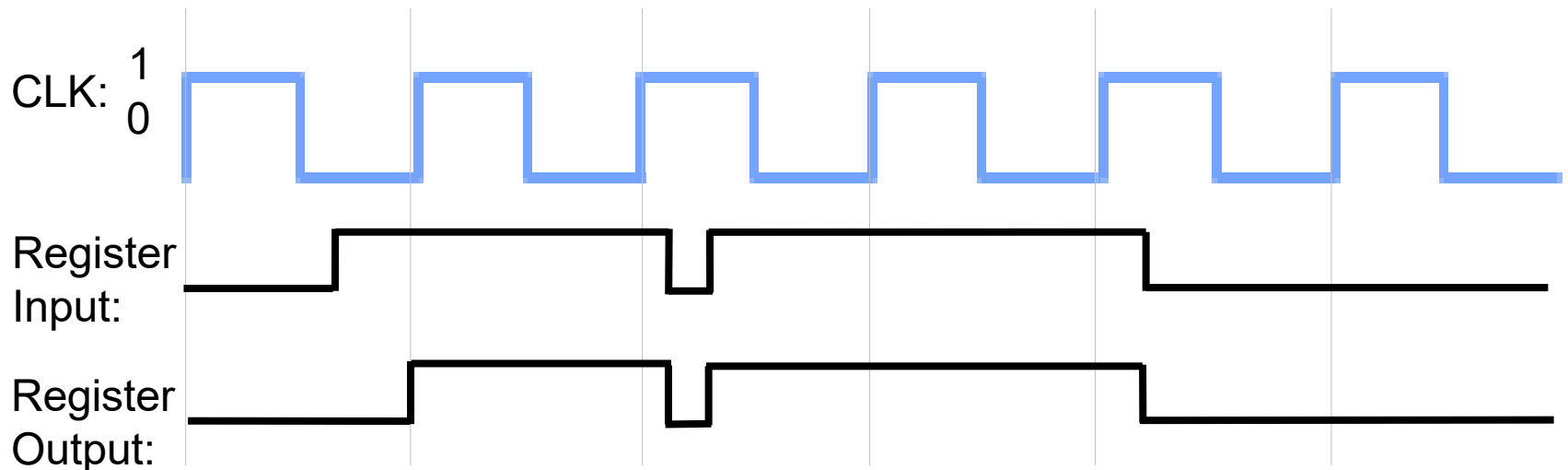


The Problem with Latches: Transparency

Recall the
Gated D Latch

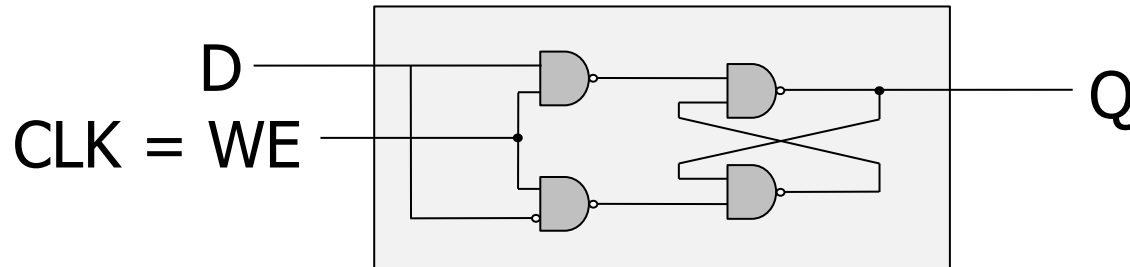


- Currently, we **cannot** simply wire a clock to WE of a latch
 - ❑ **Whenever the clock is high**, the latch propagates **D** to **Q**
 - ❑ **The latch is “transparent”**

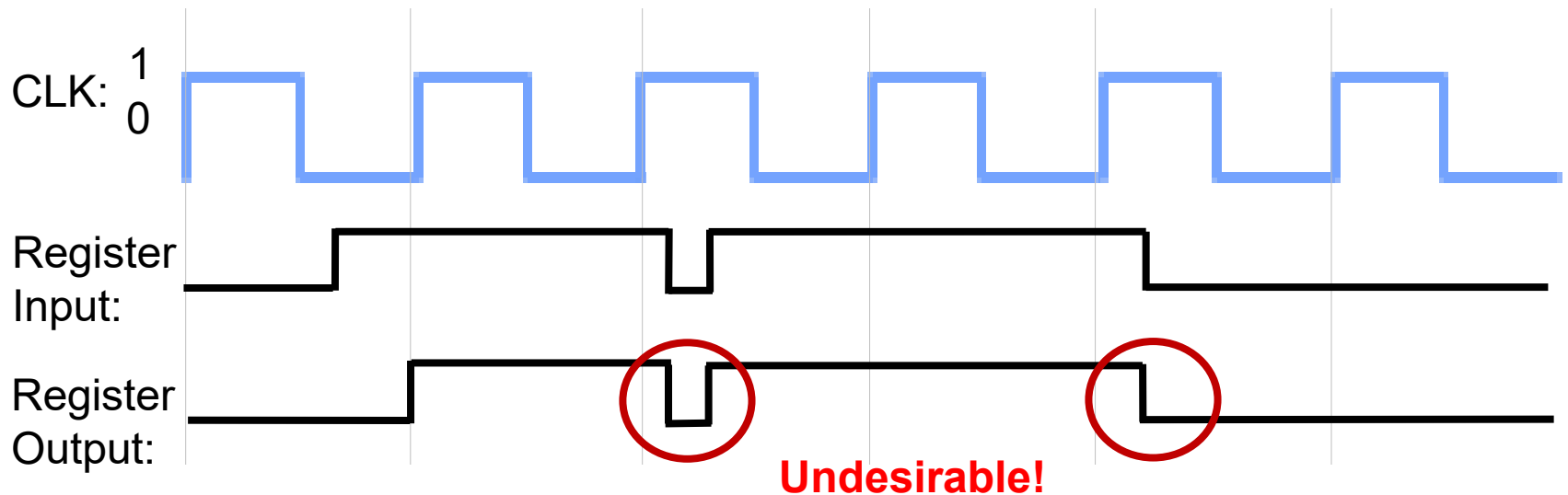


The Problem with Latches: Transparency

Recall the
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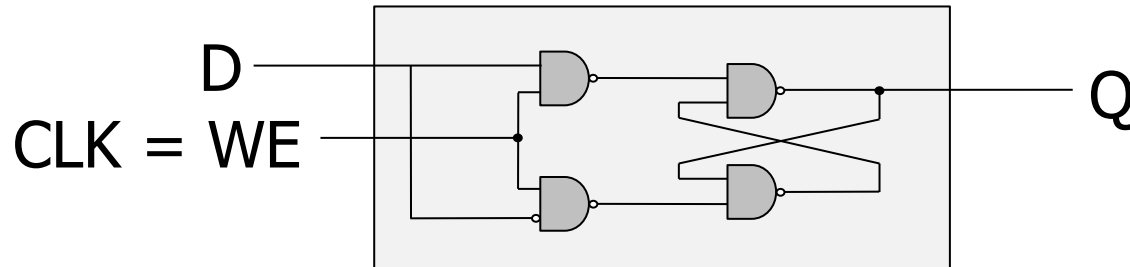


- Currently, we **cannot** simply wire a clock to WE of a latch
 - ❑ **Whenever the clock is high**, the latch propagates **D** to **Q**
 - ❑ **The latch is “transparent”**



The Problem with Latches: Transparency

Recall the
Gated D Latch



How can we change the latch, so that

1) D (input) is **observable** at **Q** (output)
only at the **beginning of next** clock cycle?

2) Q is **available for the full clock cycle**

The Need for a New Storage Element

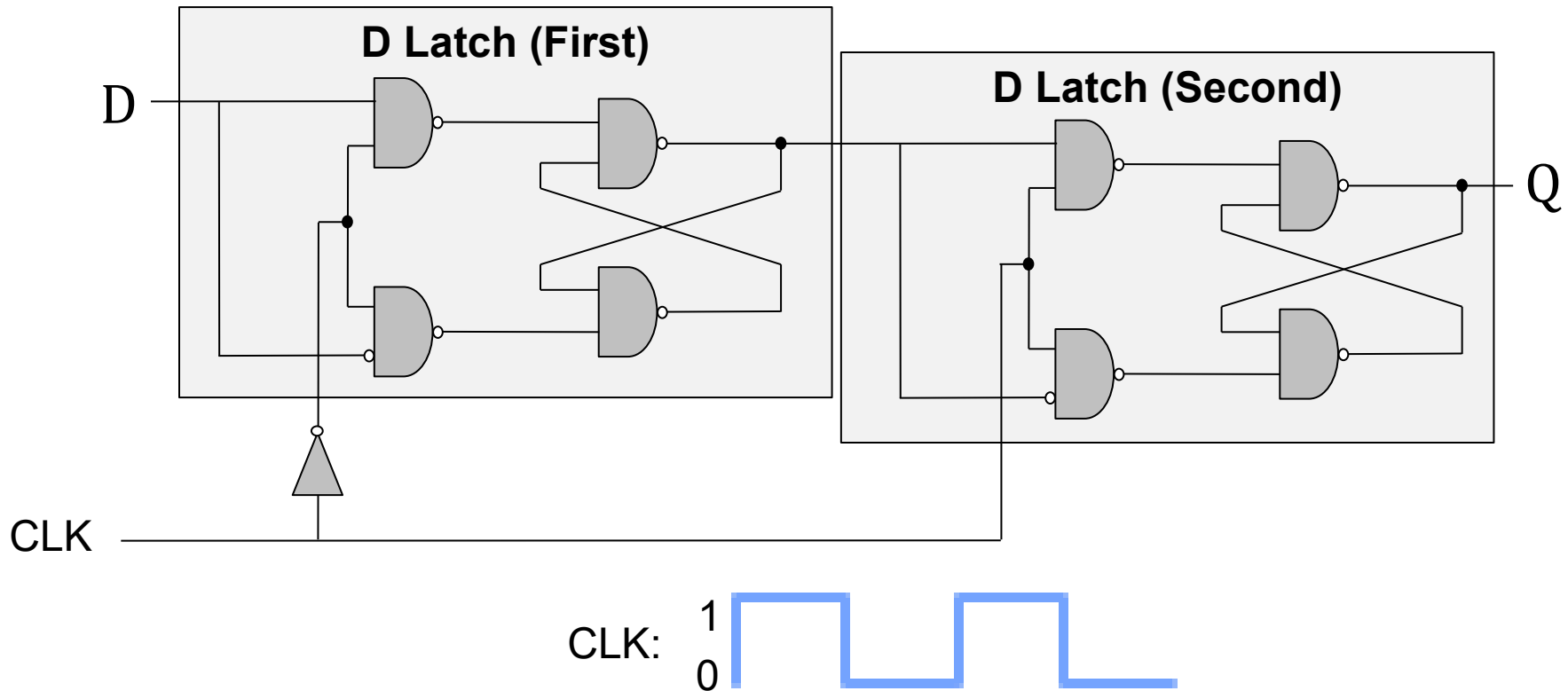
- To design viable FSMs
- We need storage elements that allow us to:
 - **read the current state** throughout the **entire current clock cycle**

AND

- **not write the next state** values into the storage elements **until** the beginning of the **next clock cycle**

The D Flip-Flop

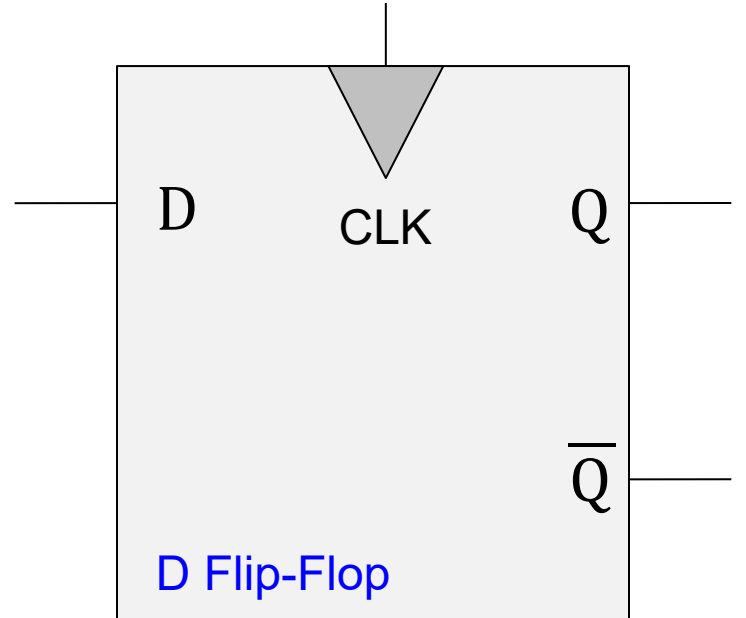
- 1) state change on clock edge, 2) data available for full cycle



- When the clock is low, 1st latch propagates **D** to the input of the 2nd (Q unchanged)
- Only when the clock is high, 2nd latch latches **D** (**Q stores D**)
 - At the rising edge of clock (clock going from 0→1), Q gets assigned D

The D Flip-Flop

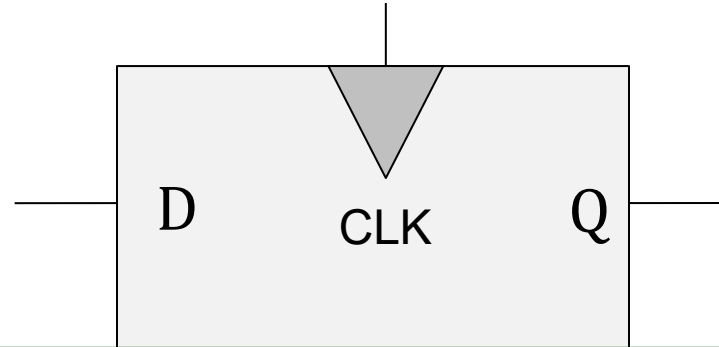
- 1) state change on clock edge, 2) data available for full cycle



- At the rising edge of clock (clock going from 0→1), **Q** gets assigned **D**
- At all other times, Q is unchanged

The D Flip-Flop

- 1) state change on clock edge, 2) data available for full cycle



We can use **D Flip-Flops**
to implement the state register

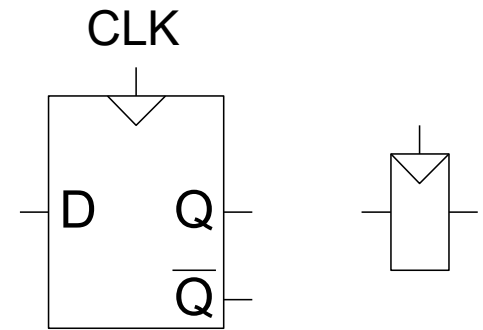
- At the rising edge of clock (clock going from 0→1), **Q** gets assigned **D**
- At all other times, **Q** is unchanged

Rising-Clock-Edge Triggered Flip-Flop

- **Two inputs:** CLK, D

- **Function**

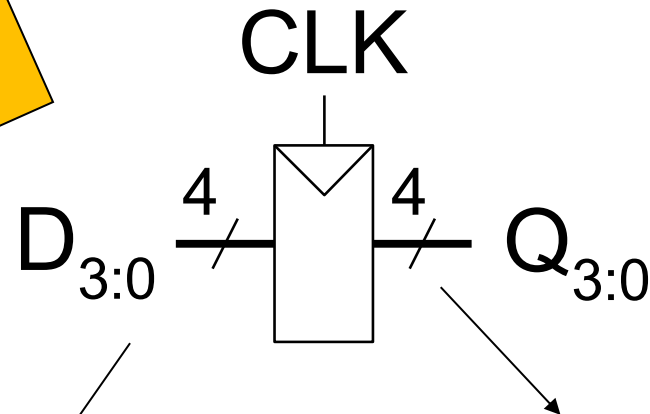
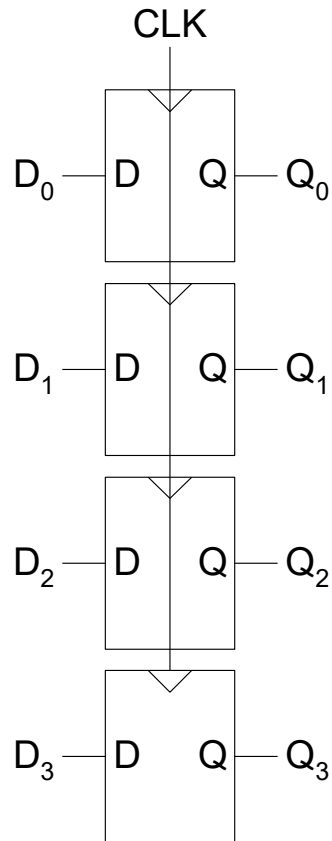
- The flip-flop “samples” **D** on the rising edge of CLK (**positive edge**)
- When CLK rises from 0 to 1, **D** passes through to **Q**
- Otherwise, **Q** holds its previous value
- **Q** changes **only** on the rising edge of CLK



- A flip-flop is called an **edge-triggered state element** because it captures data on the clock edge
 - A latch is a **level-triggered** state element

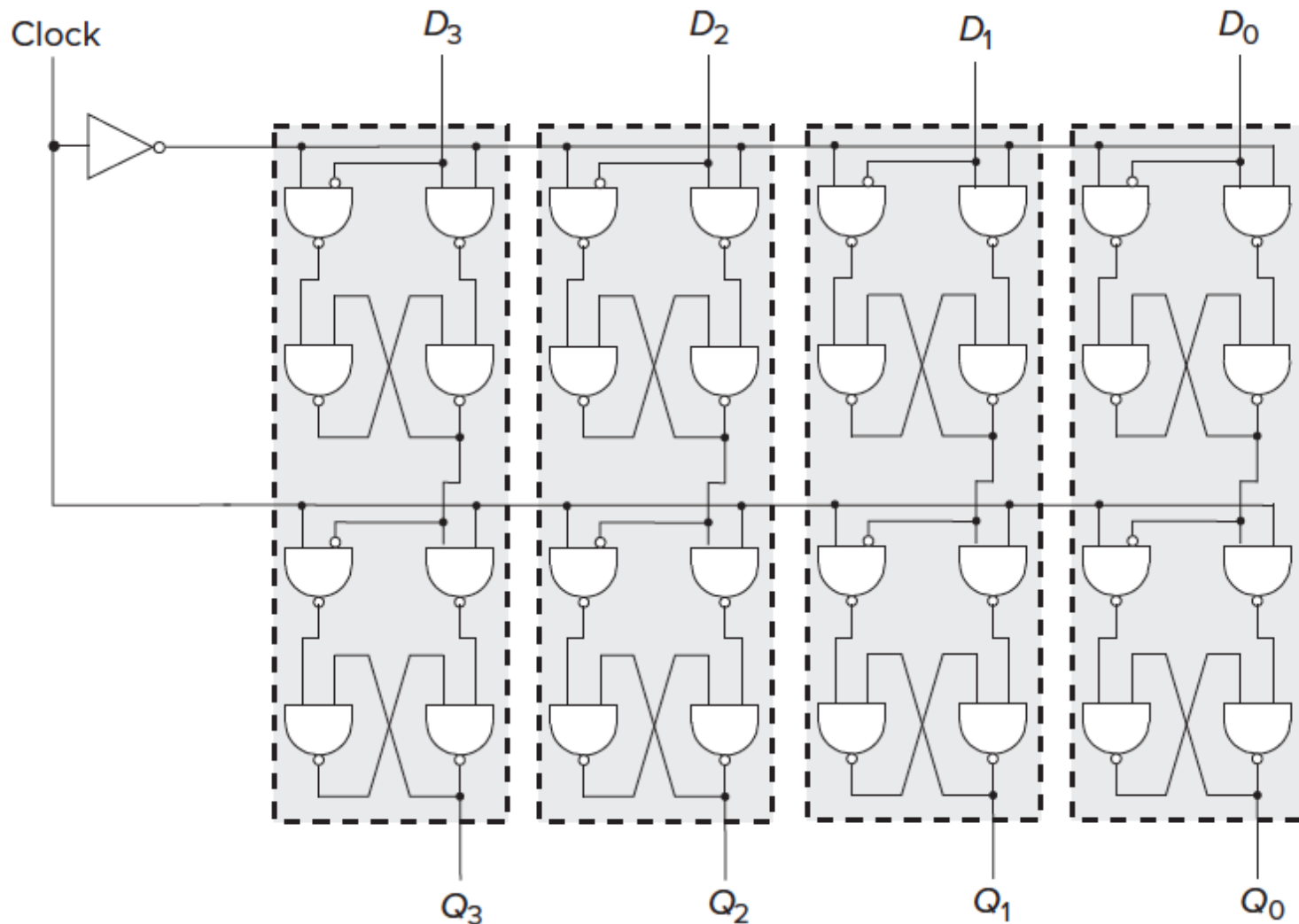
D Flip-Flop Based Register

- Multiple parallel D flip-flops, each of which storing 1 bit



This register stores 4 bits

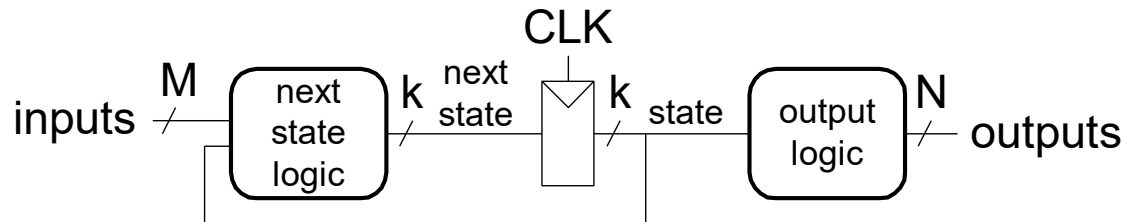
A 4-Bit D-Flip-Flop-Based Register (Internally)



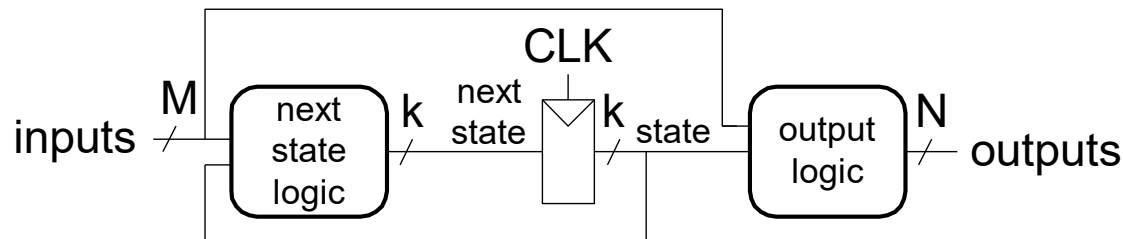
Finite State Machines (FSMs)

- Next state is determined by the current state and the inputs
- Two types of finite state machines differ in the **output logic**:
 - **Moore FSM**: outputs depend only on the current state

Moore FSM



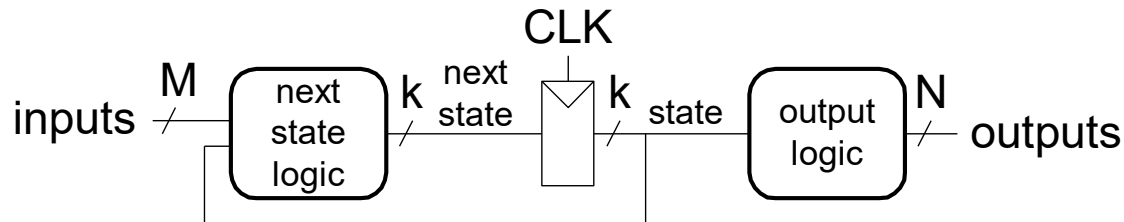
Mealy FSM



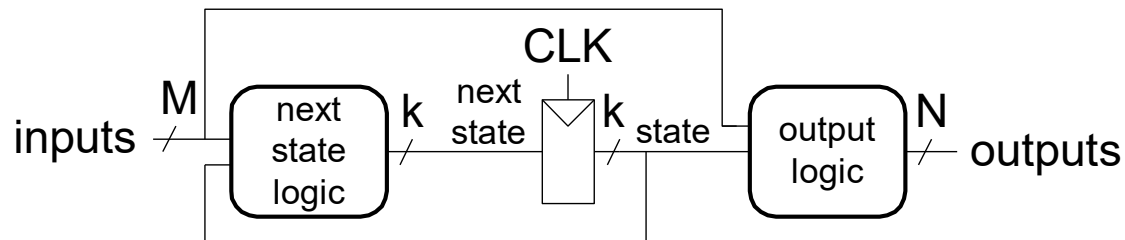
Finite State Machines (FSMs)

- Next state is determined by the current state and the inputs
- Two types of finite state machines differ in the **output logic**:
 - **Moore FSM**: outputs depend only on the current state
 - **Mealy FSM**: outputs depend on the current state and the inputs

Moore FSM

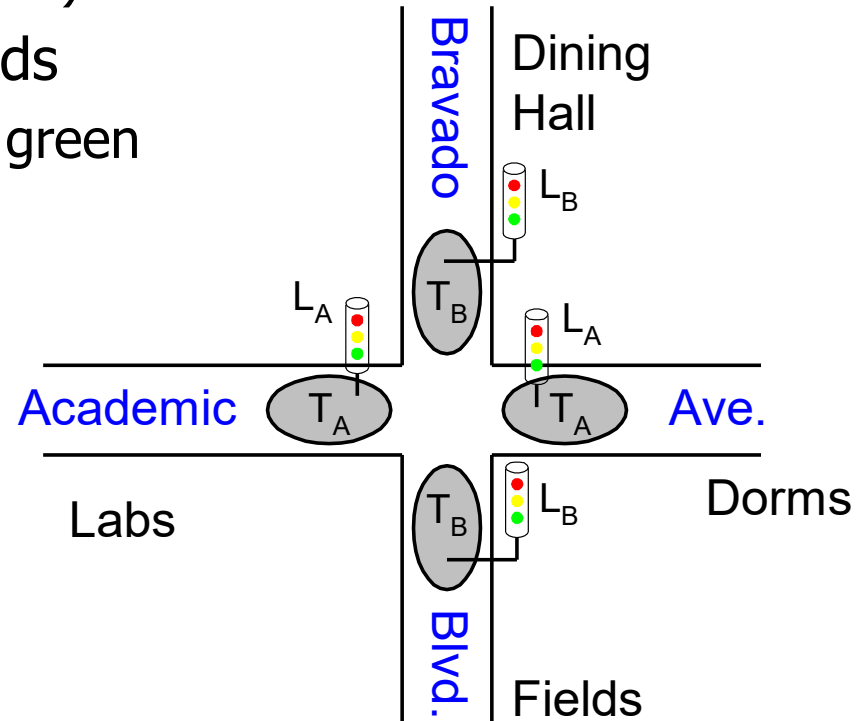


Mealy FSM



Finite State Machine Example

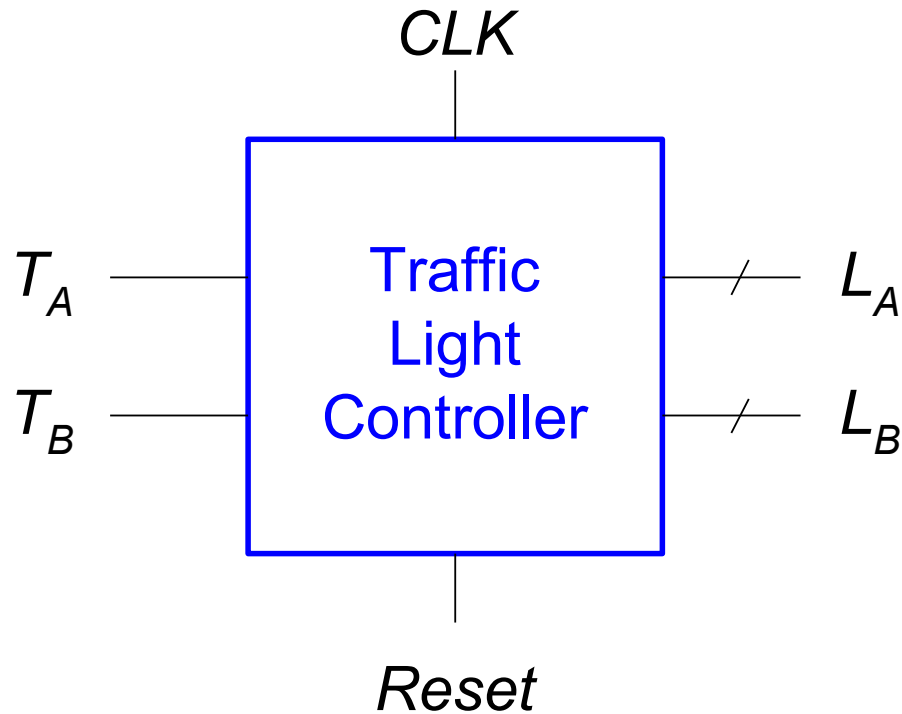
- “Smart” traffic light controller
 - **2 inputs:**
 - Traffic sensors: T_A , T_B (TRUE when there's traffic)
 - **2 outputs:**
 - Lights: L_A , L_B (Red, Yellow, Green)
 - State can change every 5 seconds
 - Except if green and traffic, stay green



From H&H Section 3.4.1

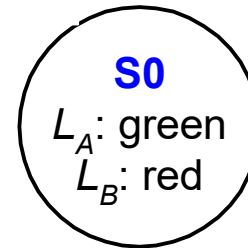
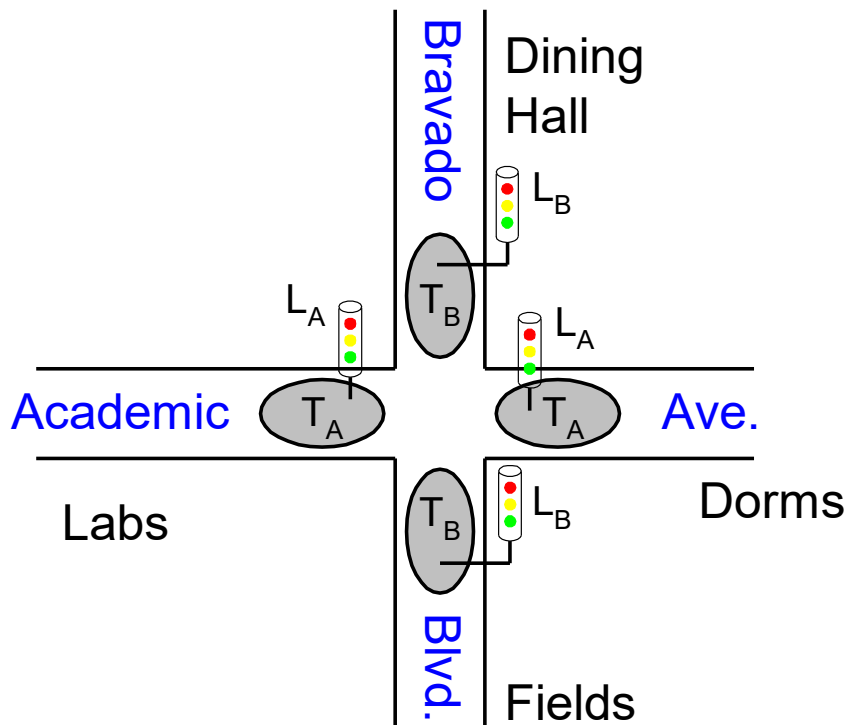
Finite State Machine Black Box

- **Inputs:** CLK, Reset, T_A , T_B
- **Outputs:** L_A , L_B



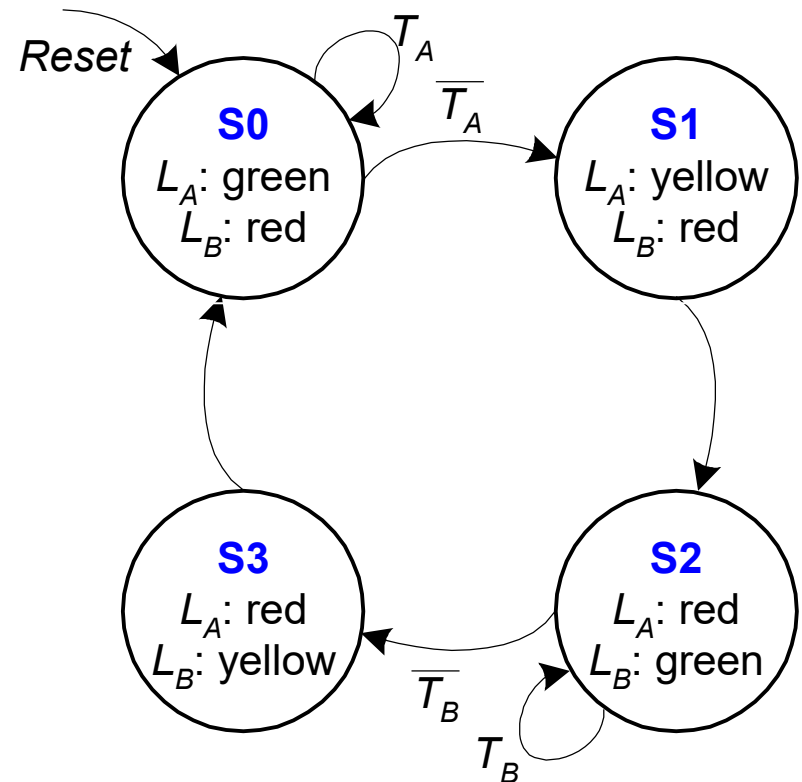
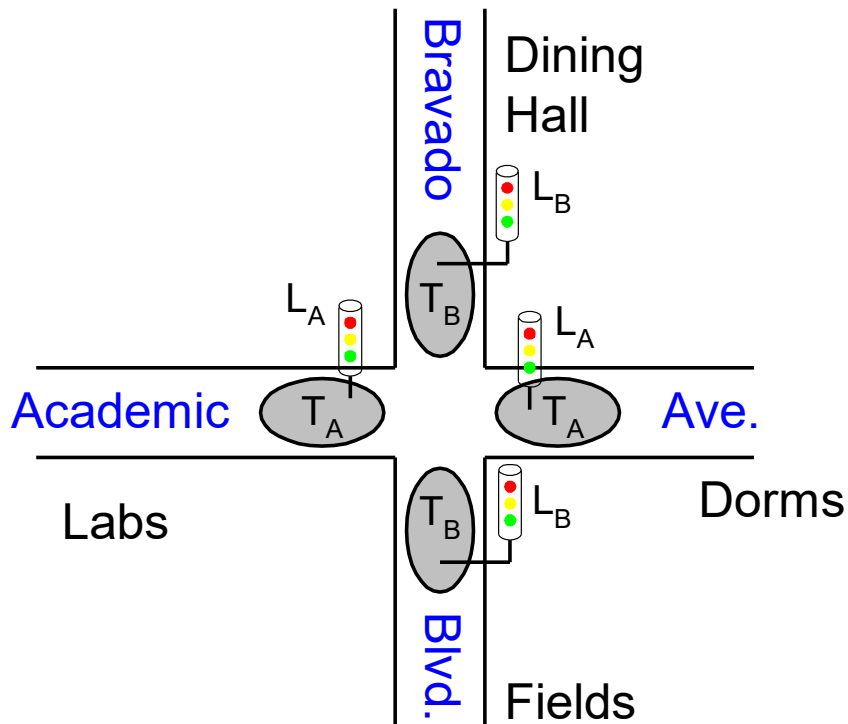
Finite State Machine Transition Diagram

- **Moore FSM:** outputs labeled in each state
 - **States:** Circles
 - **Transitions:** Arcs



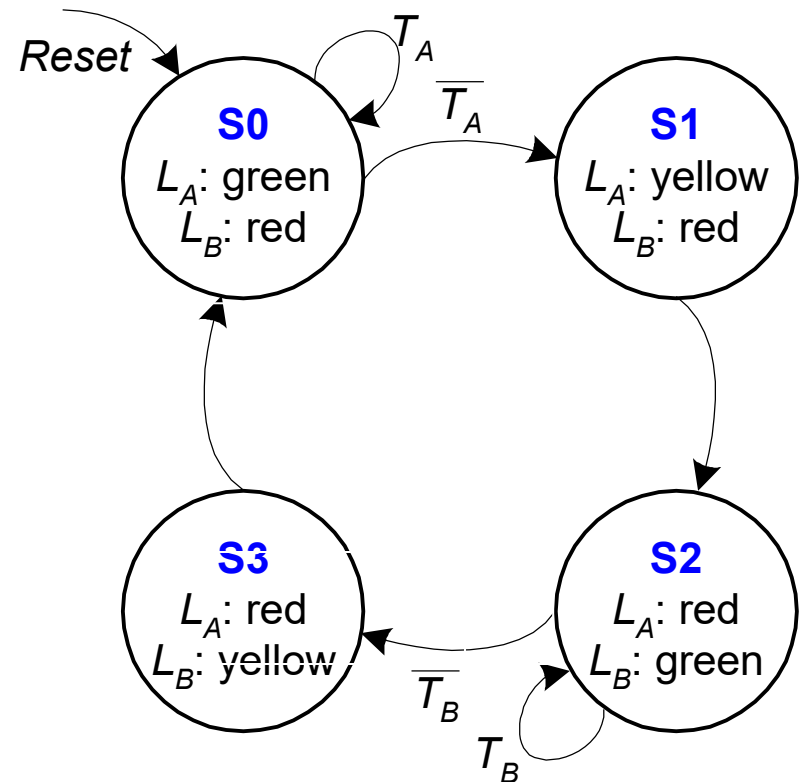
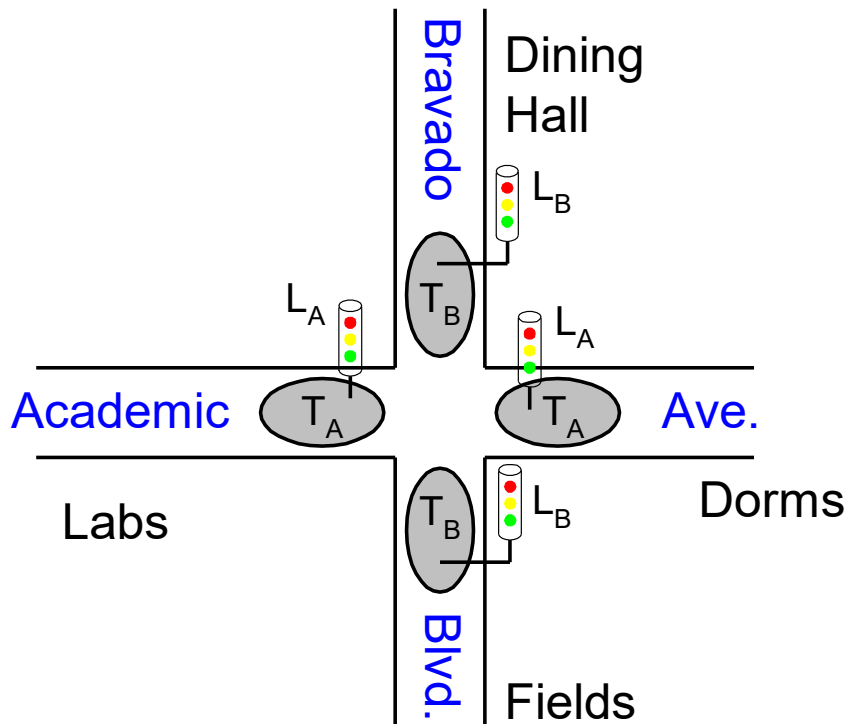
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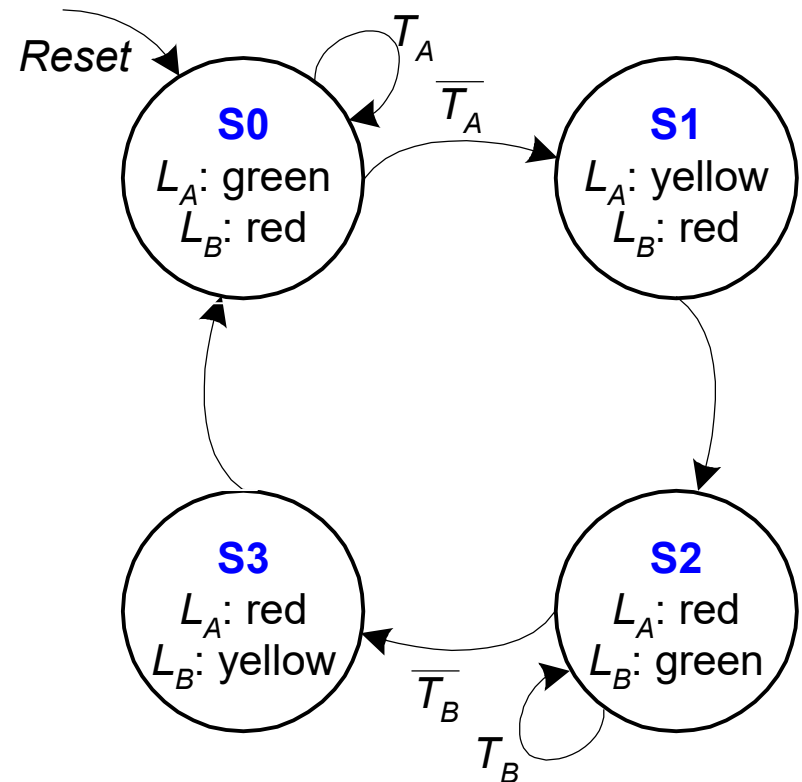
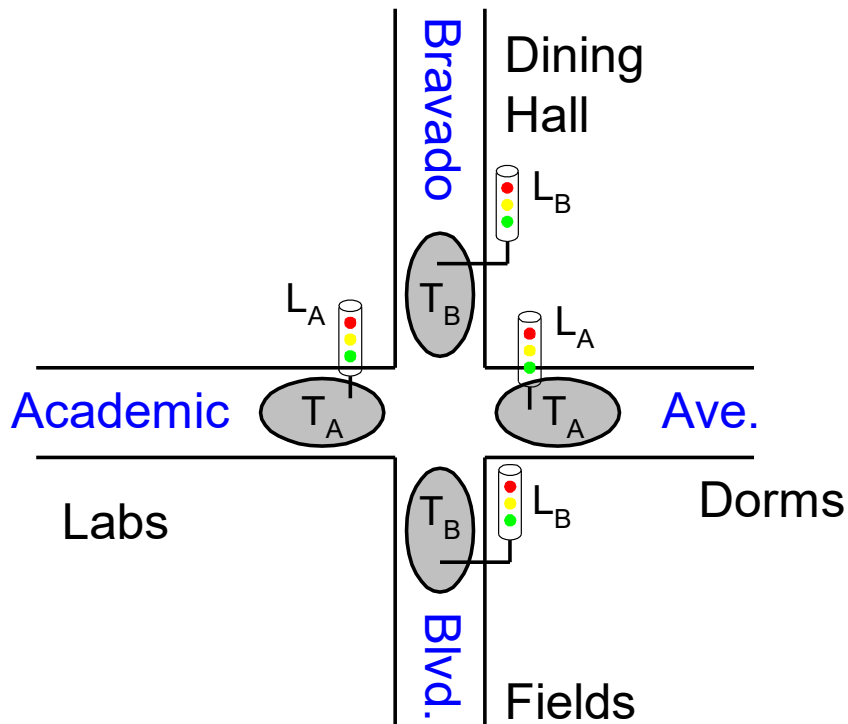
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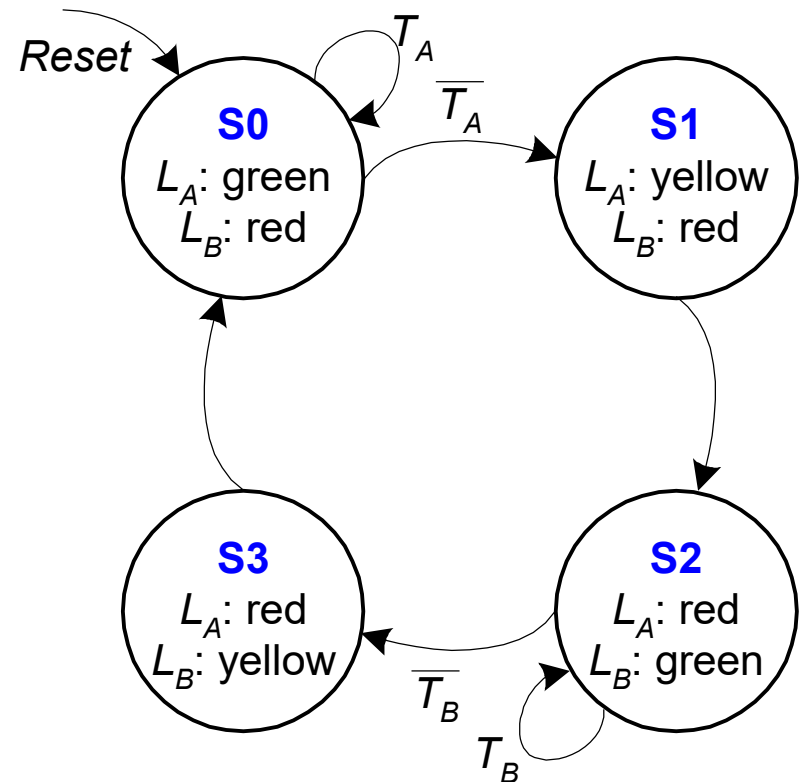
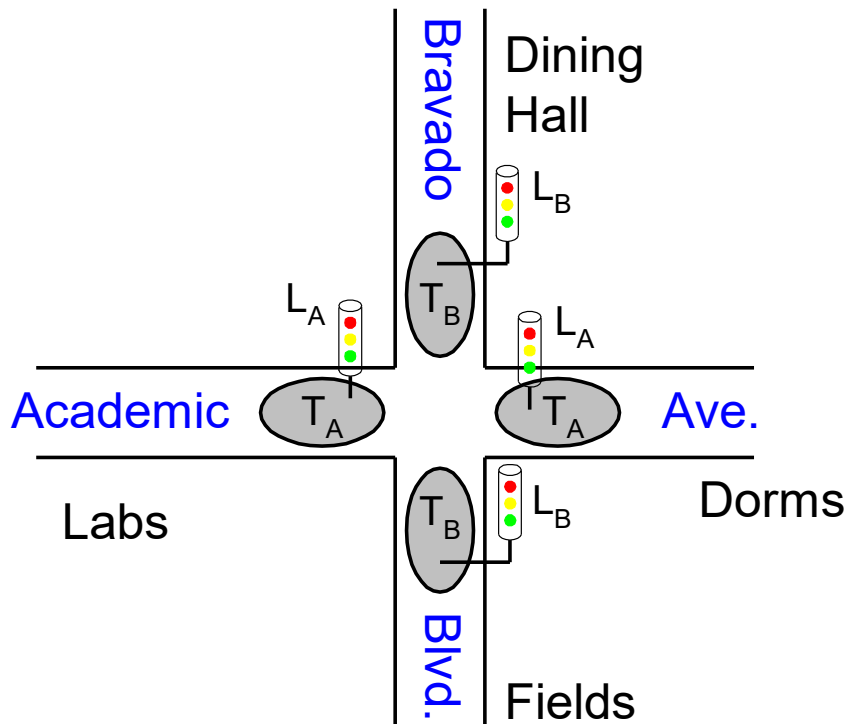
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 - **Transitions:** Arcs



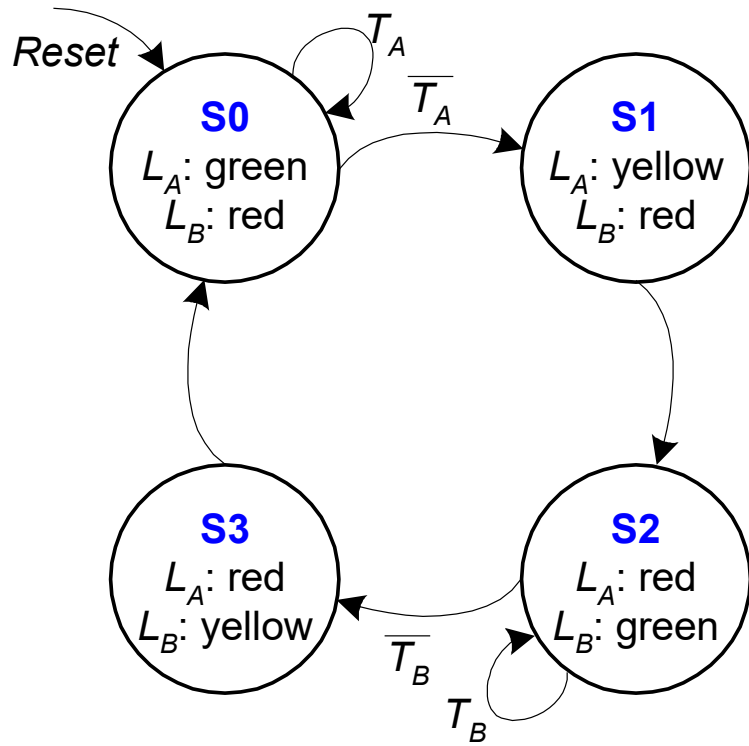
Finite State Machine Transition Diagram

- **Moore FSM:** outputs labeled in each state
 - **States:** Circles
 - **Transitions:** Arcs



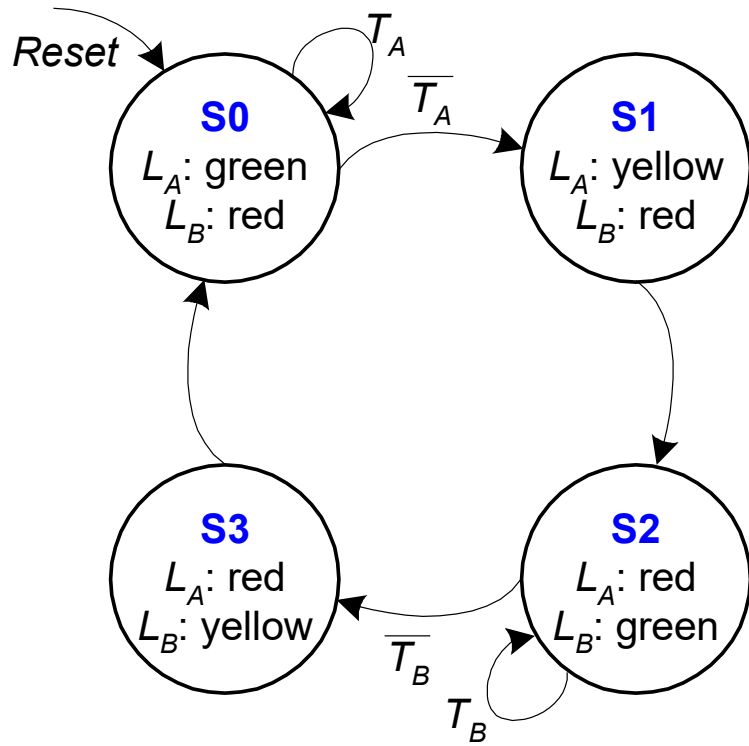
Finite State Machine: State Transition Table

FSM State Transition Table



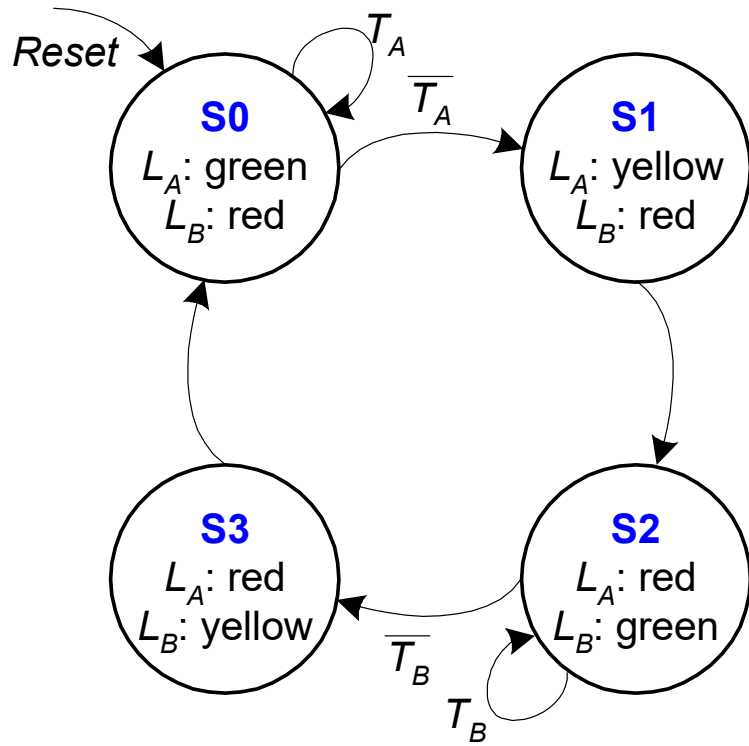
Current State	Inputs		Next State
S	T _A	T _B	S'
S0	0	X	
S0	1	X	
S1	X	X	
S2	X	0	
S2	X	1	
S3	X	X	

FSM State Transition Table



Current State	Inputs		Next State
S	T_A	T_B	S'
S0	0	X	S1
S0	1	X	S0
S1	X	X	S2
S2	X	0	S3
S2	X	1	S2
S3	X	X	S0

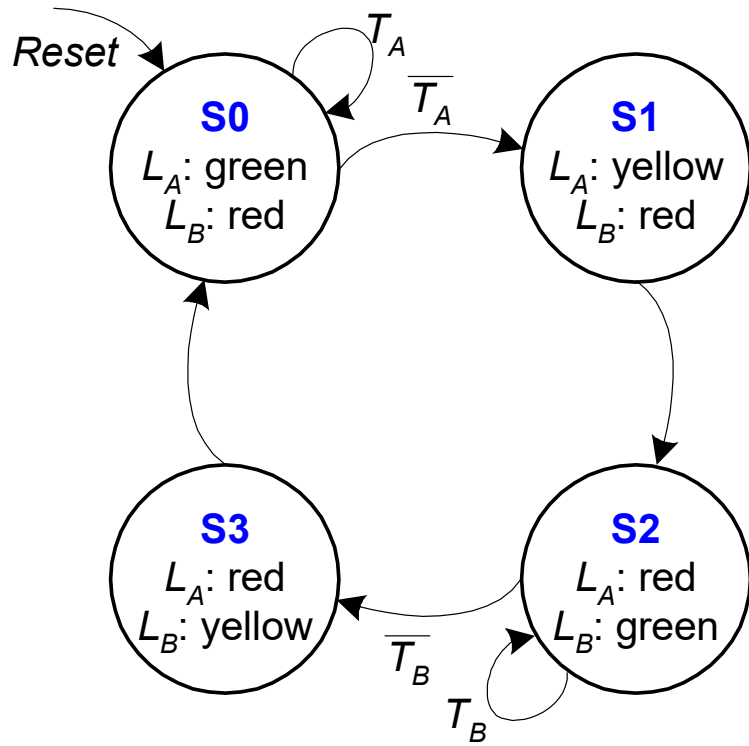
FSM State Transition Table



Current State	Inputs		Next State
S	T_A	T_B	S'
S0	0	X	S1
S0	1	X	S0
S1	X	X	S2
S2	X	0	S3
S2	X	1	S2
S3	X	X	S0

State	Encoding
S0	00
S1	01
S2	10
S3	11

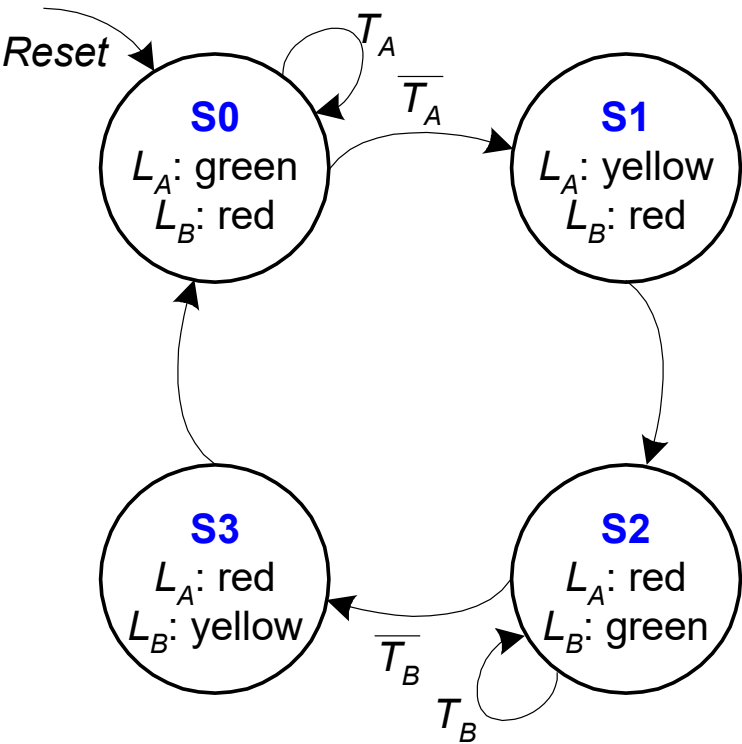
FSM State Transition Table



Current State		Inputs		Next State	
S_1	S_0	T_A	T_B	S'_1	S'_0
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

State	Encoding
S0	00
S1	01
S2	10
S3	11

FSM State Transition Table

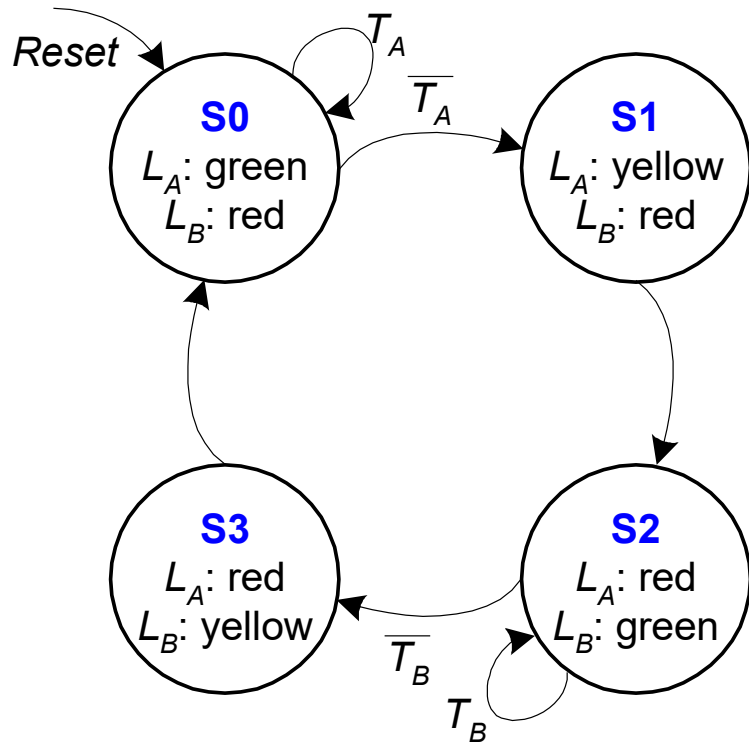


Current State		Inputs		Next State	
S_1	S_0	T_A	T_B	S'_1	S'_0
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

State	Encoding
S0	00
S1	01
S2	10
S3	11

$S'_1 = ?$

FSM State Transition Table

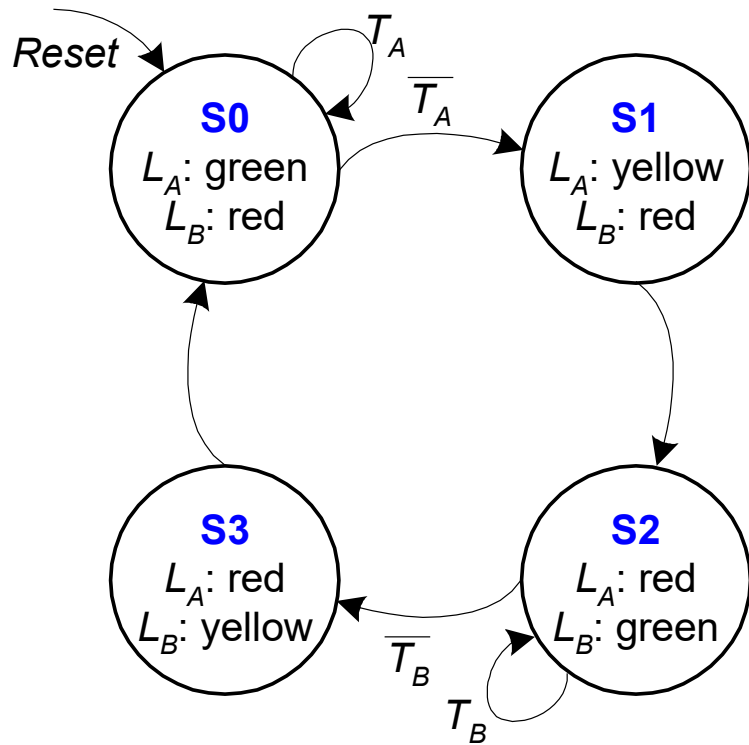


Current State		Inputs		Next State	
S_1	S_0	T_A	T_B	S'_1	S'_0
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

State	Encoding
S0	00
S1	01
S2	10
S3	11

$$S'_1 = (\overline{S_1} \cdot S_0) + (S_1 \cdot \overline{S_0} \cdot \overline{T_B}) + (S_1 \cdot \overline{S_0} \cdot T_B)$$

FSM State Transition Table



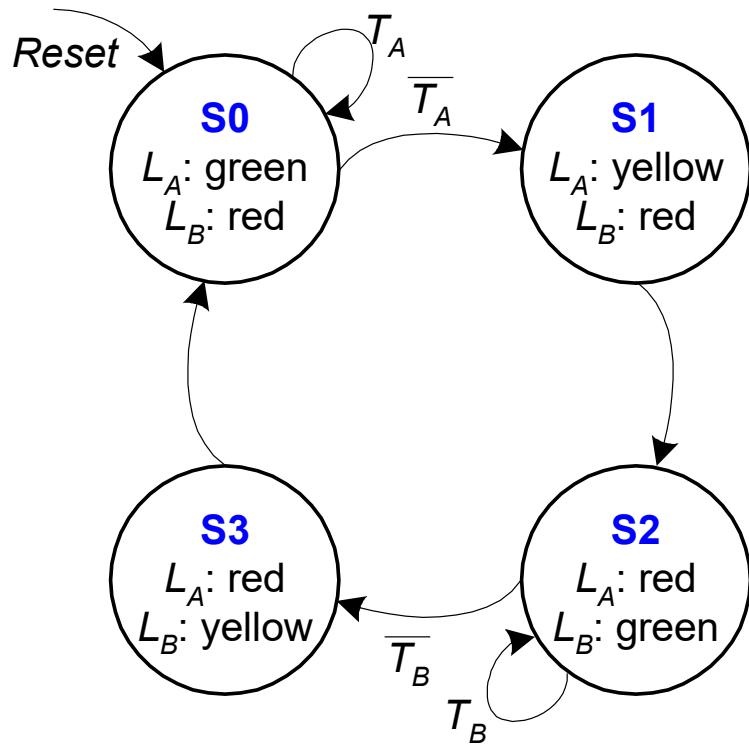
Current State		Inputs		Next State	
S_1	S_0	T_A	T_B	S'_1	S'_0
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

State	Encoding
S0	00
S1	01
S2	10
S3	11

$$S'_1 = (\overline{S_1} \cdot S_0) + (S_1 \cdot \overline{S_0} \cdot \overline{T_B}) + (S_1 \cdot \overline{S_0} \cdot T_B)$$

$$S'_0 = ?$$

FSM State Transition Table



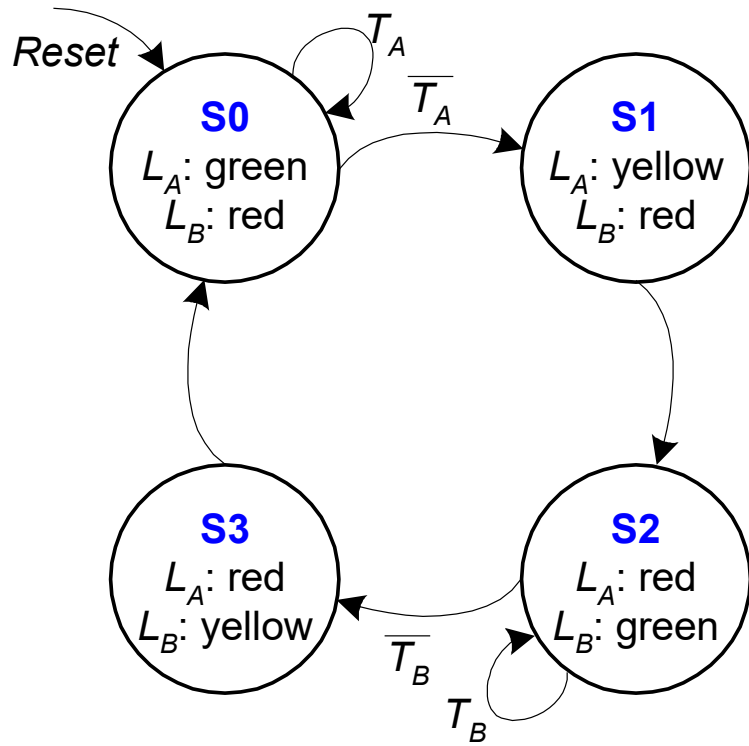
Current State		Inputs		Next State	
S_1	S_0	T_A	T_B	S'_1	S'_0
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

State	Encoding
S0	00
S1	01
S2	10
S3	11

$$S'_1 = (\overline{S_1} \cdot S_0) + (S_1 \cdot \overline{S_0} \cdot \overline{T_B}) + (S_1 \cdot \overline{S_0} \cdot T_B)$$

$$S'_0 = (\overline{S_1} \cdot \overline{S_0} \cdot \overline{T_A}) + (S_1 \cdot \overline{S_0} \cdot \overline{T_B})$$

FSM State Transition Table



Current State		Inputs		Next State	
S ₁	S ₀	T _A	T _B	S' ₁	S' ₀
0	0	0	X	0	1
0	0	1	X	0	0
0	1	X	X	1	0
1	0	X	0	1	1
1	0	X	1	1	0
1	1	X	X	0	0

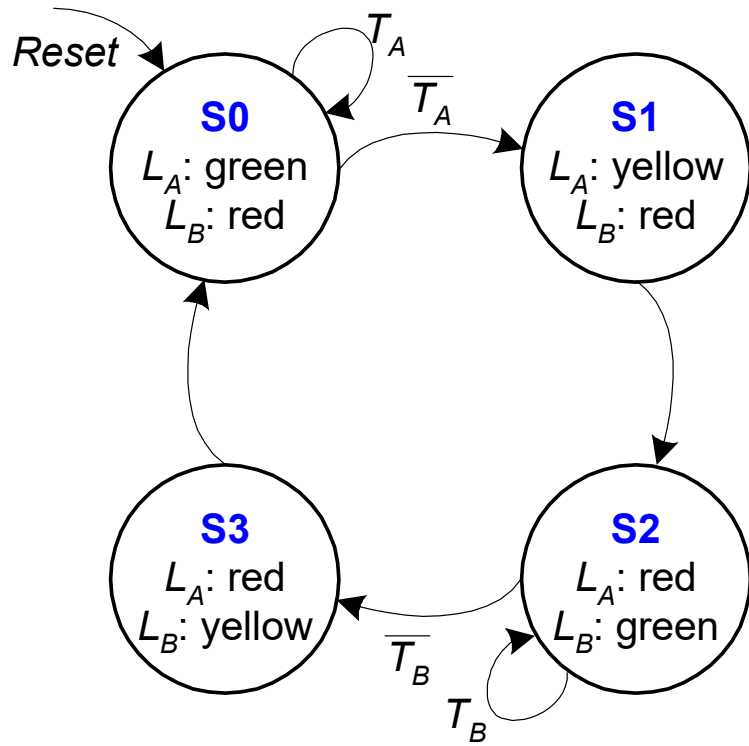
State	Encoding
S0	00
S1	01
S2	10
S3	11

$$S'_1 = S_1 \text{ xor } S_0 \quad \textbf{(Simplified)}$$

$$S'_0 = (\overline{S}_1 \cdot \overline{S}_0 \cdot \overline{T}_A) + (S_1 \cdot \overline{S}_0 \cdot \overline{T}_B)$$

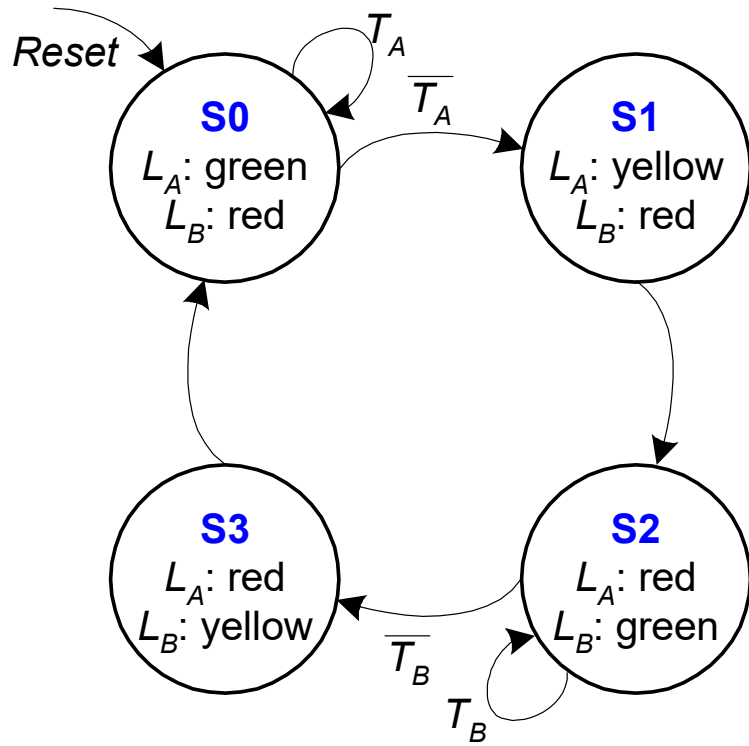
Finite State Machine: Output Table

FSM Output Table



Current State		Outputs	
S_1	S_0	L_A	L_B
0	0	green	red
0	1	yellow	red
1	0	red	green
1	1	red	yellow

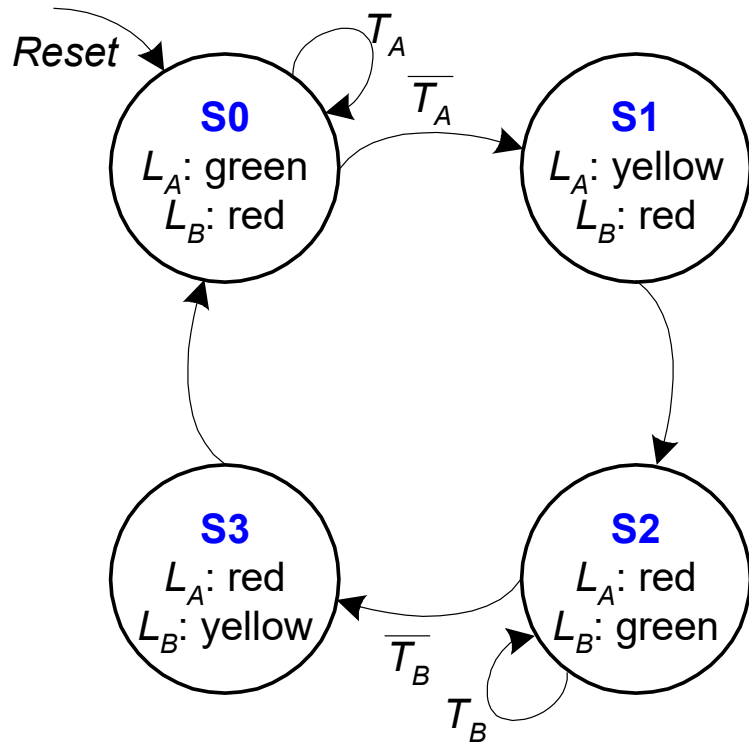
FSM Output Table



Current State		Outputs	
S_1	S_0	L_A	L_B
0	0	green	red
0	1	yellow	red
1	0	red	green
1	1	red	yellow

Output	Encoding
green	00
yellow	01
red	10

FSM Output Table

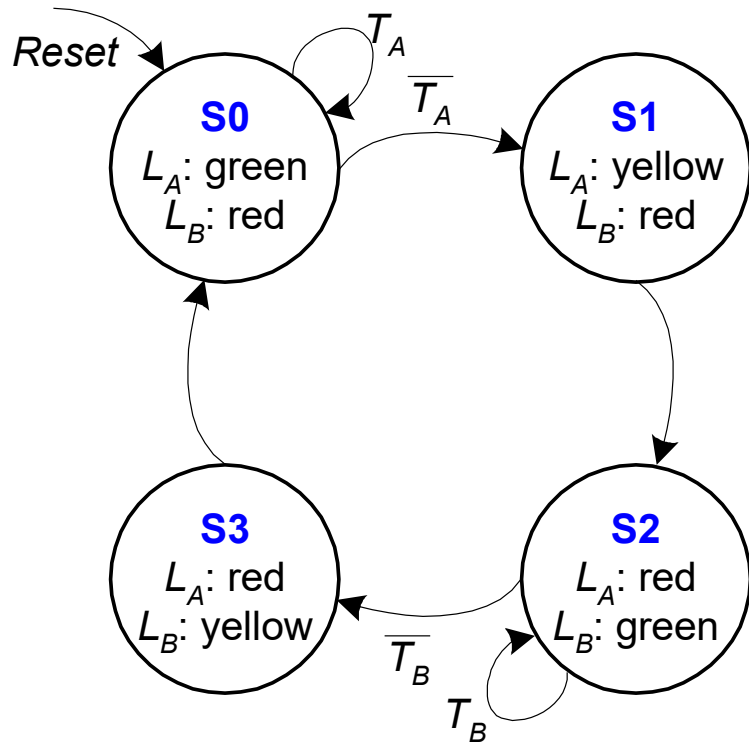


$$L_{A1} = S_1$$

Current State		Outputs			
S_1	S_0	L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	0	1	0
0	1	0	1	1	0
1	0	1	0	0	0
1	1	1	0	0	1

Output	Encoding
green	00
yellow	01
red	10

FSM Output Table



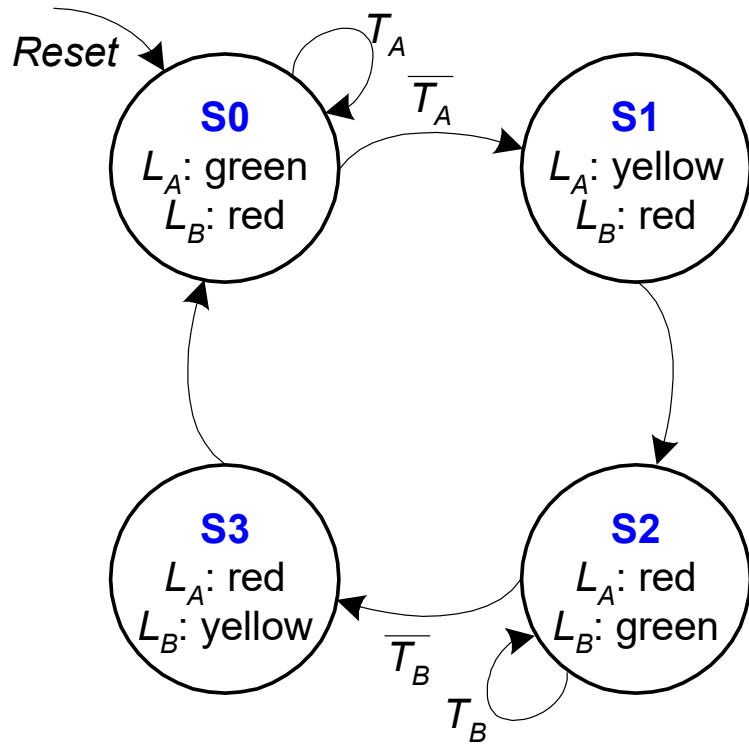
$$L_{A1} = S_1$$

$$L_{A0} = \overline{S_1} \cdot S_0$$

Current State		Outputs			
S_1	S_0	L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	0	1	0
0	1	0	1	1	0
1	0	1	0	0	0
1	1	1	0	0	1

Output	Encoding
green	00
yellow	01
red	10

FSM Output Table

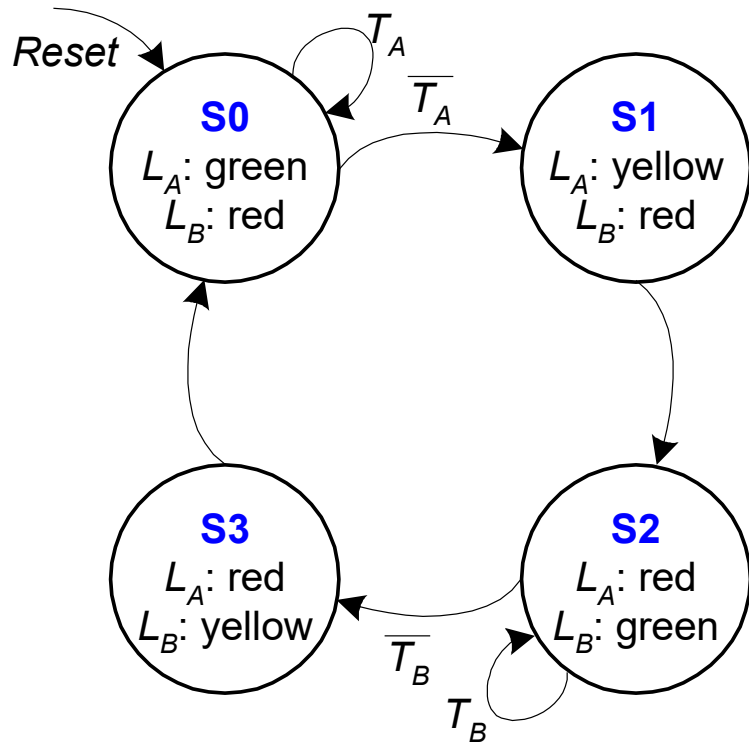


$$\begin{aligned}
 L_{A1} &= S_1 \\
 L_{A0} &= \overline{S_1} \cdot S_0 \\
 L_{B1} &= \overline{S_1}
 \end{aligned}$$

Current State		Outputs			
S_1	S_0	L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	0	1	0
0	1	0	1	1	0
1	0	1	0	0	0
1	1	1	0	0	1

Output	Encoding
green	00
yellow	01
red	10

FSM Output Table



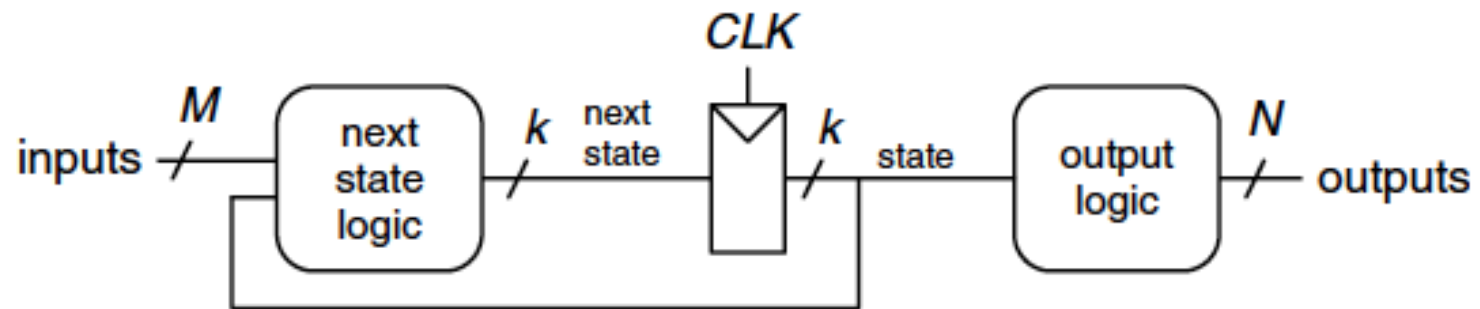
Current State		Outputs			
S_1	S_0	L_{A1}	L_{A0}	L_{B1}	L_{B0}
0	0	0	0	1	0
0	1	0	1	1	0
1	0	1	0	0	0
1	1	1	0	0	1

$$\begin{aligned}
 L_{A1} &= S_1 \\
 L_{A0} &= \overline{S_1} \cdot S_0 \\
 L_{B1} &= \overline{S_1} \\
 L_{B0} &= S_1 \cdot S_0
 \end{aligned}$$

Output	Encoding
green	00
yellow	01
red	10

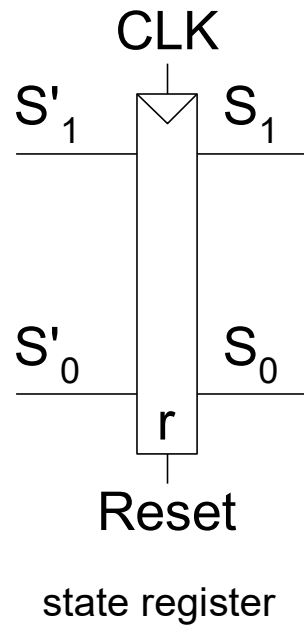
Finite State Machine: Schematic

FSM Schematic: State Register

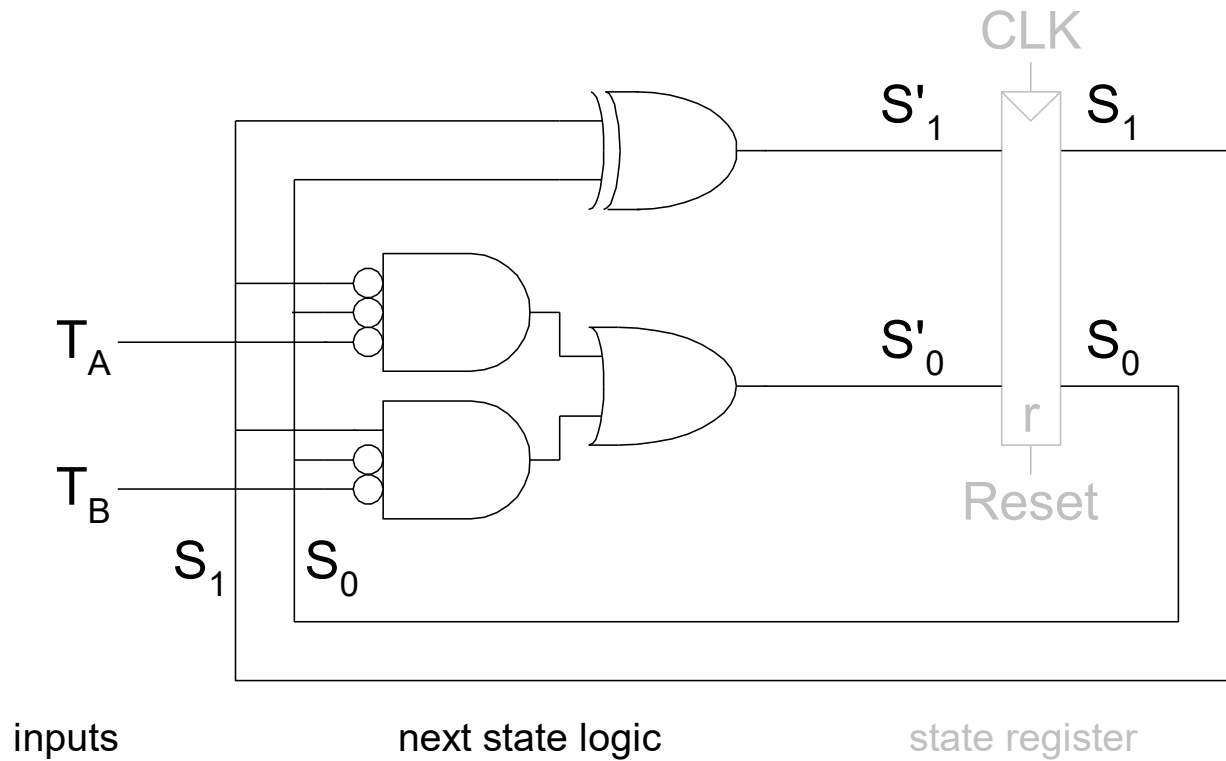


(a)

FSM Schematic: State Register



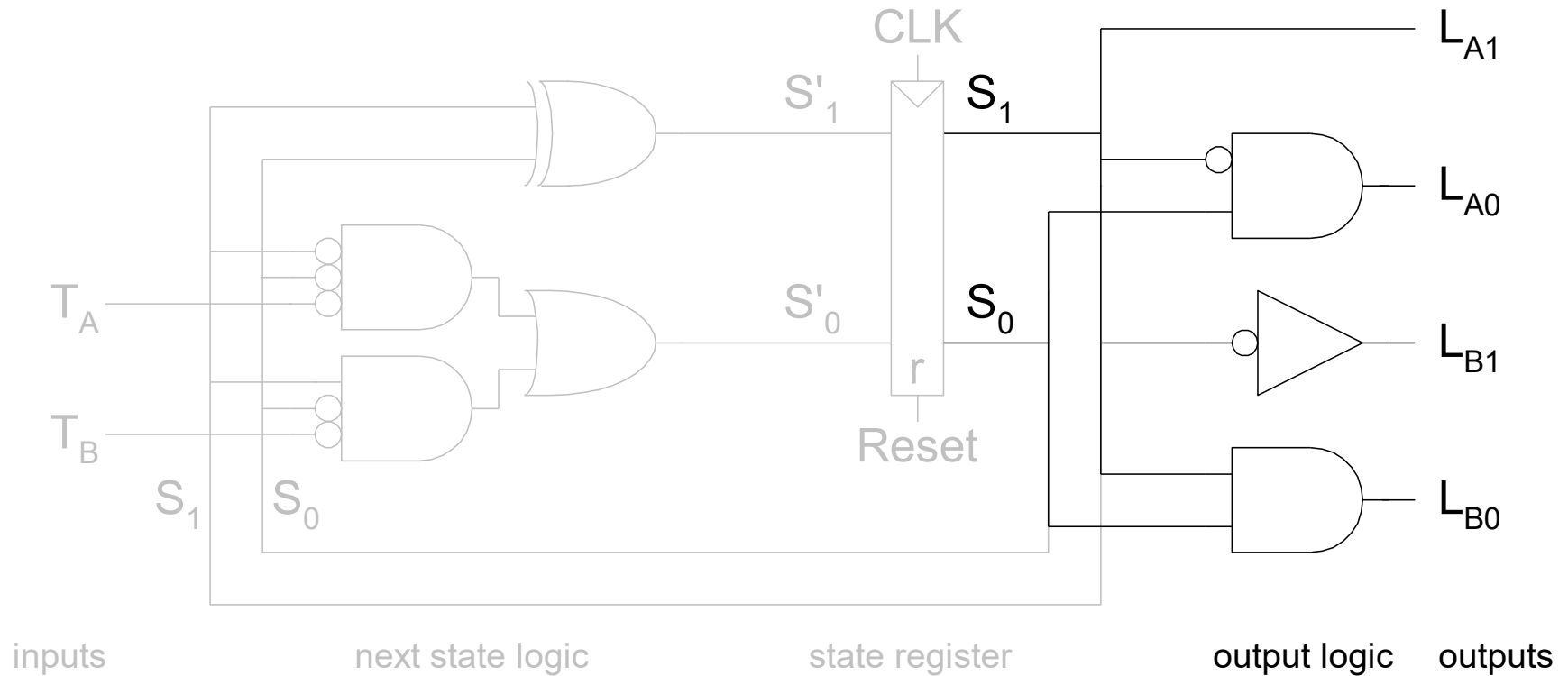
FSM Schematic: Next State Logic



$$S'_1 = S_1 \text{ xor } S_0$$

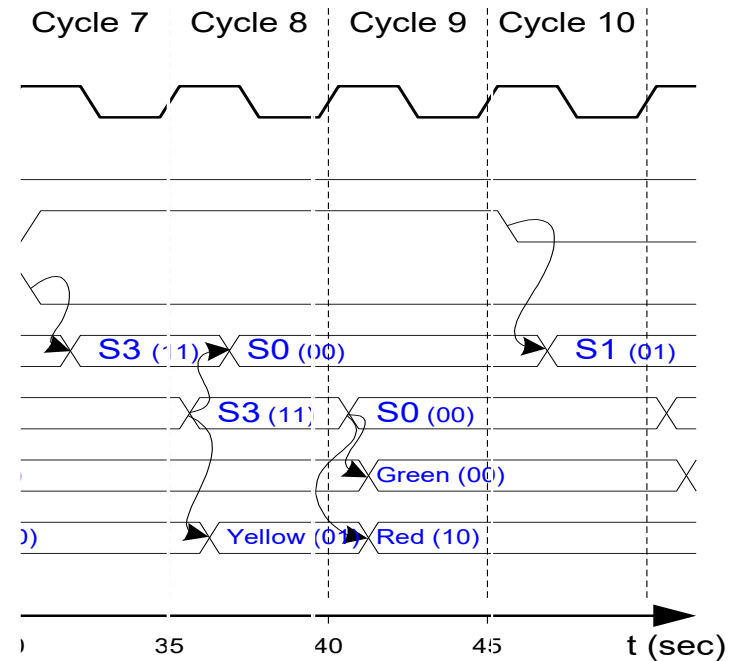
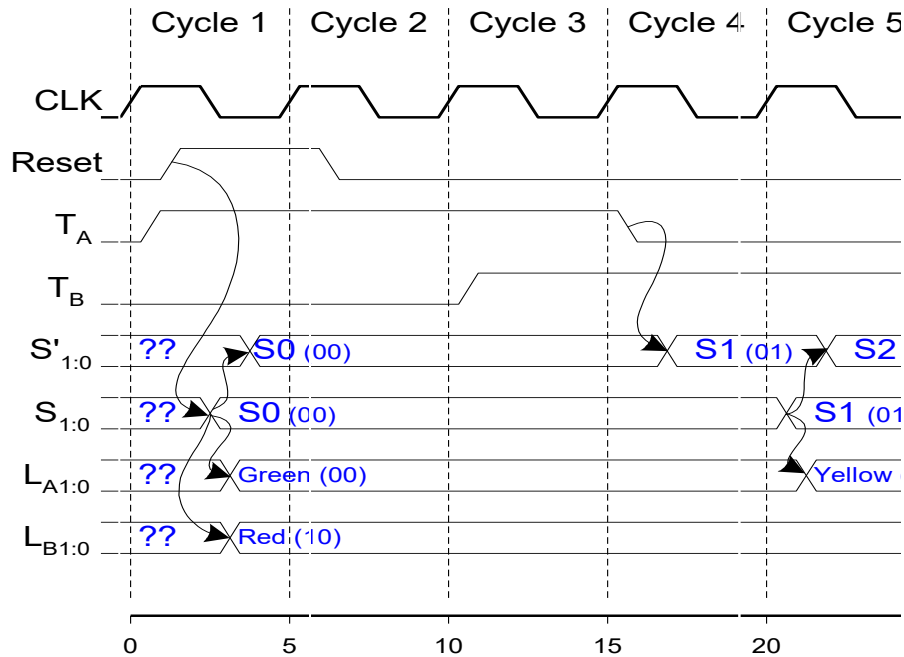
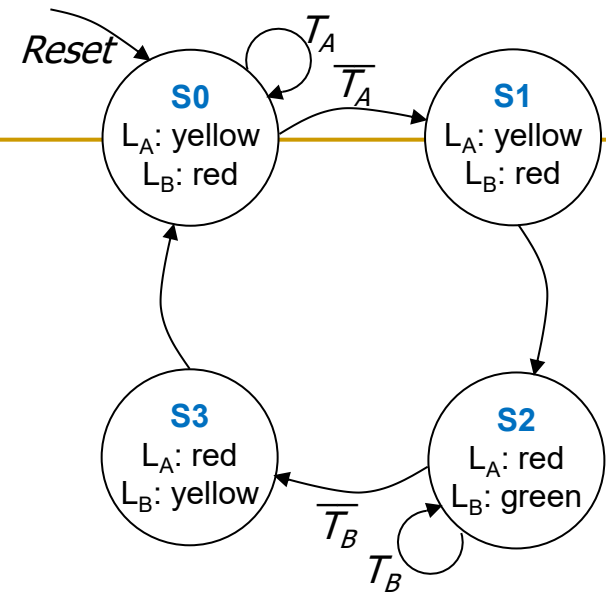
$$S'_0 = (\overline{S_1} \cdot \overline{S_0} \cdot \overline{T_A}) + (S_1 \cdot \overline{S_0} \cdot \overline{T_B})$$

FSM Schematic: Output Logic

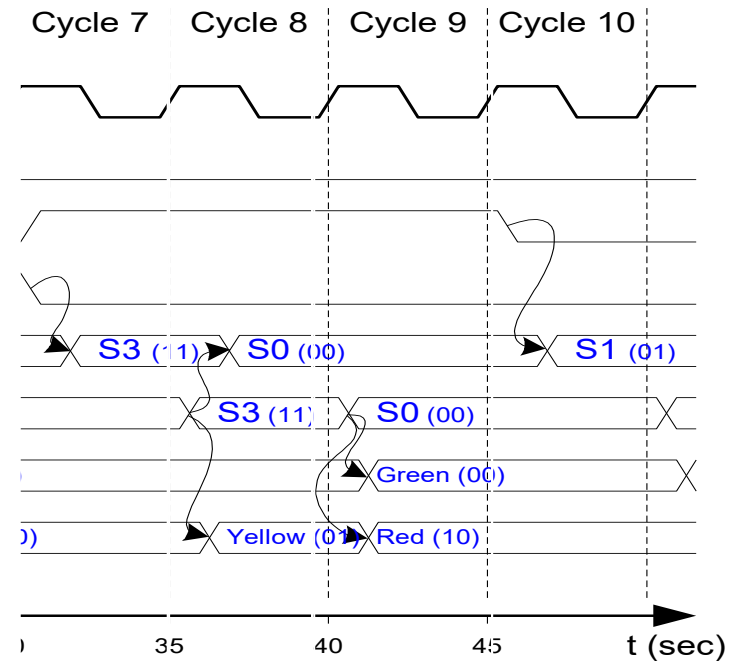
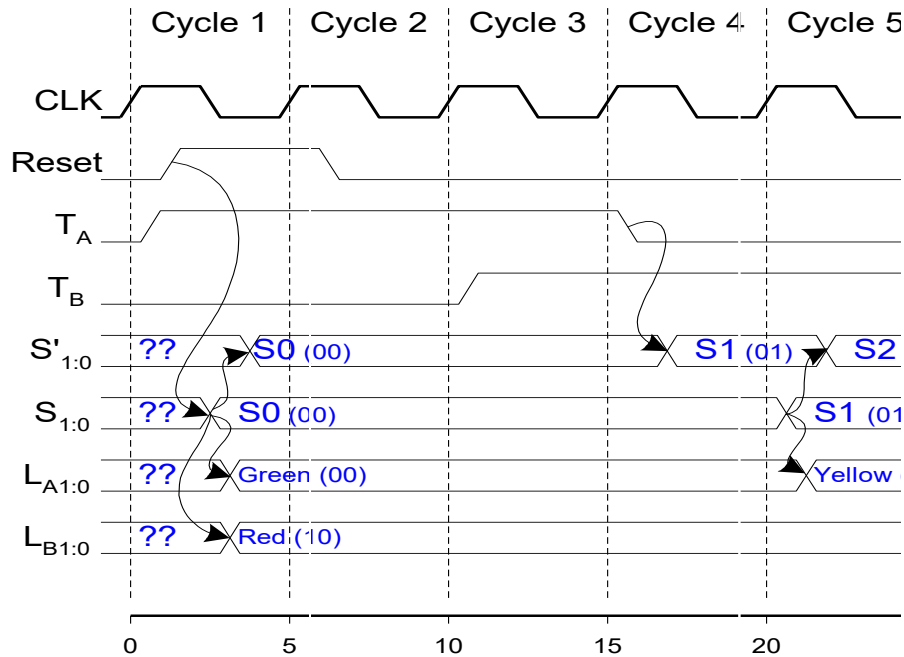
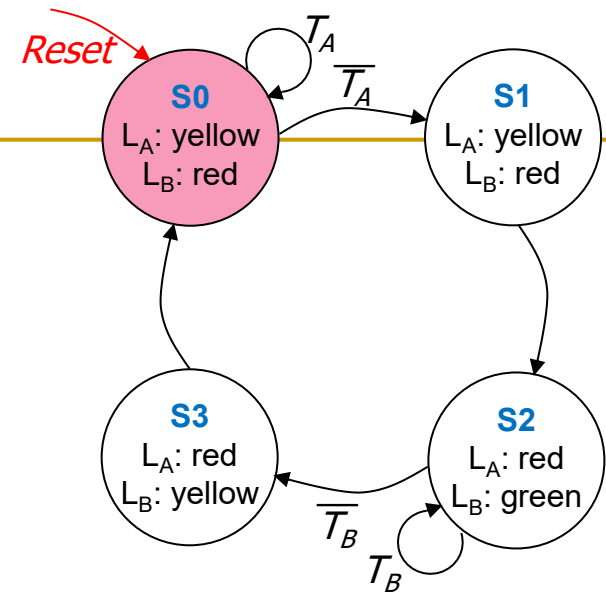


$$\begin{aligned} L_{A1} &= S_1 \\ L_{A0} &= \overline{S_1} \cdot S_0 \\ L_{B1} &= \overline{S_1} \\ L_{B0} &= S_1 \cdot S_0 \end{aligned}$$

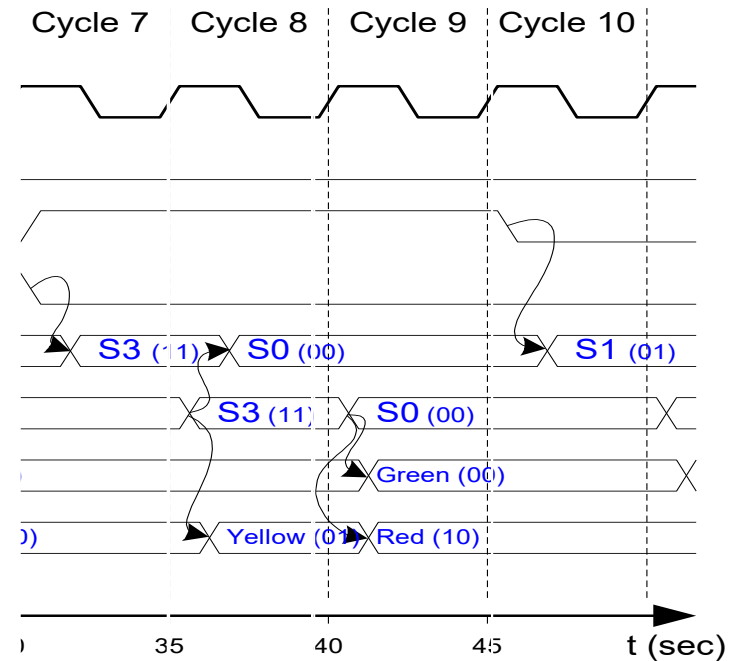
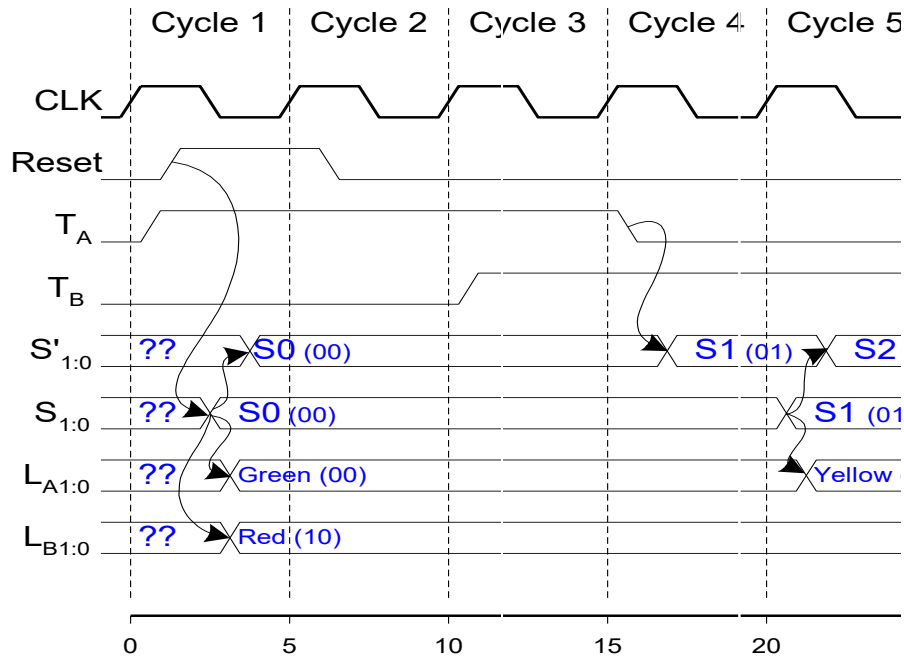
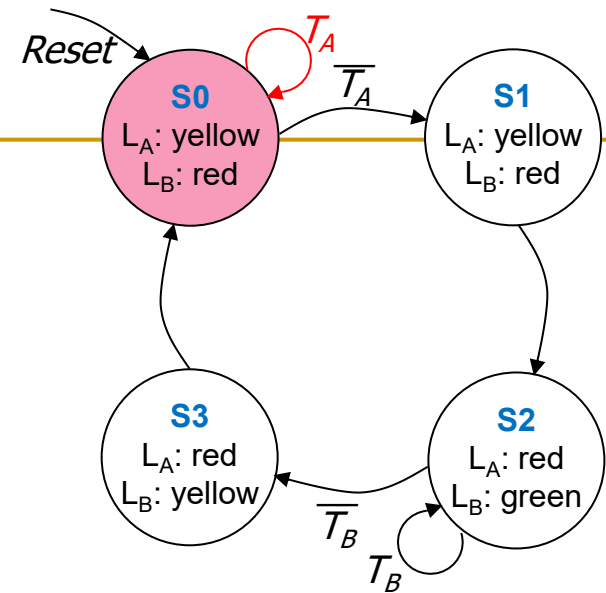
FSM Timing Diagram



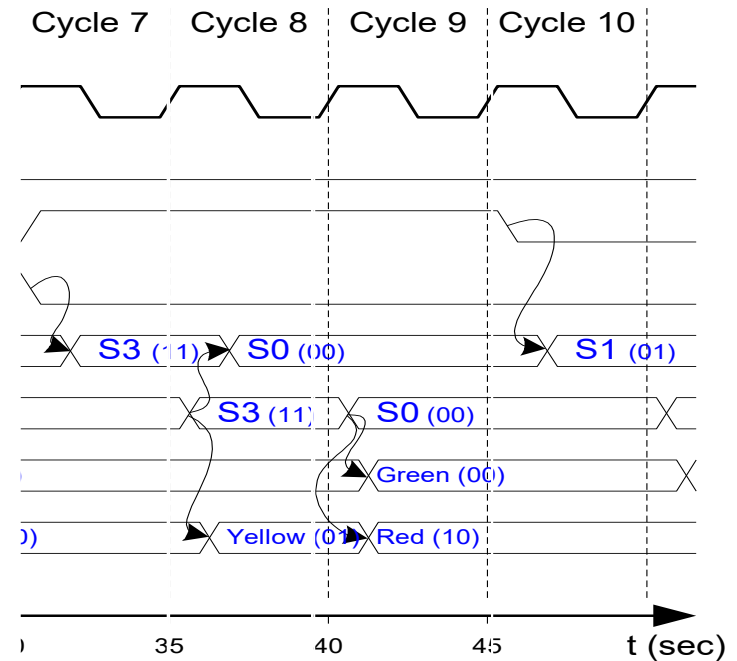
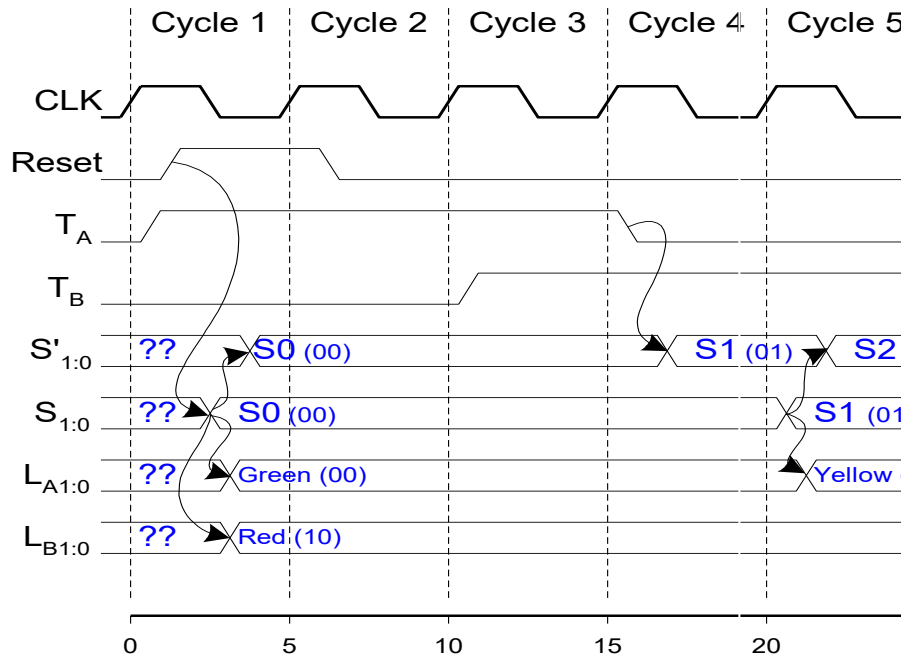
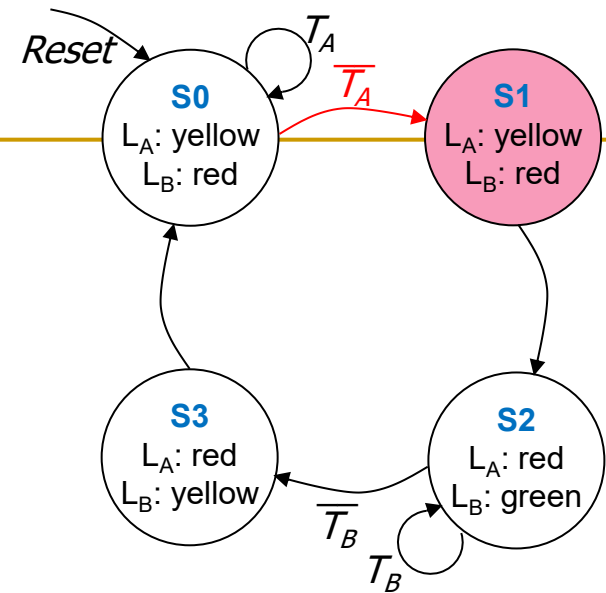
FSM Timing Diagram



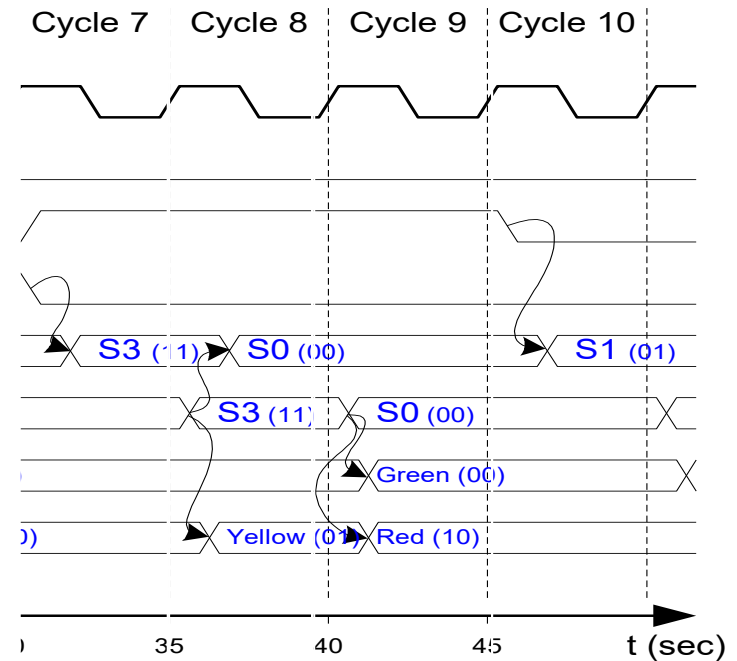
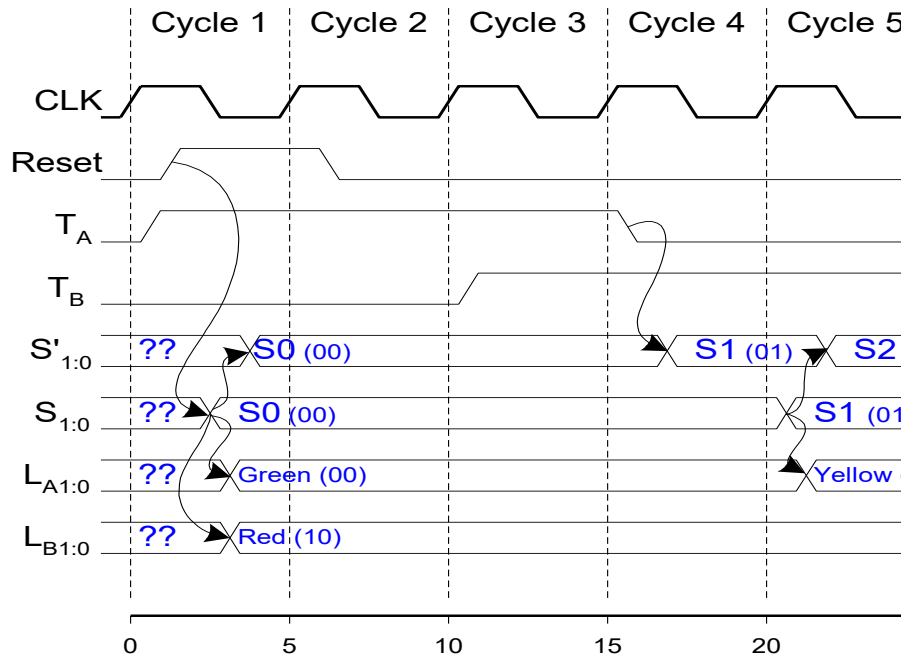
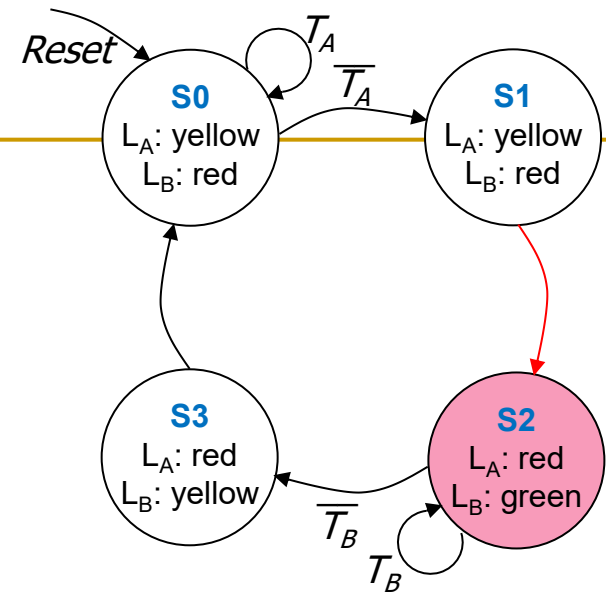
FSM Timing Diagram



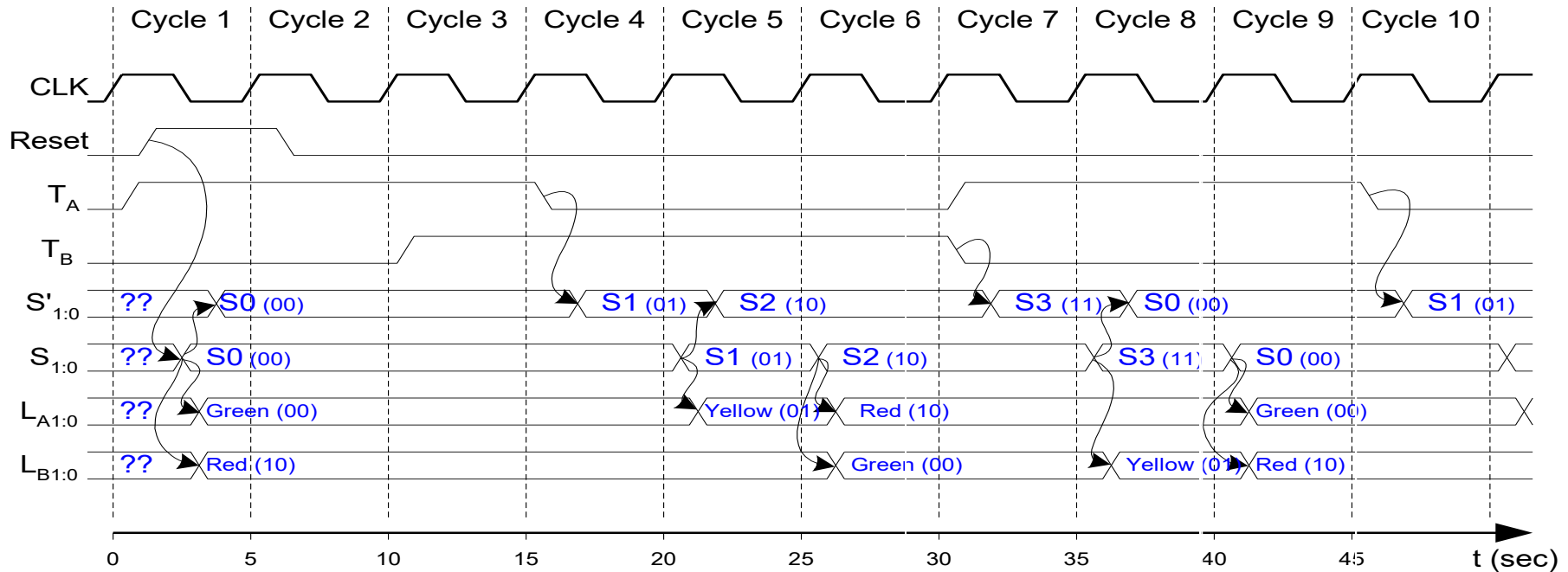
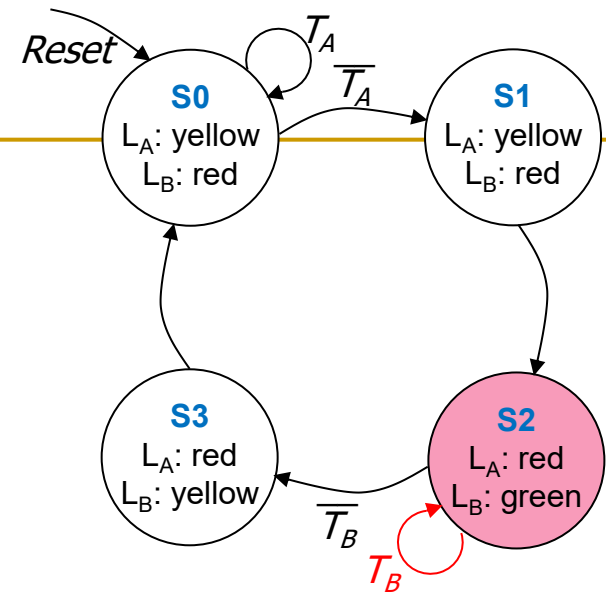
FSM Timing Diagram



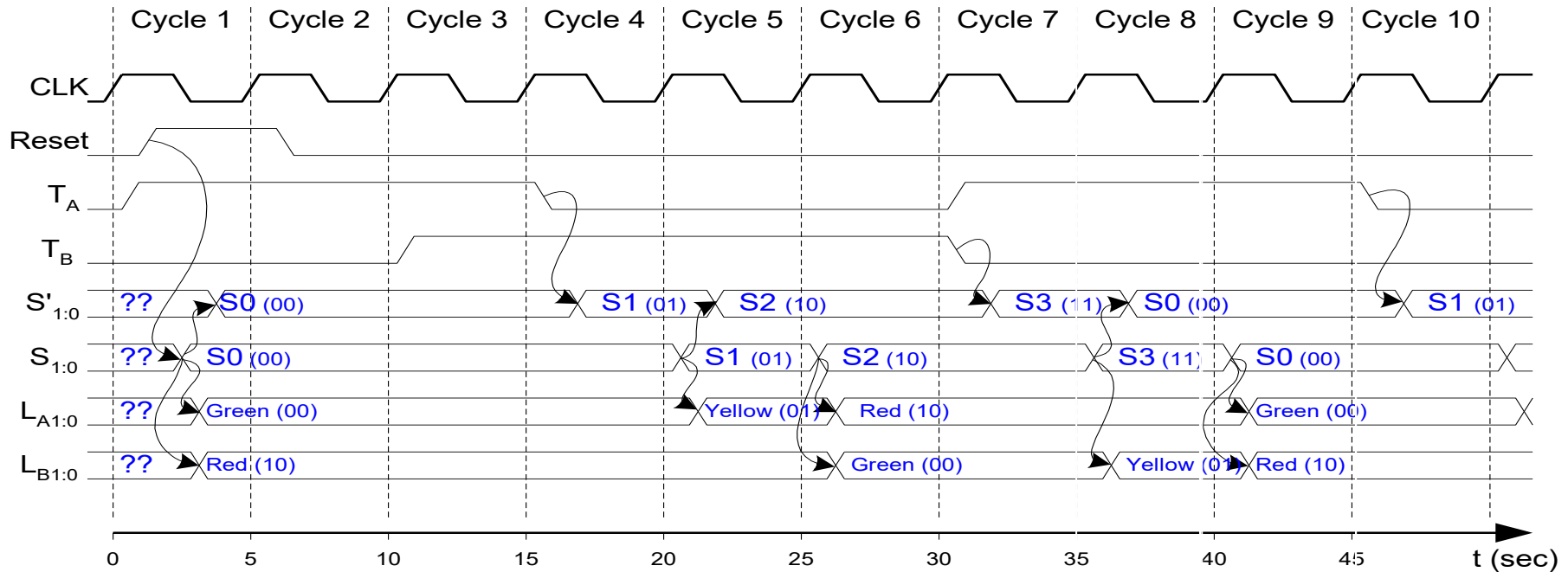
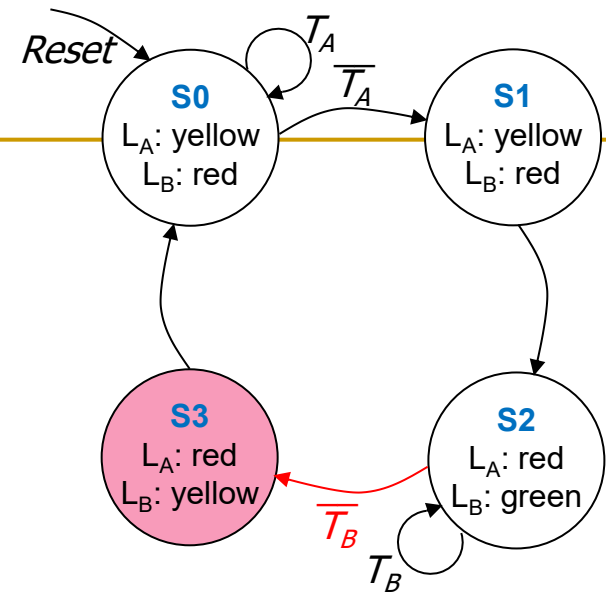
FSM Timing Diagram



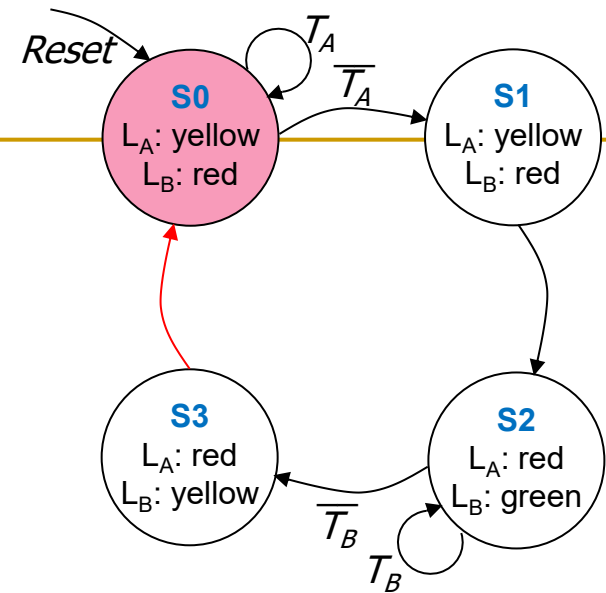
FSM Timing Diagram



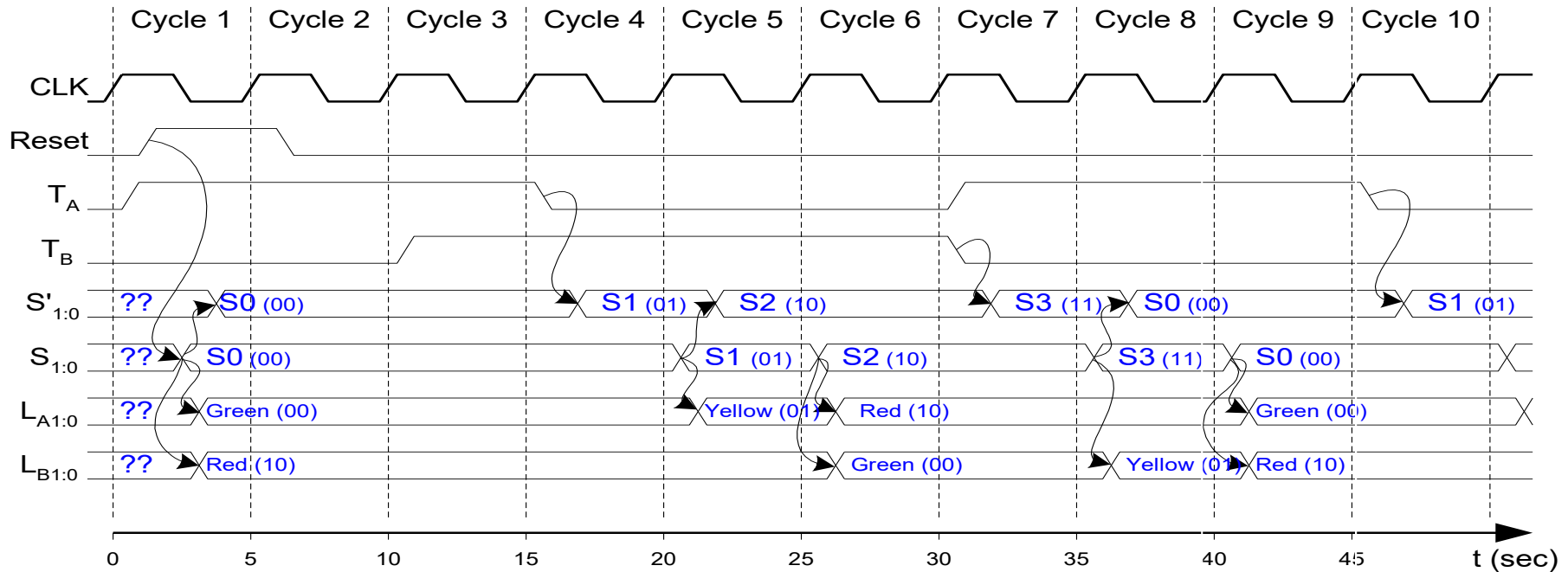
FSM Timing Diagram



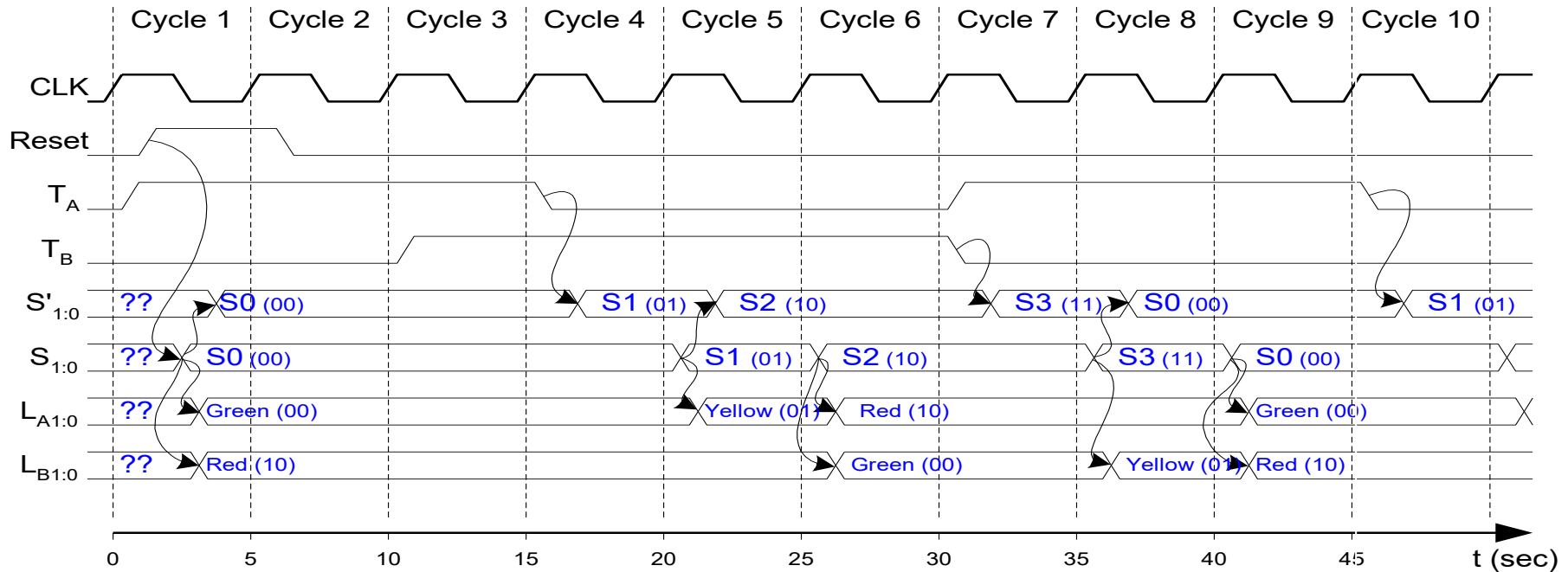
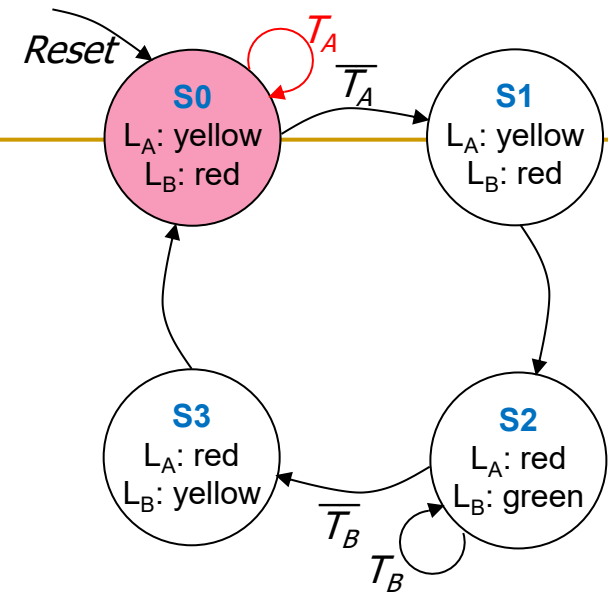
FSM Timing Diagram



This is from H&H Section 3.4.1

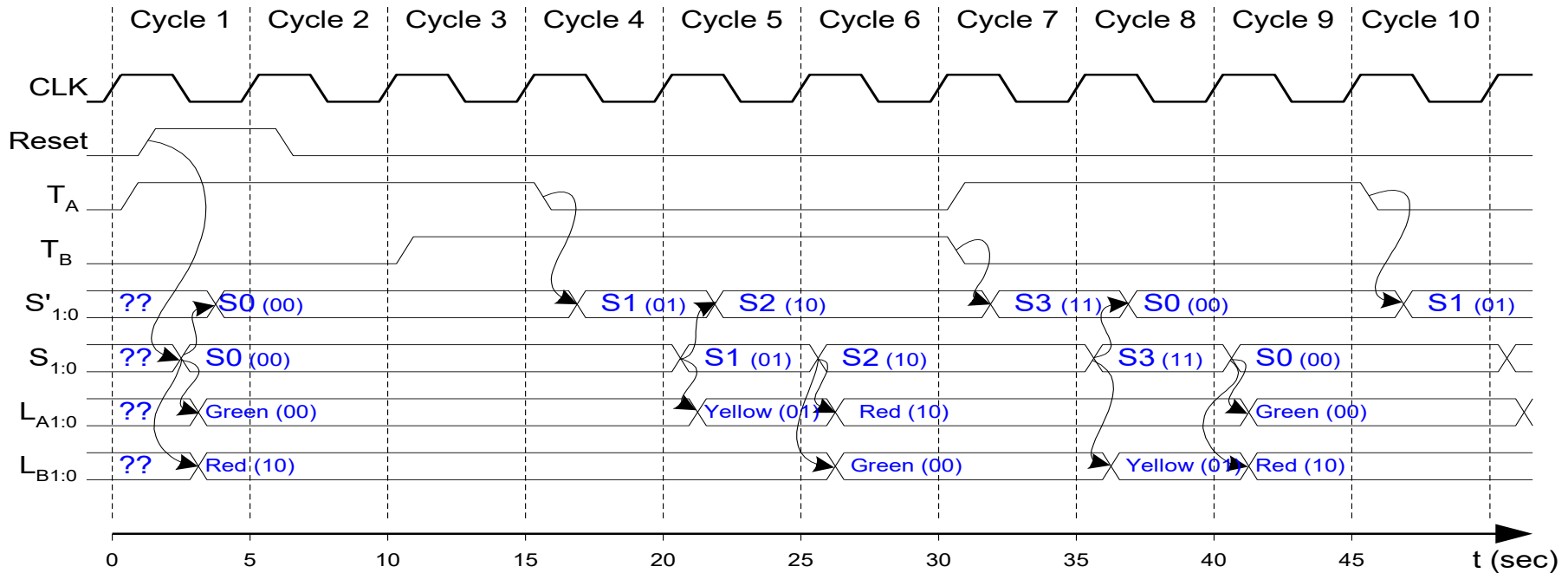
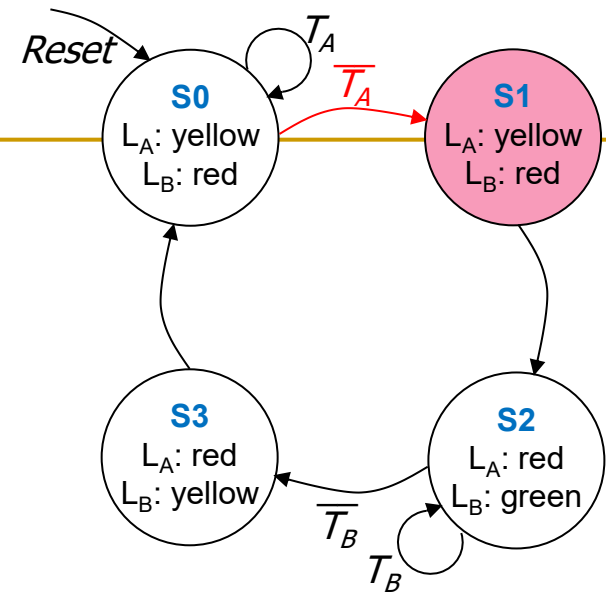


FSM Timing Diagram



FSM Timing Diagram

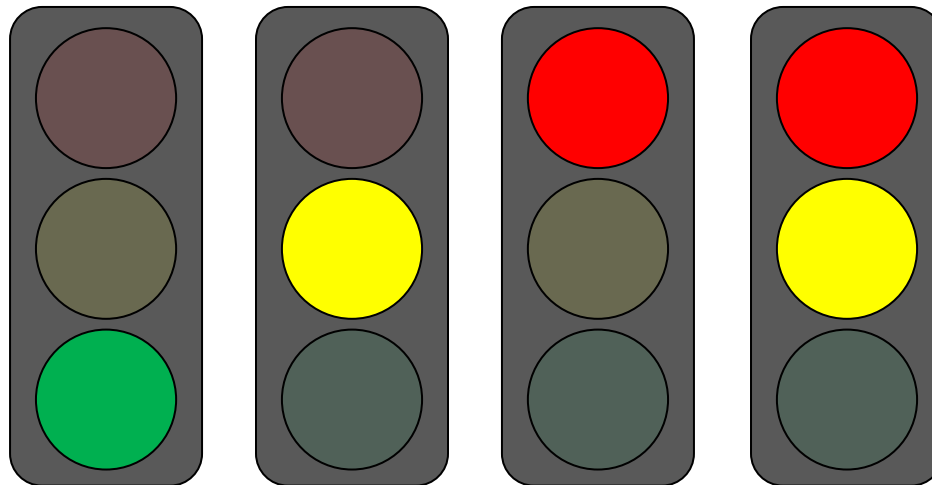
See H&H Chapter 3.4



Finite State Machine: State Encoding

FSM State Encoding

- How do we encode the state bits?
 - Three common state binary encodings with different tradeoffs
 1. **Fully Encoded**
 2. **1-Hot Encoded**
 3. **Output Encoded**
- Let's see an example **Swiss** traffic light with 4 states
 - Green, Yellow, Red, Yellow+Red



FSM State Encoding (II)

1. Binary Encoding (Full Encoding):

- ❑ Use the minimum possible number of bits
 - Use $\log_2(num_states)$ bits to represent the states
- ❑ *Example state encodings: 00, 01, 10, 11*
- ❑ **Minimizes** # flip-flops, but not necessarily output logic or next state logic

2. One-Hot Encoding:

- ❑ Each bit encodes a different state
 - Uses num_states bits to represent the states
 - Exactly 1 bit is “hot” for a given state
- ❑ *Example state encodings: 0001, 0010, 0100, 1000*
- ❑ **Simplest design process** – very automatable
- ❑ **Maximizes** # flip-flops, **minimizes** next state logic

FSM State Encoding (III)

3. Output Encoding:

- ❑ Outputs are **directly accessible** in the state encoding
- ❑ For example, since we have **3 outputs** (light color), encode state with **3 bits**, where each bit represents a color
- ❑ *Example states:* 001, 010, 100, 110
 - Bit₀ encodes **green** light output,
 - Bit₁ encodes **yellow** light output
 - Bit₂ encodes **red** light output
- ❑ **Minimizes** output logic
- ❑ Only works for Moore Machines (output function of state)

FSM State Encoding (III)

3. Output Encoding:

- ❑ Outputs are **directly accessible** in the state encoding

The **designer** must **carefully** choose an encoding scheme to **optimize** the design under given constraints

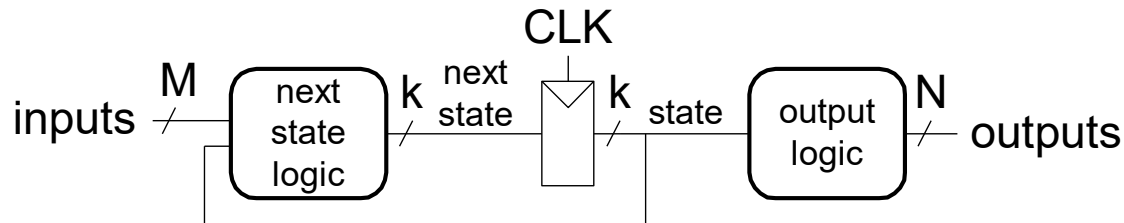
- ❑ **Minimizes** output logic
- ❑ Only works for Moore Machines (output depends only on state)

Moore vs. Mealy Machines

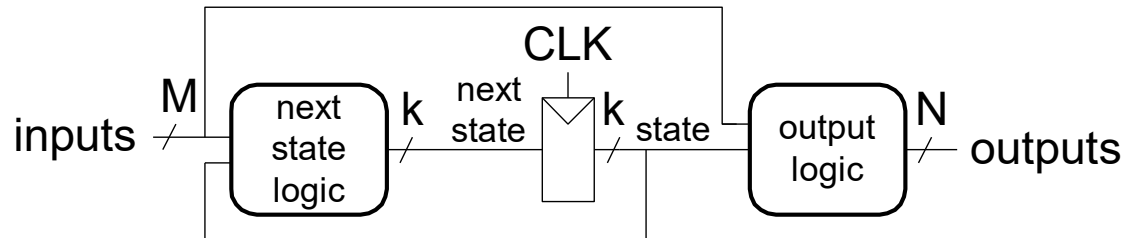
Recall: Moore vs. Mealy FSMs

- Next state is determined by the current state and the inputs
- Two types of FSMs differ in the **output logic**:
 - **Moore FSM**: outputs depend only on the current state
 - **Mealy FSM**: outputs depend on the current state and the inputs

Moore FSM



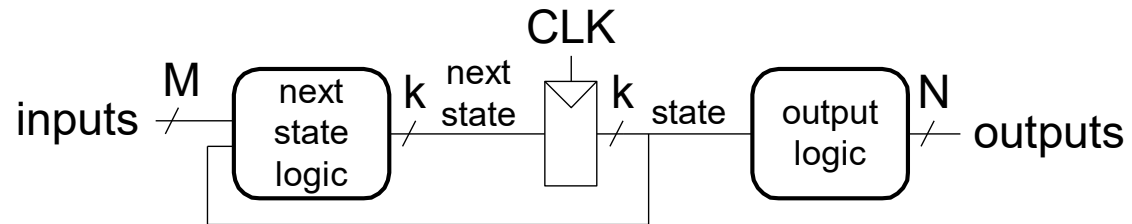
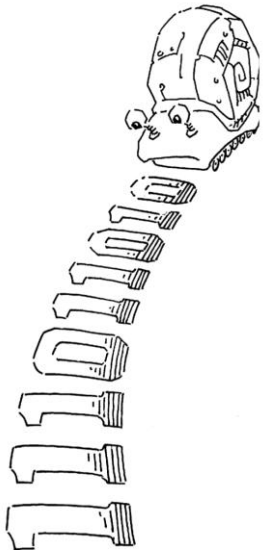
Mealy FSM



Moore vs. Mealy FSM Examples

- Alyssa P. Hacker has a snail that crawls down a paper tape with 1's and 0's on it.
- The snail smiles whenever the last four digits it has crawled over are **1101**.
- Design Moore and Mealy FSMs of the snail's brain.

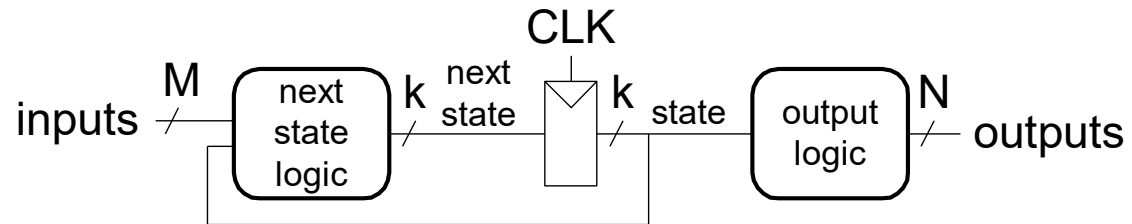
Moore FSM



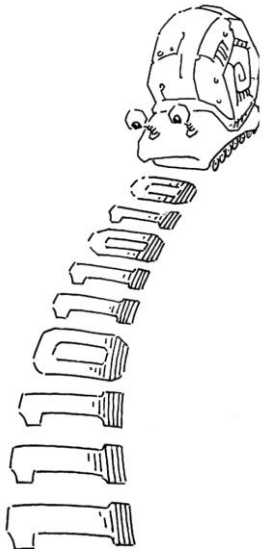
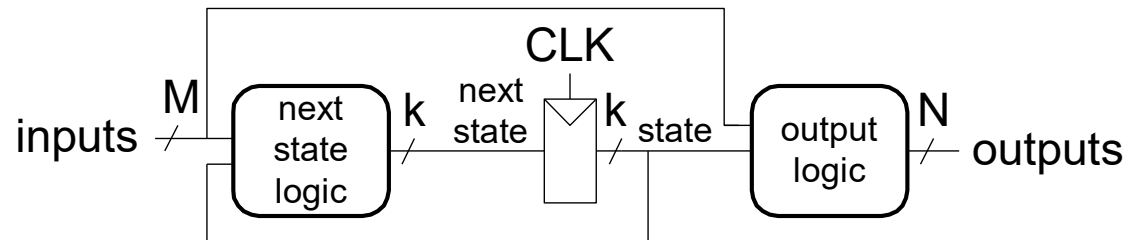
Moore vs. Mealy FSM Examples

- Alyssa P. Hacker has a snail that crawls down a paper tape with 1's and 0's on it.
- The snail smiles whenever the last four digits it has crawled over are **1101**.
- Design Moore and Mealy FSMs of the snail's brain.

Moore FSM

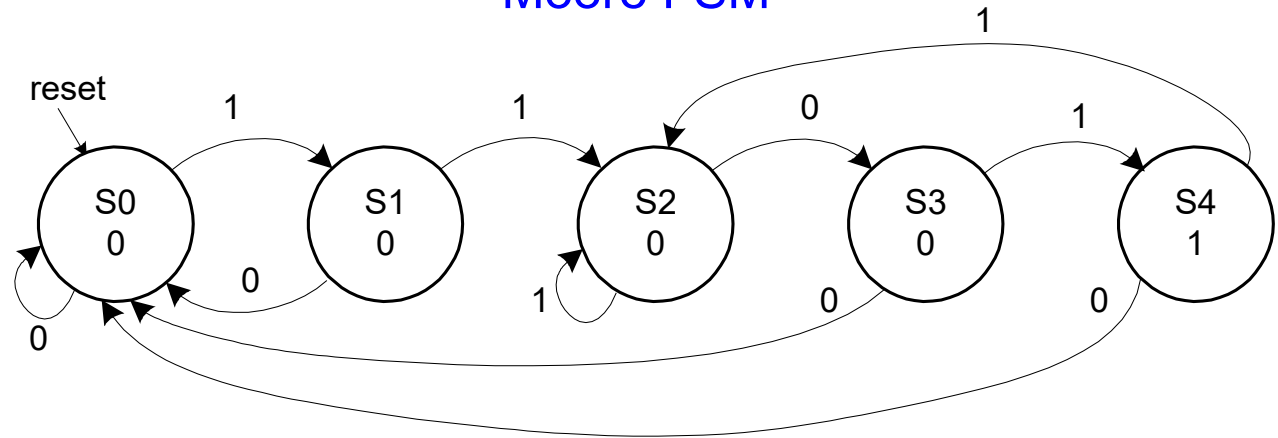


Mealy FSM



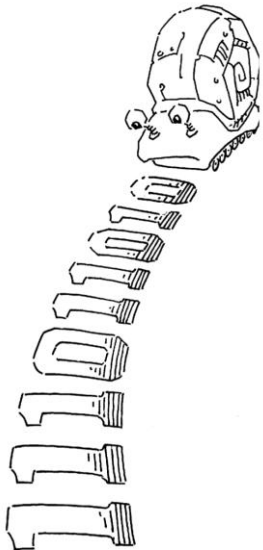
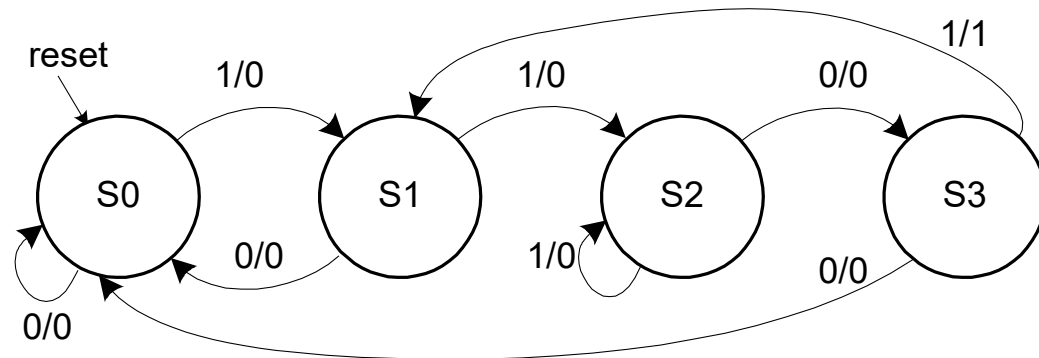
State Transition Diagrams

Moore FSM



What are the tradeoffs?

Mealy FSM



FSM Design Procedure

- **Determine** all possible states of your machine
- **Develop** a **state transition diagram**
 - Generally this is done from a textual description
 - You need to 1) determine the **inputs** and **outputs** for each **state** and 2) figure out how to get from one state to another
- **Approach**
 - Start by defining the **reset state** and what happens from it – this is typically an easy point to start from
 - Then continue to add **transitions** and **states**
 - Picking **good state names** is very important
 - Building an FSM is **like** programming (but it *is not* programming!)
 - An FSM has a sequential “control-flow” like a program with conditionals and goto’s
 - The if-then-else construct is controlled by one or more inputs
 - The outputs are controlled by the state or the inputs
 - In hardware, we typically have many concurrent FSMs

What is to Come: LC-3 Processor

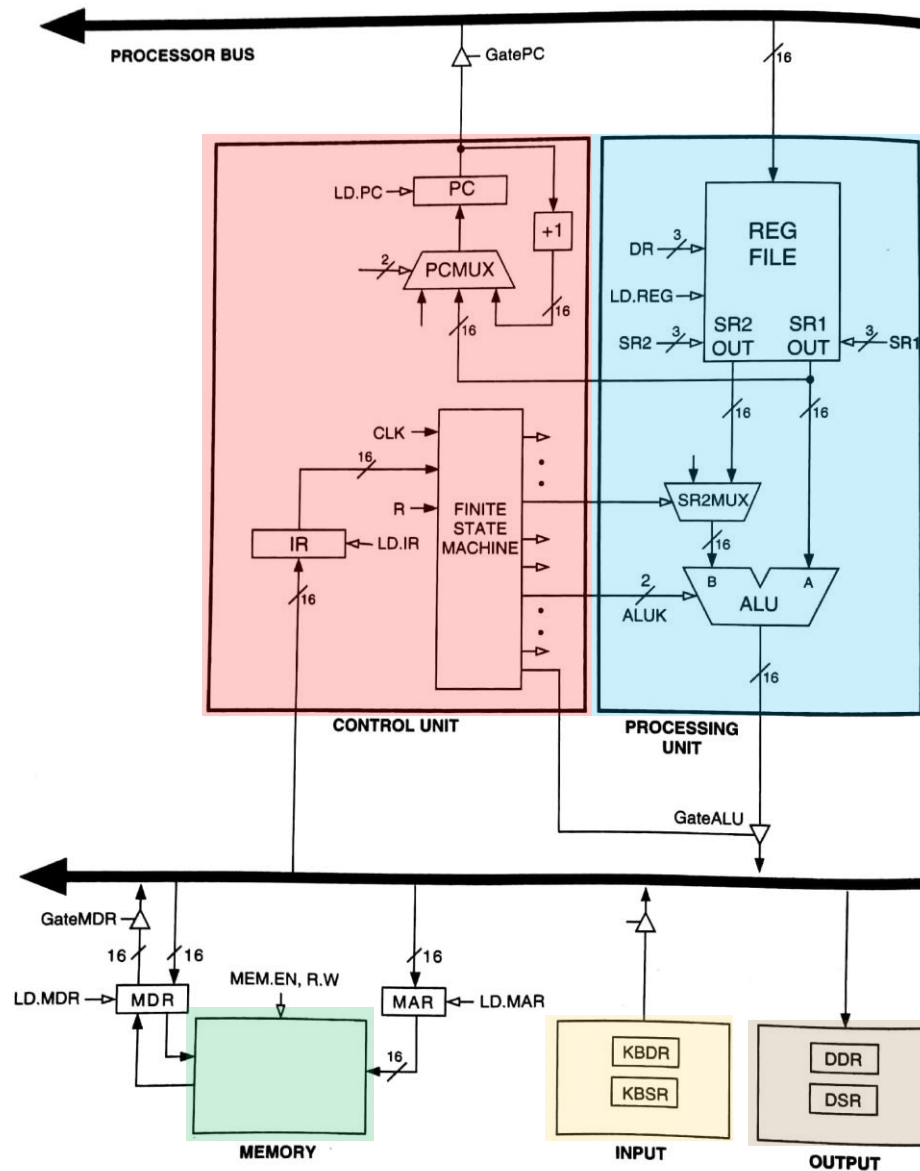
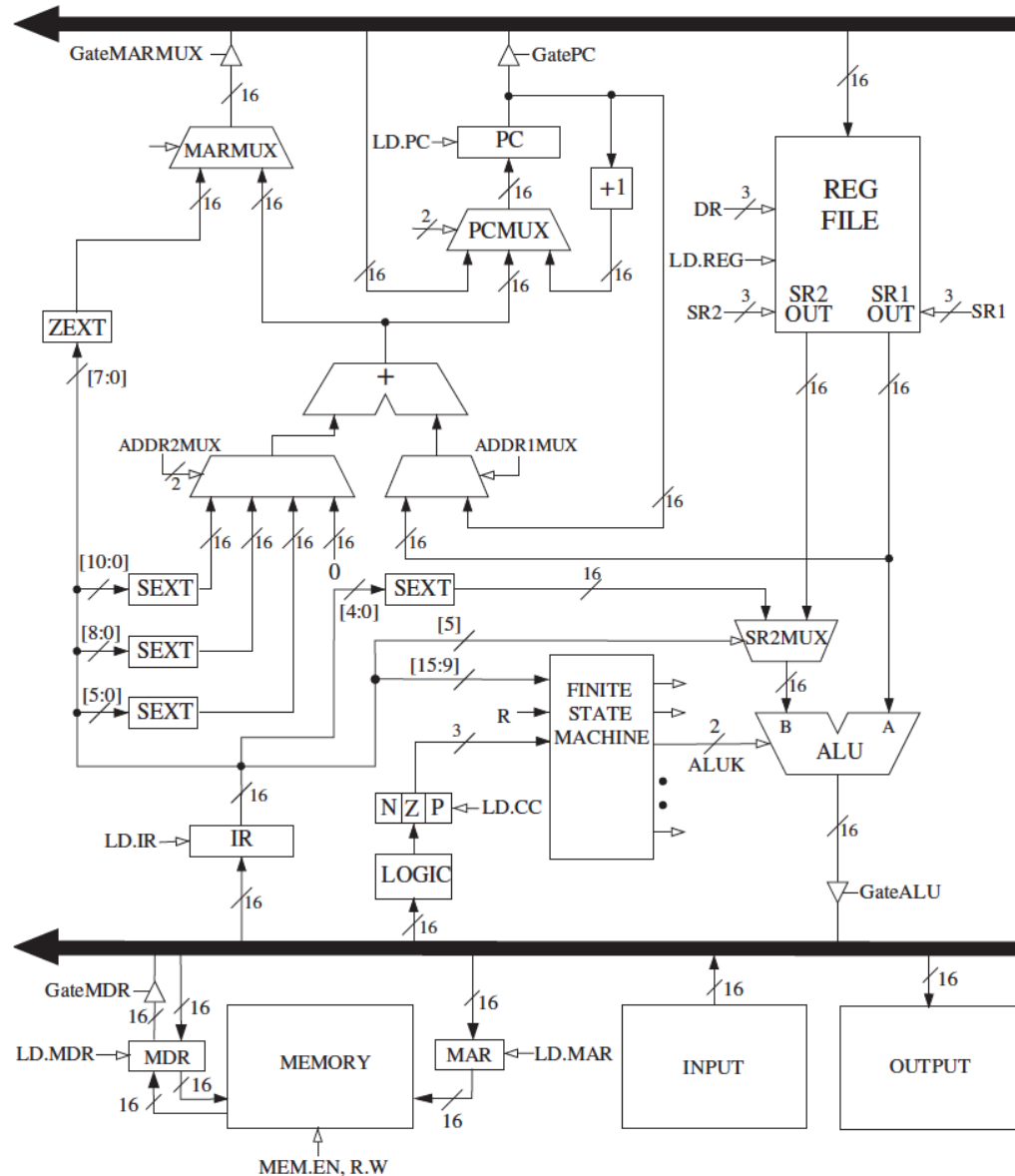


Figure 4.3 The LC-3 as an example of the von Neumann model

What is to Come: LC-3 Datapath



Digital Design & Computer Arch.

Lecture 3: Combinational Logic II and Sequential Logic

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